

INTRODUCTION TO CLINICAL RESEARCH

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OUTLINE

- What is Clinical Research?
- CR Scenario in India
- Drug development
- Phases in Clinical Research
- Clinical research Players, Pathways

CLINICAL RESEARCH

- **is scientific study that involves HUMANS**
- Individuals **volunteer** to participate in **carefully conducted studies** which ultimately uncover **improved methods and knowledge** on screening, diagnoses, treatment and prevention of disease and on the promotion of health and health care

INTERVENTIONAL	OBSERVATIONAL
• are controlled experiments, • Viz. Clinical trials, where every participant is allocated to the treatment or control group at the start of the study.	• projects where there is no investigational product assigned to participants
The WHO's International Clinical Trials Registry Platform states that 'interventions can include drugs, cells and other biological products, surgical procedures, radiologic procedures, devices, behavioural treatments, process-of-care changes, preventive care, etc'.	e.g. research on how therapies work, health economic research and qualitative clinical research such as understanding patients' experiences, etc.

TRENDS

Global Trends

- Drive for increased productivity & efficiency in the pharmaceutical industry
- Growth of BioTech sector & virtual pharma model
- Proven success of other outsourcing players

Indian Industry

- Established track record in Business process & IT outsourcing
- Amongst Top 3 in generic, API* sectors
- Evolution of IP protection

Core Competencies

- Well trained, multi-lingual, computer savvy biomedical workforce
- Cost effective workforce willing and able to adhere meticulously to protocols and processes
- LARGE HETEROGENEOUS “urban centric” patient pool

*API Active Pharmaceutical Ingredient: biologically active component of a drug product

Generic a medication with the exact same active ingredient as the brand-name drug, is taken the same way and offers the same effect.

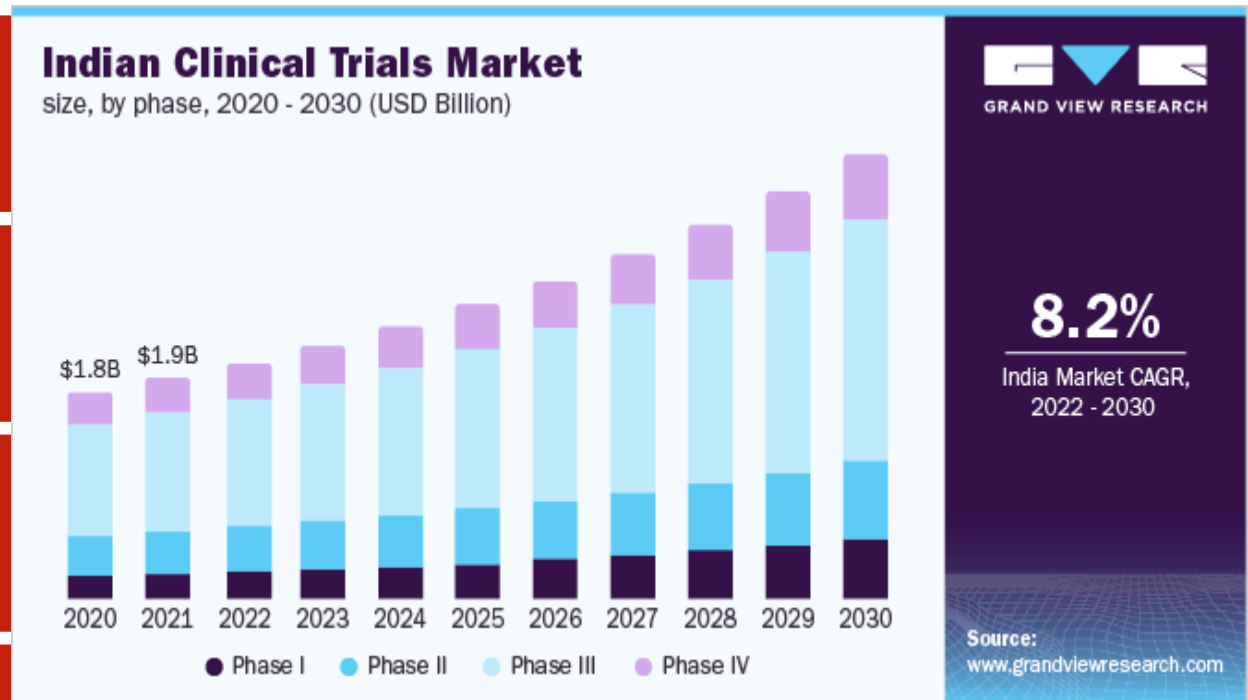
CLINICAL DEVELOPMENT SECTOR

India's clinical development sector

Revenues from contract R&D for international sponsors

Indian clinical trials market size was estimated at USD 1.93 billion in 2021 and is expected to reach USD 2.07 billion in 2022

CAGR of 8.2% from 2022 to 2030



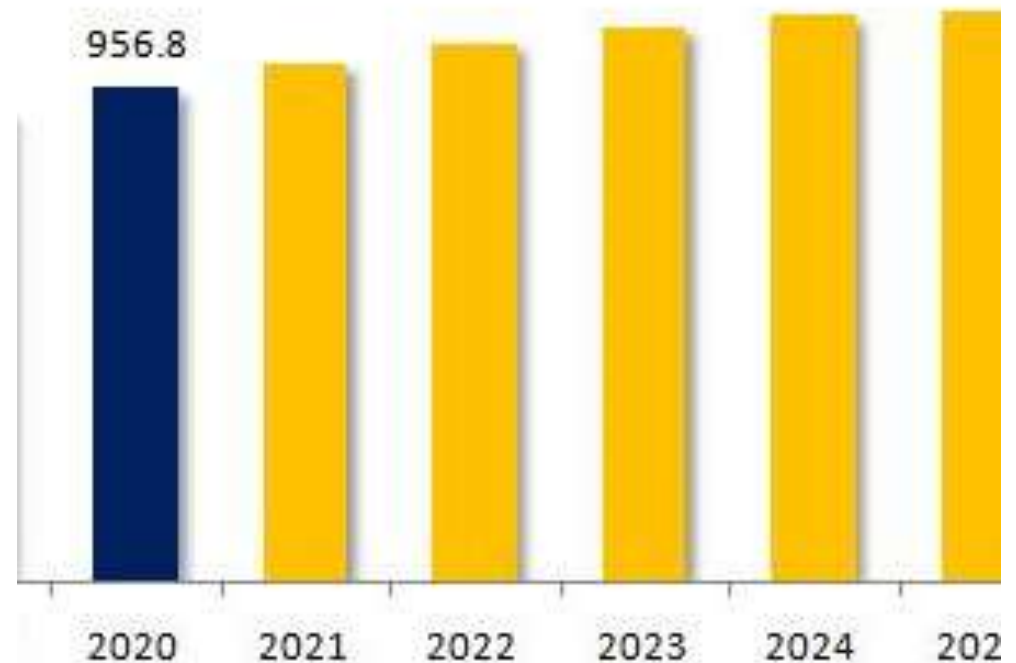
Nascent but dynamic CRO sector

India CRO Market has grown at a CAGR of 5.51%, in value terms, and is valued at **USD 956.8 million** in 2020

The market is further expected to reach USD 1,883.0 million by 2027 with a CAGR of 10.75% during the forecast period.

The major companies in the India CRO market are IQVIA Consulting and Information Services India Private Limited (Quintiles IMS Holdings, Inc.), PAREXEL International (India) Pvt Ltd., Syngene International Ltd., Pharmaceutical Product Development LLC, Siro Clinpharm Pvt.

India CRO Market, By Value (USD Million)





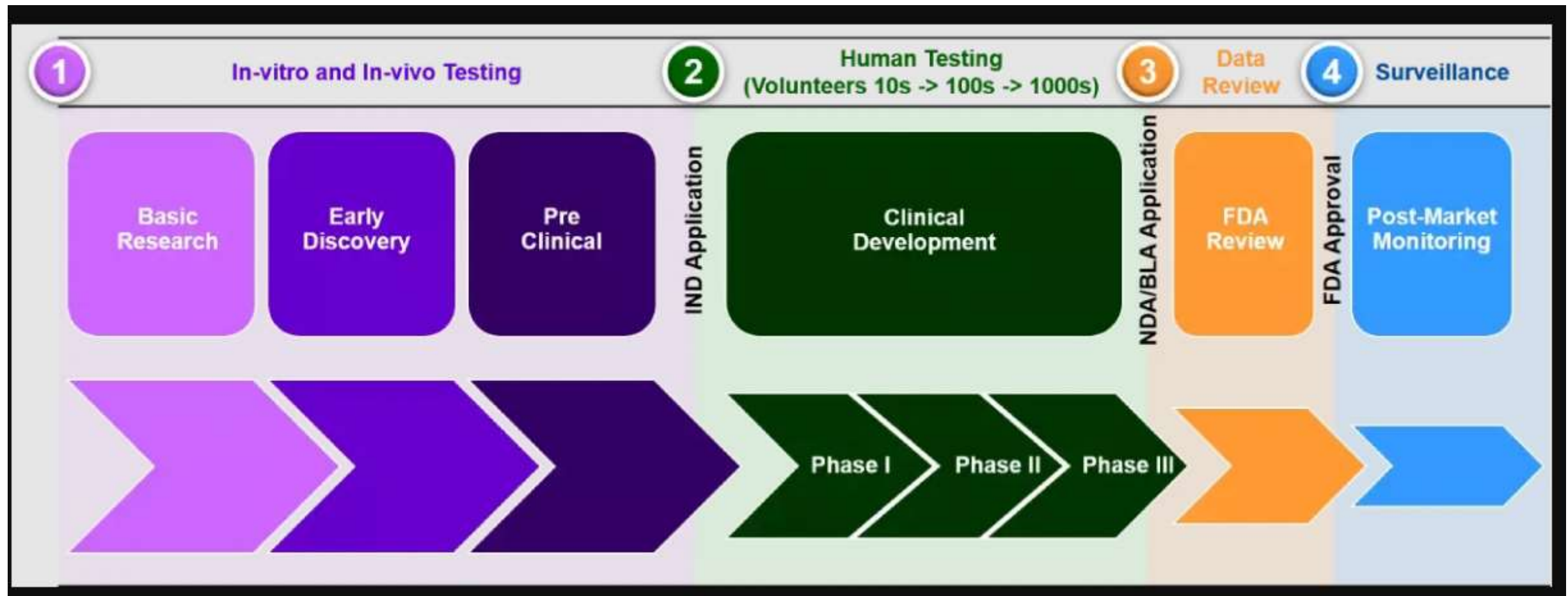
CLINICAL RESEARCH

Organized research on human beings intended to provide adequate information on the drug used as a therapeutic agent on its **safety and efficacy**

the drug's ability to produce the desired result (efficacy) and the type and likelihood of adverse effects (safety).

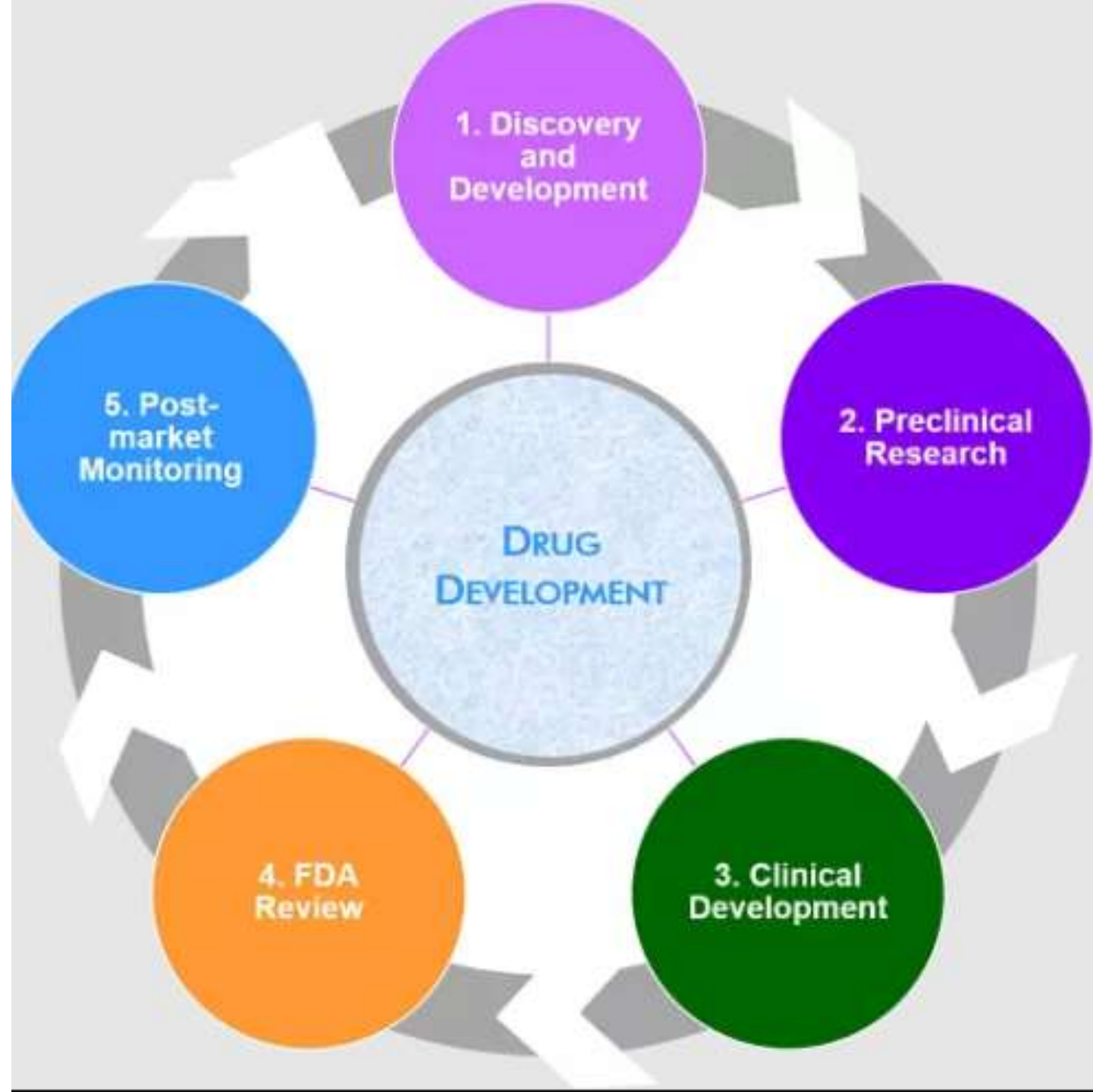
It is also referred to as **CLINICAL TRIAL MANAGEMENT**

PATHWAY TO RESEARCH



PHASES AND STAGES

- Step 1: Discovery and Development
- Step 2: Preclinical Research
- Step 3: Clinical Development
- Step 4: FDA Review
- Step 5: FDA Post-Market Safety Monitoring.



STEP 1: DISCOVERY & DEVELOPMENT

Phase	Target Discovery	Target Validation	Lead Generation & Refinement	Preclinical Development	
Goal	Find All Targets	Eliminate Wrong Targets	Generate Molecules	Eliminate Molecules	Advance Molecules
					



PROGRESS TIMELINE

- Historically, drug discovery, design, and development mostly started with identifying active ingredients from traditional medicines or purely by chance.
- Later, classical pharmacology was used to investigate chemical libraries including small molecules, natural products, or plant extracts, and find those with therapeutic effects.
- Post human DNA was sequencing, reverse pharmacology has found remedies to existing diseases through modern testing
- Disease processes, molecular compound tests, existing treatments with unanticipated effects, and new technologies spur drug discovery timeline through the cycle
- Today the steps in drug discovery and development involve screening hits, iterative medicinal chemistry, and optimization of hits to reduce potential drug side effects (increasing affinity and selectivity).
- **Efficacy or potency, metabolic stability (half-life), and oral bioavailability are also improved in these steps of the drug development process.**

- **Target Identification And Validation**

- Target identification **finds a gene or protein** (therapeutic agent) that plays a significant role in disease
- **record** the target's therapeutic characteristics
- **Criteria are:** efficacious, safe, usable, and capable of meeting clinical and commercial requirements
- **Validation of targets**, using modern tools and techniques such as disease association, bioactive molecules, cell-based models, protein interactions, signalling pathways analysis, functional analysis of genes, in vitro genetic manipulation, antibodies, and chemical genomics
- **EXAMPLE:** the Sanger Whole Genome CRISPER library and Duolink PLA are excellent sources for drug discovery targets
- https://www.sanger.ac.uk/news_item/2015-01-29-crispr-collaboration-aims-to-speed-drug-discovery/

- **Hit Discovery Process**

- Following target validation, compound screening assays are developed.

- **Assay Development And Screening**

- Assay development in drug discovery is a crucial component of drug discovery workflow. Assays are test systems that evaluate the effects of the new drug candidate at the cellular, molecular, and biochemical levels.

- **High Throughput Screening**

- High Throughput Screening (HTS) uses robotics, data processing/control software, liquid handling devices, and sensitive detectors to rapidly conduct millions of pharmacological, chemical, and genetic tests, eliminating hours of painstaking testing by scientists. HTS identifies active compounds, genes, or antibodies that affect human molecules.

- **Hit To Lead**

- In the Hit to Lead (H2L) process, small molecule hits from an HTS are evaluated and optimized in a limited way into lead compounds. These compounds then move on to the lead optimization process.

- **Lead Optimization**

- In the lead optimization (LO) process, the lead compounds discovered in the H2L process are synthesized and modified to improve potency and reduce side effects. Lead optimization conducts experimental testing using animal efficacy models and ADMET tools, designing the drug candidate.

- **Active Pharmaceutical Ingredients**

- Active pharmaceutical ingredients (APIs) are biologically active ingredients in a drug candidate that produce desired effects.
- All drugs are made up of the API or APIs and excipients. Excipients are inactive substances that deliver the drug into the human system.
- High Potency Active Pharmaceutical Ingredients (HP APIs) are molecules that are effective at much smaller dosage levels than standard APIs.
- They are classified based on toxicity, pharmacological potency, and occupational exposure limits (OELs), and used in complex drug development involving more than ten steps.

- **The drug discovery process gets narrowed when one lead compound is found for a drug candidate, and the process of drug development starts.**

STEP 2: PRECLINICAL RESEARCH

Once a lead compound is found,
**preclinical phase of drug
development** begins with in vivo
research to determine the efficacy
and safety of the drug.



Researchers determine the following
about the drug:

Absorption, distribution, metabolization, and excretion information	Potential benefits and mechanisms of action	Best dosage, and administration route	Side effects/adverse events	Effects on gender, race, or ethnicity groups	Interaction with other treatments	Effectiveness compared to similar drugs
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Preclinical Trials test the new drug on non-human subjects for efficacy, toxicity, and pharmacokinetic (PK) information.

These trials are conducted by scientists in vitro and in vivo with unrestricted dosages.

Absorption, Distribution, Disposition, Metabolism, & Excretion

- Absorption, Distribution, Disposition, Metabolism, & Excretion (ADME) is a Pharmacokinetic (PK) process of measuring the ways the new drug affects the body.
- ADME involves mathematical descriptions of each effect.

Proof Of Principle / Proof Of Concept

- Proof of Principle (PoP) are studies that are successful in preclinical trials and early safety testing.
- Proof of Concept (PoC) terminology is used almost interchangeably with PoP in drug discovery and development projects. Successful PoP/PoC studies lead to program advancement to the Phase II studies of dosages.

In Vivo, In Vitro, And Ex Vivo Assays

These three types of studies are conducted on the whole, living organisms or cells, including animals and humans; or using non-living organisms or tissue extract.

In vivo, preclinical research examples are the development of new drugs using mice, rat, and dog models.

In vitro is research conducted in a laboratory.

Ex vivo uses animal cells or tissues from a non-living animal.

- Examples of **ex vivo research** assays are finding effective cancer treatment agents; measurements of tissue properties (physical, thermal, electrical, and optical); and realistic modelling for new surgical procedures. In an ex vivo assay, a cell is always used as the basis for small explant cultures that provide a dynamic, controlled, and sterile environment.

In Silico Assays

- In silico assays are test systems or biological experiments performed on a computer or via computer simulation. These are expected to become increasingly popular with the ongoing improvements in computational power, and behavioral understanding of molecular dynamics and cell biology

Drug Delivery

- New drug delivery methods include oral, topical, membrane, intravenous, and inhalation. Drug delivery systems are used for targeted delivery or controlled release of new drugs. Physiological barriers in animal or human bodies may prevent drugs from reaching the targeted area or releasing when they should. The goal is to prevent the drug from interacting with healthy tissues while still being effective.

Oral: Oral delivery of medications is reliable, cost-effective, and convenient for patients.

- Oral drug delivery may not monitor precise dosages to the desired area but is ideal for prophylactic vaccinations and nutritional regimens. Delayed action, stomach enzyme destruction, absorption inconsistencies, or patients with gastrointestinal issues or upset can occur, and patients must be conscious during administration.

Topical: Topical drug delivery involves ointments, creams, lotions, or transdermal patches that deliver a drug by absorption into the body.

- Topical delivery is more useful for patient skin or muscular conditions — it is preferred by patients due to non-invasive delivery and their ability to self-administer the medicine.

Parenteral (IM, SC or LP Membrane): Parenteral drug delivery utilizes bodily membranes, including intramuscular (IM), intraperitoneal (IP), or subcutaneous or (SC).

- It is often used for unconscious patients and avoids epithelial barriers that are difficult for drugs to cross.

Parenteral (Intravenous):

Intravenous injection is one of the fastest drug delivery absorption methods. IV injection ensures entire doses of drugs enter the bloodstream

- it is more effective than IM, SC, or LP membrane methods.

Parenteral (Inhalation): Inhalation drug delivery gets the drug rapidly absorbed into the mucosal lungs, nasal passages, throat, or mouth.

- Problems with inhalation delivery include difficulty delivering the optimum dosage due to small mucosal surface areas and patient discomfort. Pulmonary inhalation drug

Formulation Optimization & Improving Bioavailability

- Formulation optimization is ongoing throughout pre-clinical and clinical stages. It ensures drugs are delivered to the proper place at the right time and in the right concentration. Optimization may include overcome solubility

STEP 3: CLINICAL DRUG DEVELOPMENT PROCESS

- preclinical research is complete, researchers move on to clinical drug development, including clinical trials and volunteer studies to finetune the drug for human use
- **Complexity Of Study Design, Associated Cost & Implementation Issues**
- The complexity of Clinical Trial design and its associated costs and implementation issues may affect trials carried out during this phase. Trials must be safe and efficacious and be completed under the drug development budget, using a methodology to ensure the drug works as well as possible for its intended purpose. This rigorous process must be set up correctly and enroll many volunteers to be effective.
- **Clinical Trials– Dose Escalation, Single Ascending & Multiple Dose Studies**
- Proper dosing determines medication effectiveness, and clinical trial examine dose escalation, single ascending, and multiple dose studies to determine the best patient dosage.



PHASE I HEALTHY VOLUNTEER STUDY

This phase is the **first time the drug is tested on humans**

Number of volunteers: less than 100 volunteers

Benefits: will help researchers assess the safety and pharmacokinetics, absorption, metabolic, and elimination effects on the body, as well as any side effects for safe dosage ranges.



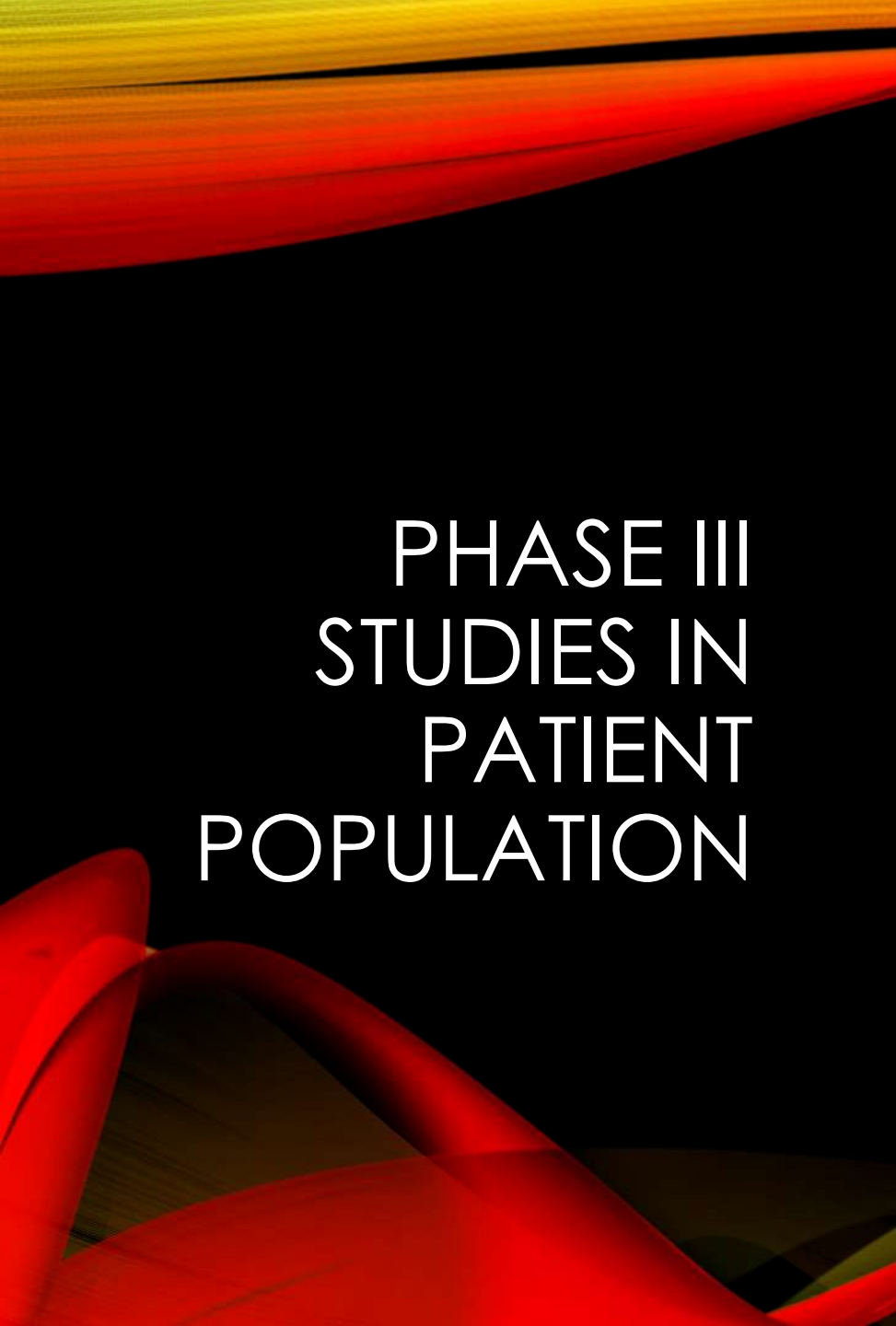
PHASE II STUDIES IN PATIENT POPULATION

assesses drug safety and efficacy

Number of patients: 100-500 patients

Trial based: population receives may receive a placebo or standard drug previously used as treatment.

Analysis of optimal dose strength helps **create schedules** while adverse events and risks are recorded.



PHASE III STUDIES IN PATIENT POPULATION

enabling medication labelling and instructions for proper drug use.

Patient pool: 1,000-5,000 patients

Trials require **extensive collaboration, organization**, and Independent Ethics Committee (IEC) or Institutional Review Board (IRB) coordination and regulation in anticipation of full-scale production following drug approval.

Biological Samples Collection, Storage & Shipment

- During clinical trials, biological samples are collected, stored, and shipped from testing sites according to global standards and regulations. Transport containers of biological samples may include dry ice packs or other temperature stabilizing methods.
- Different requirements apply to different types of biological samples.

Pharmacodynamic (PD) Biomarkers

- PD biomarkers are molecular indicators of the drug's effects on the target human area, and link drug regimen and biological responses.
- This data can help select rational combinations of targeted agents and optimize drug regimens and schedules. Rationality and hypothesis-testing power are increased through the use of PD endpoints in human trials.

Pharmacokinetic Analysis

- Pharmacokinetic Analysis is an experimental trial that determines the theory of how a new drug behaves in the human body.
- The volume of distribution, clearance, and terminal half-life are defined through compartmental modeling.

Bioanalytical Method Development And Validation

- Bioanalytical methods detect analytes and metabolites such as drug or biomarkers in biological or human samples to determine drug efficacy and safety.
- The complete bioanalytical assay consists of sample collection, clean-up, analysis, and detection.

Drug (Analyte) & Metabolite Stability In Biological Samples

- Stability is important in determining human drug efficacy, and biological samples are required.
- Drug and drug metabolites are susceptible to degradation, which can lower drug concentration over the life of the drug.

Blood, Plasma, Urine & Feces Sample Analysis For Drug And Metabolites

- Biological samples used in clinical trials include blood, plasma, urine, and feces to determine and analyze various properties and effects of the drug and its metabolites on humans.

Patient Protection – GCP, HIPAA, & Adverse Event Reporting

- Human patients must always be protected during clinical trials, and Good Clinical Practices (GCP), the Health Insurance Portability and Accountability Act (HIPAA), and adverse event reporting to IEC/IRB regulates and ensures their safety.

STEP 4: FDA REVIEW

Once the new drug has been formulated for its best efficacy and safety, and the results from clinical trials are available, it's advanced forward for wholistic FDA review. At this time, the FDA reviews and approves, or does not approve, the drug application submitted by the drug development company.



Regulatory Approval Timeline

The new drug regulatory approval timeline may be standard, fast track, breakthrough, accelerated approval, or priority review depending on its applications and necessity for patients.

If standard or priority review is required, the approval timeline may be up to an year. Fast track, breakthrough, or accelerated approvals may occur sooner.



IND Application

IND applications are submitted to the FDA before starting clinical trials.

If clinical trials are ready to be conducted, and the FDA has not responded negatively about the drug, developers may start the trials.

- **Accelerated Approval**

- New drugs may be granted accelerated approval if there is strong evidence of positive impact on a surrogate endpoint instead of evidence of impact on actual clinical benefits the drug provides.
- Expedition of approval means the medication can help treat severe or life-threatening conditions.

- **Reasons For Drug Failure**

- New drug applications may fail for a variety of reasons, including toxicity, efficacy, PH properties, bioavailability, or inadequate drug performance.
 - **Toxicity:** If the toxicity of a new drug is too high in human or animal patients, the drug may be rejected due to safety concerns about its use following manufacture.
 - **Efficacy:** If a new drug's efficacy is not high enough or evidence is inconclusive, the FDA may reject it.
 - **PK Properties or Bioavailability:** PK properties or poor bioavailability due to low aqueous solubility, or high first-pass metabolism, may also cause a drug to fail FDA review. PK causes of drug failure include inadequate action duration and unanticipated human drug interactions.
 - **Inadequate Drug Performance:** If the new drug performs the desired function, but only at a shallow level, the FDA may reject the application in favor of a formulation that performs better.

- **NDA / ANDA / BLA Applications**

- An NDA abbreviated new drug application (ANDA), or Biologics License Application (BLA) is submitted to the FDA after clinical trials demonstrate drug safety and efficacy.
- The FDA reviews study data and decides whether to grant approval or not. Additional research or an expert advisory panel may be required before a final decision is made.

- **Orphan Drug**

- An orphan drug is intended to treat disease so rare that financial sponsors are unwilling to develop it under standard marketing conditions. These drugs may not be approved quickly or at all.

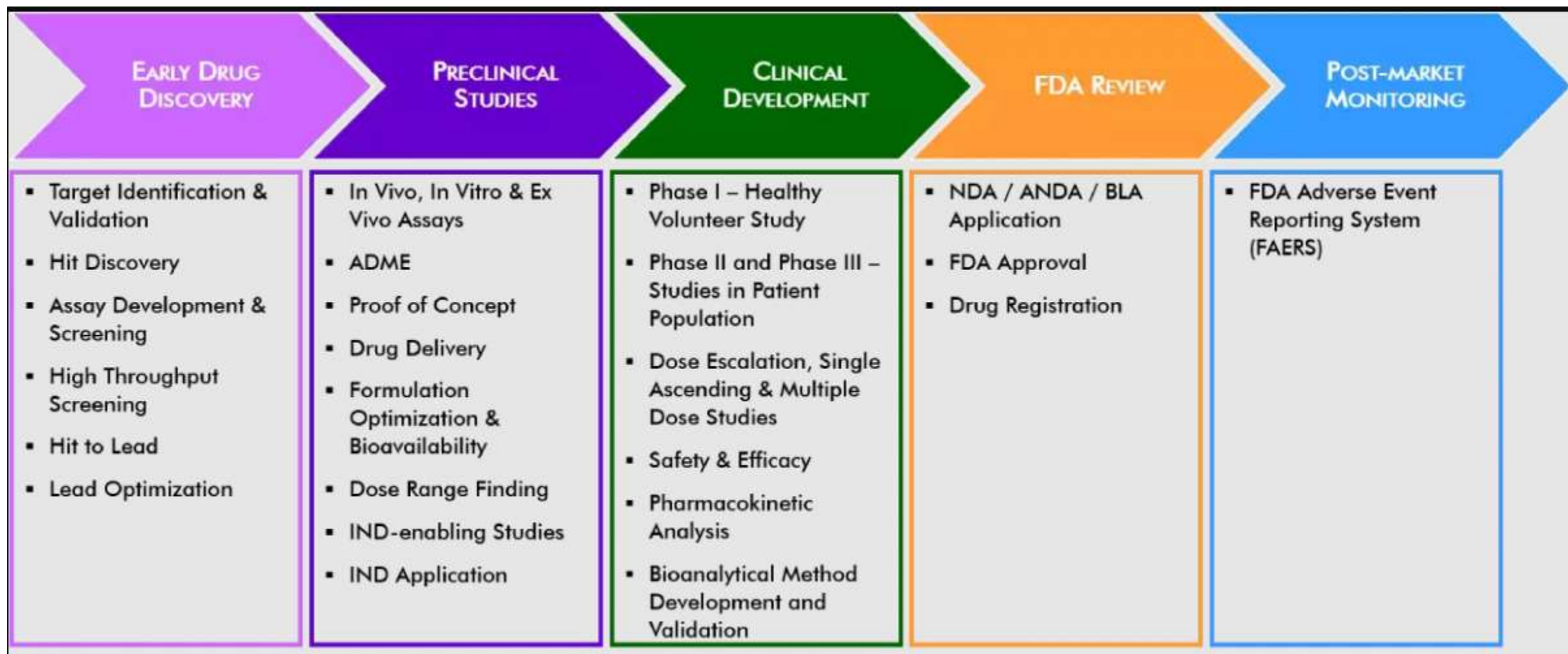


STEP 5: POST-MARKET MONITORING

Following drug approval and manufacturing, the FDA requires drug companies to monitor the safety of its drug using the **FDA Adverse Event Reporting System (FAERS) database**.

FAERS helps FDA implement its **post-marketing safety surveillance program**.

Through this program, manufacturers, health professionals, and consumers report problems with approved drugs.





NEED FOR CLINICAL TRIALS

New drugs

New uses of existing drugs

Medical devices

New drug delivery systems

Evidence to prove the efficacy and safety in Humans

Only a well designed clinical study on a defined population can give meaningful results (positive or negative) about any therapeutic intervention

SCOPE OF CLINICAL RESEARCH

Total cost of drug development

- cost per drug developed and approved by a single-drug company was \$350 million

Time to develop a new drug

- 9-10 years

2/3rd of the development time and cost is spent on the Clinical trials



REGULATORY AFFAIRS

Drugs and comestics Act 1947

SCHEDULE-Y

Indian GCP Guidelines

ICMR Guidelines

WHO Guidelines

ICH-GCP Guidelines



A MULTIDISCIPLINARY APPROACH

Sponsor

Investigators

Monitors

Auditors and QC personnel

Biostatisticians

Data Management group

Regulatory affairs

Reporting and Documentation medical writing

Business development group

PLAYERS IN CR INDUSTRY



Pharma companies



Clinical/ Contract research Organizations
(CROs)



Biotech companies



Central diagnostic laboratories



Clinical research Training institutes

CLINICAL TRIALS



CONDUCTS

Investigators
Principal
Co-
investigator



PARTICIPATES

Healthy
volunteers
Patients

CR MAJOR PLAYERS

Some prominent players in the Indian clinical trials market include:

IQVIA HOLDINGS INC.

PAREXEL International Corporation

Pharmaceutical Product Development (PPD) LLC

Charles River Laboratory

ICON PLC

PRA Health Inc.

Chiltern International LTD.

Syneos Health Inc.

SGS SA

Syngene International Limited

Aurigene Discovery Technologies Limited

Aragen Life Sciences

Abiogenesis Clinpharm Pvt Ltd

Accelsiors

Accutest Global.

Clario

Cliantha Research

Cliniminds

CliniRx

JSS Medical Research

CR CAREER PATHS

Clinical operations/ CROs/Pharma Co.	Site Management Organisations	Data management	CR training institutes
Clinical Trial Assistant (CTA)	Clinical Research Coordinator (CRC) / Study Coordinator	Data entry operator	Trainer-CTM
Clinical research associate (CRA)	Principal & Co Investigators	Data Manager	Trainer- Data Mgt
Senior CRA	Medical Monitors	Data Validation Executive	Trainer – BioStats
Clinical Team Leader	Project Manager	QA Executive	Trainer- GCP regulation
Project Manager	Medical & regulatory Manager	QA Manager	Trainer- Proj Mgt
Senior Proj Manager	Business Dev Manager	Programmer	Training Coordinators
QA Manager	Medical director	Statistician	Training Director
Medical Director	Associate director Clinical	Date Reviewer	
Associate Director- clinical	Associate director Proj	Database designer	
Associate Director- projects	Director/Head Clinical Ops	Medical writer	
Director- Business Dev	General Manager/CEO/President	Head data management	
Director Clinical Ops			
General Manager/CEO/President			



SUMMARY

Clinical research is an integral part of drug development

To date, it has gained a unique position due to the regulatory requirements and ethical guidelines available globally

Designing, conducting, monitoring, appropriate QA and data management determine the success

Abstract geometric lines in the top left corner of the slide, consisting of several thin, light gray lines that intersect to form various polygons and shapes.

PRECLINICAL TESTING

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WHAT IS DEFINED AS PRECLINICAL TESTING ?

- Research using **animals** to test if a drug, procedure, or treatment is likely to be useful
- Preclinical studies take place **prior** to testing in/on humans
- In drug development, **preclinical development**, also termed **preclinical studies** or nonclinical studies, is a stage of research that begins before clinical trials and during which important **feasibility, iterative testing and drug safety data** are collected, typically in laboratory animals.

OVERVIEW

- The drug development process has several stages which can take decades to complete.
- Understanding the stages novel drugs must go through is essential to understand how new therapeutics and vaccines enter the market.
- Initially, pharmaceutical companies focused on curing diseases rather than treating them
- In the advent of systematic drug development, active ingredients for therapeutics came into the picture and companies started to manufacture them at scale
- pharmaceutical product development accelerated, allowing treatment for chronic diseases or those without a cure
- drug development is a hugely expensive business costing upto to USD 350M to USD 2B
- only around 4% of the drugs launched into the marketplace make a return on their initial investment.

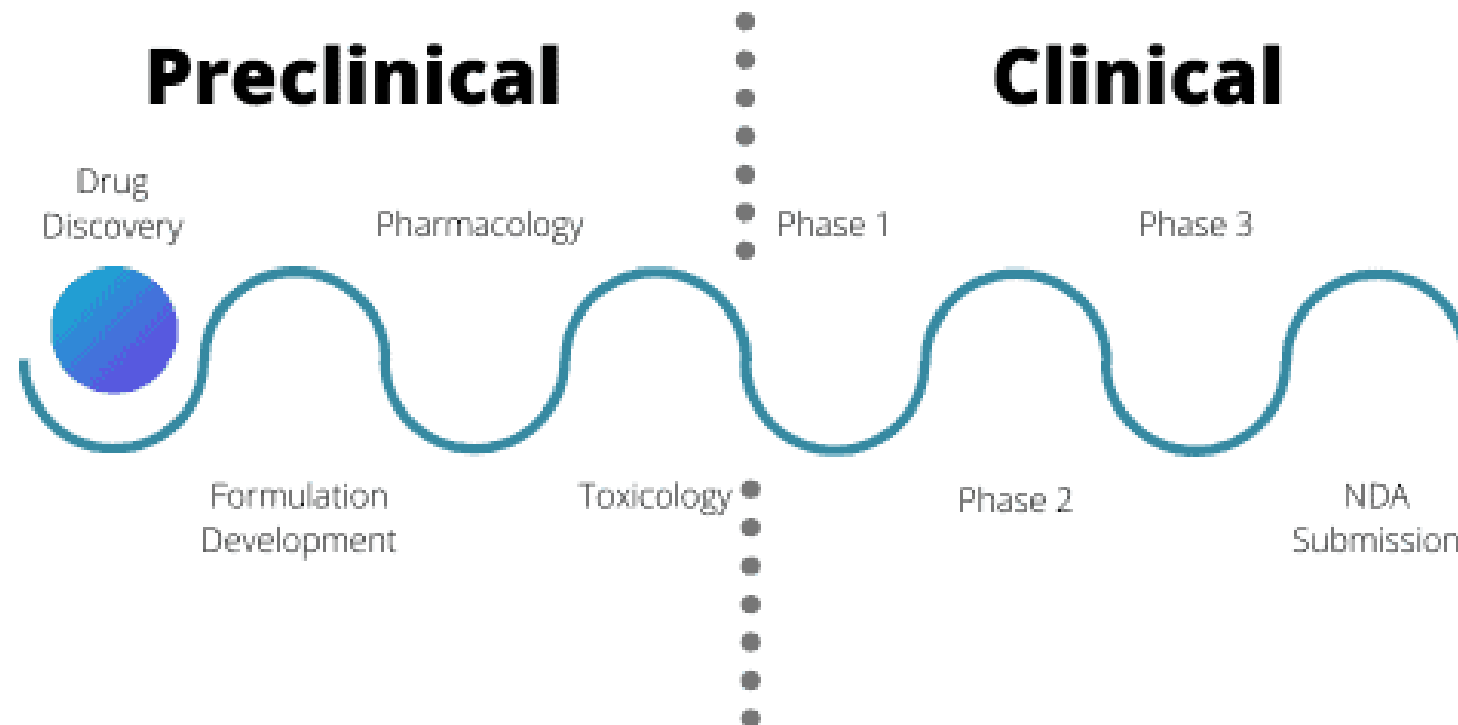
THE DRUG DEVELOPMENT CYCLE

There are two broad sections to the drug development cycle: the **preclinical phase** and the **clinical phase**.

Generally, biotechnological companies specialise in the preclinical phase and develop numerous drug candidates, running preliminary studies to test viability.

Once a compound has reached a specific stage, they will partner with pharmaceutical companies to develop the candidates further or sell the assets on.

DRUG DISCOVERY



DRUG DISCOVERY

- Much of the boom in drug discovery and pharmaceutical product development is due to advances in chemistry and compound screening development.
- Combinatorial chemistry is responsible for the generation of vast compound libraries that can be screened.
- These libraries are searchable with High-throughput screening (HTS): an automated process to look for hits, where each screen is an assay for a particular target, such as a protein
- HTS generates a vast amount of data and can collect millions of data points.
- The analysis at this stage is often to detect an increase or decrease from the baseline normal for the target.
- For protein and biomarker targets, expression can be altered and up- or down-regulated. The 'hit' gives a starting point for which compounds to focus on.

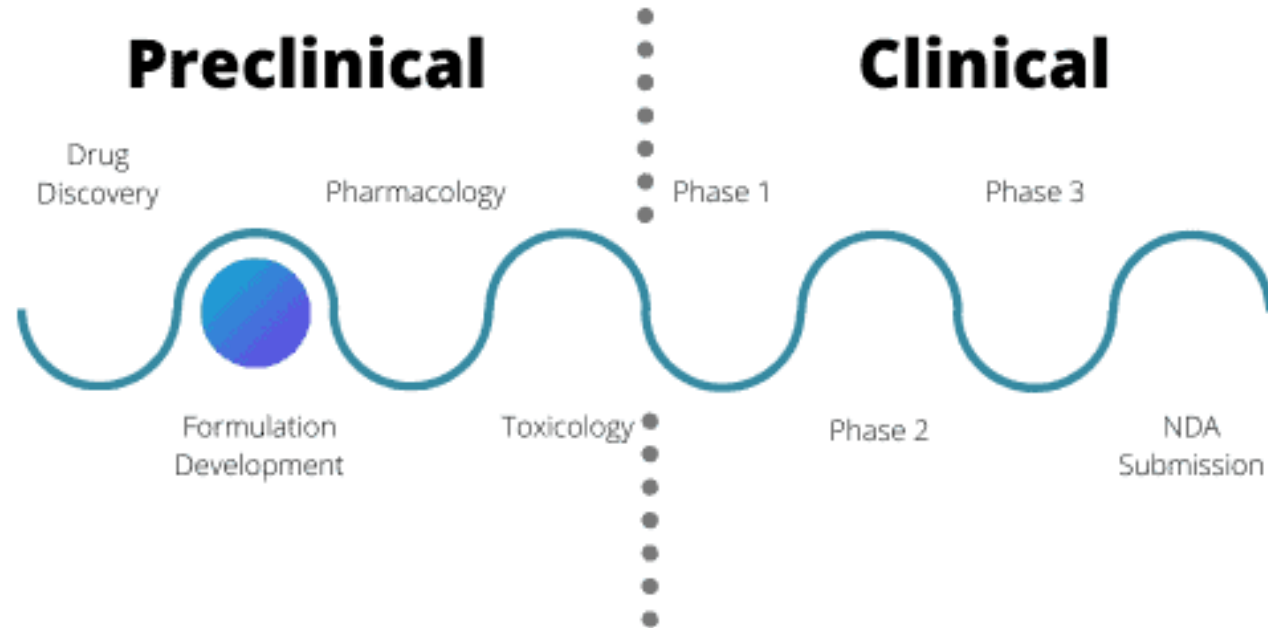
DRUG DISCOVERY

- HTS is, however, a high effort, low yield activity in the preclinical phase. This is because researchers use it to test thousands of compounds against the targets to find potentially hundreds of candidates, with no statement on their efficacy.
- pharmaceutical companies create pipelines which are built around the drugs created rather than pipelines for a specific target disease
- researchers identify surrogate markers for an illness, such as specific but unrelated biomarkers, instead of biologically relevant ones.
- HTS use has decreased in favour of more rational drug design including knowledge of biological targets, so drugs can be designed in the preclinical phase with specific biomarkers in mind

PRECLINICAL DEVELOPMENT

- Post identification a potential drug via the research and development stages, preclinical testing can occur
- For preclinical and clinical testing, agencies worldwide regulate and monitor the industry.
- An **Investigational New Drug Application (IND)** is a request from a clinical study sponsor to obtain authorization from the **Food and Drug Administration (FDA)** to administer an investigational drug or biological product to humans.
- These agencies process Investigational New Drug (IND) applications which, once approved, allow clinical trials to proceed.
- There are three parts in IND applications:
 - formulation development
 - pharmacology
 - toxicology

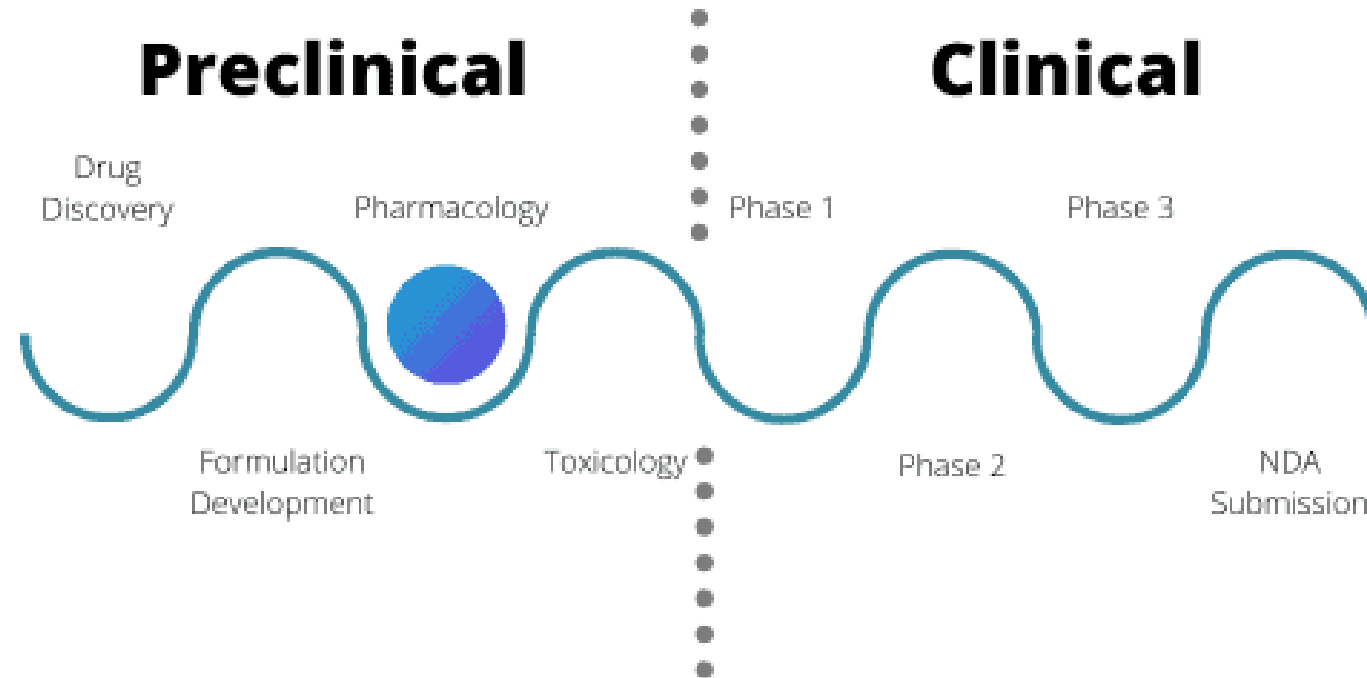
FORMULATION DEVELOPMENT IN PRECLINICAL PHASE



FORMULATION DEVELOPMENT IN PRECLINICAL PHASE

- Formulation development assesses the best way to prepare a drug in the preclinical phase for its intended clinical use in patients
- Factors such as solubility, frequency and mode of administration, stability of the formula, and palatability are all assessed.
- This takes into consideration where the disease or problem occurs, such as the sinuses or tumour cells
- Developing the formula is only half the story – manufacturing is also vital here.
- At the R&D stage, a drug may only be produced in milligram quantities, but for development, kilograms are needed.
- Manufacturing needs to adhere to Good Manufacturing Procedures (GMP) and must test every batch of the potential drug to ensure the purity, quality and potency.

PHARMACOLOGY



PHARMACOLOGY

- The pharmacology stage assesses the safety of a drug as well as its ADME – Absorption; Distribution; Metabolism; and Excretion.
- ADME is the backbone of pharmacology.

ADME

- **Absorption** concerns the bioavailability of the drug once administered. Before most drugs can achieve their goal, the bloodstream carries them away.
- The bioavailability of the drug measures the fraction of an administered drug dose that reaches the target system, while also measuring the uptake in the target cells and organs.
- Once the body has absorbed a drug, the latter distributes from one part of the body to another.
- **Metabolism** studies the metabolites produced by the drug's breakdown. It then evaluates whether they are active or harmful, and the breakdown's location in the body
- **Excretion** looks at how the drugs and the metabolites produced exit the body.
- While ADME is not a regulatory requirement for a first-time, in man (FTIM) study, having conducted it helps improve the quality of the safety information and give further information to **pharmacology and toxicology studies**.

PROOF OF PRINCIPLE/ CONCEPT

- At the early stage of a program, human genetic evidence linking a proposed target to a disease provides **powerful evidence supporting 'proof of concept'**.
- PoP are an early stage of clinical drug development when a compound shows potential in animal models and early safety testing
- This can also assist in defining initial patient populations for study.
- There are several elements needed to ensure “proof of concept” has been obtained in preclinical in vitro and in vivo studies. These involve the following areas:

Experimental design:

- Ensuring experiments are performed rigorously– hypothesizing after results (HARKing) or p-hacking
 - Pre-clinical target validation studies should demonstrate
 - (i) that the target of interest has potential for a biologically meaningful effect on modulating disease onset or progression,
 - (ii) that the target can be modulated therapeutically.
- Subjective assays performed by blinded investigators
- Having the appropriate controls (negative/vehicle and positive controls)

PROOF OF PRINCIPLE/ CONCEPT

- Experiments are repeated
- All the data is shown and analysed
- Appropriate statistical analysis
- Using the appropriate tool molecule for testing (confidence in potency and selectivity)

***In vivo* studies require:**

- Randomisation of animals
- Understanding pharmacokinetics [PK] (dose, schedule and exposure)
- Delineating the pharmacokinetics and pharmacodynamics (PKPD) relationship
- Ensuring a suitable formulation
- Ensuring adequate sample size as determined by pre-specified power calculation
- Ensuring there is adequate sampling (samples should also be taken for PK, tumour biomarker assessment and other tissue biomarkers)
- Using the appropriate tool molecule for testing (confidence in potency and selectivity)

PROOF OF PRINCIPLE/ CONCEPT

Considerations regarding choice of most appropriate *in vivo* preclinical model:

- Model should be chosen based on the molecule's mode of action and potential clinical indication
- Model should be well characterised (gene expression, mutational analysis, immunophenotype)
- The molecule required for “bench-marking”/differentiation has previously been examined in the chosen model
- Will this therapy be developed as a monotherapy or part of a combination therapy?
- When using human-drugs for testing or bench-marking in animal studies, have those reagents been demonstrated to be active and/or equi-potent on the target in the proposed test species as for the human target?
- Testing targeted therapies require models demonstrated to be expressing relevant levels of the target of interest (e.g. cell line derived or patient-derived xenograft models)
- Testing immune therapies require models with a functional immune system (e.g. syngeneic or humanised models)
- The model of choice must be widely understood and accepted as having reliably predicted human clinical responses making it clear how to interpret and use the data generated

SAFETY

- The safety assessment of a drug monitors both pharmacodynamic (PD) and pharmacokinetic (PK) interactions
- PD interactions are where the drug administered can affect the actions of another specified drug without affecting its concentration, such as warfarin and antibiotics used in tandem
- PK interactions are where the drug administered can affect the actions of another specified drug by affecting its concentration or that of its metabolites, such as the relationship between alcohol and paracetamol.
- Safety plays a huge part in the preclinical stage of pharmaceutical product development: researchers must test any drugs vying for IND status extensively before allowing human testing.
- Researchers can use computer modelling and simulations to an extent, however, they still require animal models to assess safety.
- This is because they will assess the pharmacodynamics of the drug on a number of the major body systems such as:
 - the cardiovascular and respiratory systems
 - renal functionality and intestinal transit

TOXICOLOGY

- Since researchers need to conduct IND tests in animal models, they must conduct studies in two species
- This part of the testing is highly regulated
- Researchers must use one species that is **non-rodent** and their use of **primates** as an animal model is heavily restricted
- To study the effect of longer-term drug exposure on the body, including repeat dose studies
- Pharmaceutical companies deliberately conduct studies over longer time periods than a human would take the drug for, with the duration of the study dictated by the anticipated exposure time that a human would face
- For example, if the drug regime calls for daily doses for a seven-day period in humans, animal models undergo the treatment for four weeks. Similarly, a 30-day repeated dose in humans would call for a three-month trial in both animal species
- During these studies, researchers assess parameters such as food and water consumption, body weight, hematology and urine analysis. These aim to monitor for any adverse effects that could arise as a result of taking the drug. Researchers will also monitor immune responses, such as infection occurrences, tumour incidences and histological changes in the immune system.
- **60% of potential drugs fail in the preclinical phase of pharmaceutical product development**

ANALYSIS AND BIOINFORMATICS

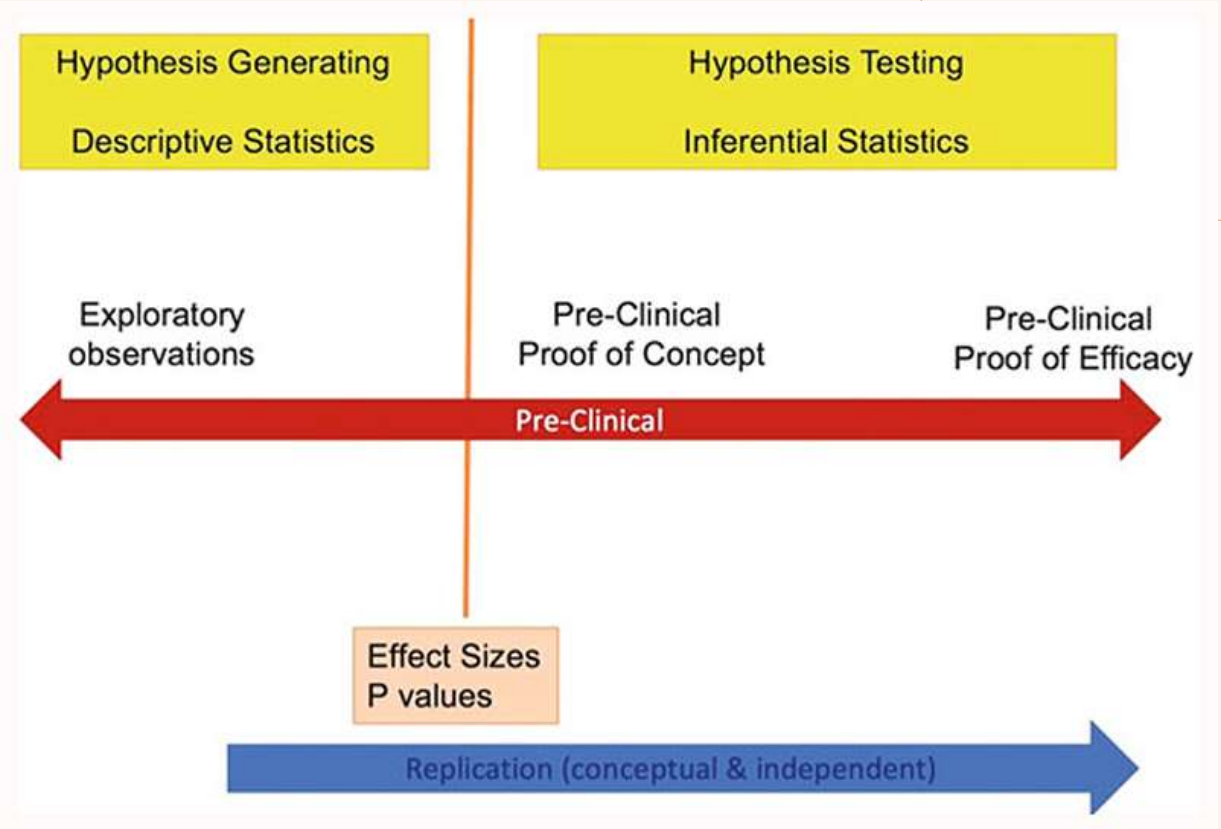
- Bioinformatics is core to every stage of the drug development process
- For example, **data landscaping or mining** of available databases or analysis of in-house HTS records can massively help at the R&D stage.
- For preclinical development on the other hand, bioinformatics can help give more insight into the method of action of a therapeutic
- Toxicology testing provides researchers with context for up- and down-regulated genes – whether the change in regulation could be harmful, whether there are associations that need further investigation.
- Additionally, they can glean more detailed information at this stage to guide research and phased clinical trials.

IN VIVO & IN VITRO STUDIES

<i>In vivo (Animal-based)</i>	<i>In Vitro (Cell-based)</i>
take place within a living organism	in vitro studies, cell lines are derived from either humans or non-human animals and are introduced to the new pharmaceutical under development within a Petri dish or test tube
they are able to demonstrate the impact of a pharmaceutical on the body as a whole	Demonstrate impact of a pharmaceutical isolated cells
better visualization of potential interactions, which can improve its predictions of safety, toxicity, and efficacy.	do not cause harm to the animal or person that the cell cultures
This helps scientists predict the impact of candidate drugs on human disease	limited as they cannot model how a pharmaceutical compound may interact with all the molecules and cell types that exist within a complex organ
Drawbacks: significant ethical concerns, particularly for preclinical studies where just animal models are permitted	free of the drawbacks of animal testing.
They are considerable expensive	relative cost effective in set up and running; they are also reliable and efficient, and produce robust results

PRECLINICAL STUDY DESI

- preclinical research can be classified into two distinct categories: **“hypothesis generating” (exploratory)** and **“hypothesis testing” (confirmatory)** research



Comparison of exploratory (hypothesis generating) and confirmatory (hypothesis testing) preclinical studies.

Descriptive statistics describes data and provides descriptions of the population, using numerical calculations, graphs, and tables. In contrast, inferential statistics predicts and infers

about a population using a sample of data from the population, therefore one can take data from samples and make generalisation about a population

PRECLINICAL STUDY DESIGN

- Preclinical studies using animals to study the potential of a therapeutic drug or strategy are important steps before translation to clinical trials
- However, **poor quality in the design and conduct of these studies** has not only impeded clinical translation but also led to significant waste of valuable research resources.
- guidelines such as **ARRIVE (Animal Research: Reporting of In Vivo Experiments) and reporting checklists (such as the Nature Research Reporting Summary**, which is included in every research article published by Nature Biomedical Engineering) prompt authors to provide details about **study design, methods, protocols, materials, animal models, data acquisition and processing, and software used.**
 - <https://arriveguidelines.org/arrive-guidelines/study-design>
 - <https://www.nature.com/documents/nr-reporting-summary-flat.pdf>

REGULATORY REQUIREMENTS

- What are the requirements for preclinical and clinical trial protocols? Who must approve the protocols?
 - The preclinical trials (or studies on animals) are regulated by **The Prevention of Cruelty to Animals Act, 1960 (“PCA Act”)** and **The Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998 (“BEACS Rules”)** made under the PCA Act.
 - The BEACS Rules stipulate that any protocol for preclinical studies must ensure that
 - animals lowest on the phylogenetic scale which may give scientifically valid results should be first considered for any experimental procedure
 - the experiment should be designed using minimum number of animals to give statistically valid results at 95% degree of confidence,
 - replacement alternatives not involving experiments on animals have been given due and full consideration
 - sound justification is provided where alternatives not involving experiments on animals, though available, are not used
 - Before conducting any experiment animals, an experiment specific permission of the Committee for Purpose of Control and Supervision of Experiments on Animals (“CPCSEA”) or an Institutional Animals Ethics Committee (“IAEC”) recognized by CPCSEA is required
 - The CPCSEA or IAEC reviews each protocol for meeting the requirements stipulated above

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Two thin orange lines intersect on the left side of the slide. One line is horizontal, and the other is diagonal, crossing it.

SUMMARY

- Preclinical testing involves **lab and animal studies** that evaluate the drug's **toxic and pharmacologic effects**.
- The **IND application** must contain information in three broad areas: **Animal Pharmacology and Toxicology Studies** - Preclinical data to permit an assessment as to whether the product is reasonably safe for initial testing in humans.
- The preclinical trials (or studies on animals) are regulated by **The Prevention of Cruelty to Animals Act, 1960 (“PCA Act”)** and **The Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998 (“BEACS Rules”)** made under the PCA Act.

Abstract geometric lines in a light gray color, forming various polygons and overlapping shapes on the left side of the slide.

THANK YOU



TERMINOLOGIES & DEFINITIONS IN CLINICAL RESEARCH

Ms. Sonali Correa

Ph.D., M.Sc.

- **Abbreviated protocol**

A summary protocol containing brief information about the study which may be used as a concept document to **get internal approval to start the planning phase of a study.**

It may also be used as an **initial discussion document for a meeting with a prospective Principal Investigator.**

- **Accepts healthy volunteers**

A type of **eligibility criteria** that indicates whether people who do not have the condition/disease being studied can participate in that clinical study.

- **Active comparator arm**

An arm type in which a **group of participants receives** an intervention/treatment considered to be effective (or active) by health care providers.

- **Adverse event**

An unfavorable change in the health of a participant, including abnormal laboratory findings, that happens during a clinical study or within a certain amount of time after the study has ended. This change may or may not be caused by the intervention/treatment being studied.

- **Age or age group**

A type of **eligibility criteria** that indicates the age a person must be to participate in a clinical study. This may be indicated by a specific age or the following age groups:

The age groups are:

Child (birth-17)

Adult (18-64)

Older Adult (65+)

- **Admission criteria**

Basis for **selecting target population** for a clinical trial.

Subjects must be screened to ensure that their **characteristics match** a list of admission criteria and that none of their characteristics match any single one of the exclusion criteria set up for the study.

- **All-cause mortality**

A **measure of all deaths**, due to any cause, that occur during a clinical study.

- **Action Letter**

The **letter from regulatory authority** to the **sponsor company**, in response to an application submission to do a clinical study. It may be an “Approval letter” with some minor modifications recommended or a “Non Approval Letter” indicating that significant changes will be needed before the application can be considered.

- **Adverse Drug Reaction (ADR)**

In case of **approved pharmaceutical products**:

A noxious and unintended response at doses normally used or tested in human.

In case of new unregistered pharmaceutical products (or those products which are not yet approved for the medical condition where they are being tested):

A noxious and unintended response at any dose(s).

In clinical studies, an untoward medical occurrence seemingly caused by overdosing, abuse / dependence and interactions with other medicinal products is also considered as an ADR.

- **Adverse Event (AE)**

Any untoward medical occurrence (including a symptom/ disease or an abnormal laboratory finding) during treatment with a pharmaceutical product in a patient or a human volunteer that does not necessarily have a relationship with the treatment being given.

When an Adverse Event has been assessed and there is reasonable grounds for suspicion that it is causally related to the investigational product(s) established, it must be considered as an ADR.

- **Aim of Study**

The **objective of the clinical study**.

Each study should only have one main objective.

If additional objectives are required then a separate study should be undertaken for each objective. Amendment

A **document that describes** changes or additions to an **Investigational New Drug** (IND) filed with the Regulatory Authority a supplement to a pending New Drug Application (NDA) such as a **safety or data update or changes to a existing protocol**.

- **Ascending Doses**

Subjects are dosed with **increasingly higher doses** of a drug until the maximum tolerated dose is reached.

This usually applies to a **Phase- I study**.

Archiving of Clinical Study Data According to GCP Guidelines, clinical study data should be stored for at least five years by the investigator/ CRO and for the lifetime of the drug by the pharmaceutical company.

- **Assent Agreement**

by an individual not competent to give legally valid informed consent (e.g., a child or cognitively impaired person) to participate in research.

Assurance A formal written, binding commitment that is submitted to a Regulatory Agency in which an institution promises to comply with applicable regulations governing research with human subjects and stipulates the procedures through which compliance will be achieved.

- **Audit of a Study**

A systematic verification of the study, carried out by persons not directly involved, such as:

Study related activities to determine consistency with the Protocol.

Studies data to ensure that there are no contradictions on Source Documents.

The audit should also compare data on the Source Documents with the interim or final report. Compliance with the adopted SOPs.

- **Baseline Measurements**

that are taken at the beginning of a study to serve as a reference for subsequent measurements or observations.

Belmont Report A statement of basic ethical principles governing research involving human subjects issued by the U.S. National Commission for the Protection of Human Subjects in 1978.

- **Bias**

When a point of view prevents impartial judgment on issue relating to the subject of that point of view.

In clinical studies, this is controlled by blinding and randomization.

- **Beneficence**

Doing no harm. Maximizing benefits while minimizing risks.

- **Blinding**

The process through which study subjects, the investigator and/or others involved parties in a clinical trial are kept unaware of the treatment assignments of study subjects

- **Case History Record**

The hospital chart or medical office file containing medical and demographic information on the study subject.

- **Causality**

The relationship between the adverse experience and the test agent in terms defined in the protocol

- **Case Report/Record Form (CRF)**

A **printed, optical, or electronic document designed** to record all of the protocol required information to be reported to the sponsor on each trial subject.

A document designed in consonance with the Protocol, to record data and other information on each study subject.

The Case Record form should be in such a **form and format that allows accurate input, presentation, verification, audit and inspection of the recorded data.**

A CRF may be in printed or electronic format.

- **Certificate of Destruction**

All returned and unused medication has to be destroyed at the end of the study.

Normally this material is returned to the sponsor together with the appropriate inventory. Certificate of destruction is issued which lists the drugs and the quantity destroyed on a given date.

This certificate is filed in the clinical study Master File.

- **Clean database (or file)**

One from which errors have been eliminated and in which measurements and other values are provided in the same units.

- **Clinical research associate (CRA)**

Person employed by a sponsor, or by a contract research organization acting on a sponsor's behalf, who monitors the progress of investigator sites participating in a clinical study.

At some (primarily academic) sites, clinical research coordinators are called CRAs.

- **Clinical research coordinator (CRC)**

Person who handles most of the administrative responsibilities of a clinical trial, acts as liaison between investigative site and sponsor, and reviews all data and records before a monitor's visit.

Synonyms: trial coordinator, study coordinator, research coordinator, clinical coordinator, research nurse, protocol nurse.

- **Certified Clinical Investigator (CCI)**

A clinical investigator who meets required experience and educational levels and has earned certification by passing an exam

- **Certified Clinical Coordinator (CCRC)**

CRC with more than 2 years experience and certification earned by passing an exam

- **Clinical trial**

Systematic study of a test article (treatment, drug, device) In one or more human subjects.

Synonyms: clinical study, clinical investigation. Confidentiality Prevention of disclosure, to other than authorized individuals, of a sponsor's proprietary information or of a subject's identity.

- **Contract research organization (CRO)**

A person or an organization (commercial, academic, or other) contracted by the sponsor to perform one or more of a sponsor's trial-related duties and functions.

- **Control group**

The group of subjects in a controlled study that **receives no treatment, a standard treatment, or a placebo.**

- **Controlled study**

A study in which a **test article is compared with a treatment that has known effects.**

The control group may receive no treatment, standard treatment, or placebo.

- **Database**

Data stored in computer form for **retrieval, processing, and/or analysis.**

- **Demographic Data**

The **characteristics of study subjects** including age, sex, medical history and other information relevant to the study

- **Data monitoring**

Process by which CRFs are examined for **completeness, consistency, and accuracy.**

- **Device**

An instrument, apparatus, implement, machine, contrivance, implant, in-vitro reagent or other similar related article, including any component, part or accessory, which is intended for use in the diagnosis, cure, treatment or prevention of a disease.

A device does not achieve its intended purpose through chemical action in the body and is not dependent upon being metabolized to achieve its purpose

- **Drug**

An article (other than food) intended for use in the diagnosis, cure, mitigation, treatment or prevention of a disease in man or other animals

- **Declaration of Helsinki**

A set of recommendations or basic principles that guide medical doctors in the conduct of biomedical research involving human subjects. (PTO)

It was originally adopted by the 18th World Medical Assembly (Helsinki, Finland, 1964);
Demographic data Characteristics of subjects or study populations, which include such information as age, sex, family history of the disease or condition for which they are being treated, and other characteristics relevant to the study in which they are participating.

- **Double-blind study**

A study in which **neither the subject(s) nor the investigator(s)** know what treatment a subject is receiving.

- **Ethics Committee**

An independent review board or committee comprising of medical, scientific and non-medical/non-scientific members, **whose responsibility is to verify the protection of the rights, safety and wellbeing of human subjects involved in the study.**

- **Efficacy**

A test product's **ability to produce a beneficial effect** on the duration or course of a disease

- **Generic Drug**

A **medicinal product with the same active ingredient(s)** as a brand name drug

- **Good Clinical Practice (GCP)**

The regulations and guidelines that specify the responsibilities of sponsors, investigators, monitors and IRBs involved in clinical trials.

They are meant to protect the safety, rights and welfare of the subjects in addition to ensuring the accuracy of the data collected during the trial

- **Food and Drug Administration (FDA)**

The United States regulatory authority charged with, among other responsibilities, granting IND and NDA approvals.

- **Informed consent**

A process by which a subject voluntarily confirms his or her willingness to participate in a particular trial, after having been informed of all aspects of the trial that are relevant to the subject's decision to participate.

- **Impartial witness**

A person, who is independent of the trial, who cannot be unfairly influenced by people involved with the trial, who attends the informed consent process if the subject or the subject's legally acceptable representative cannot read, and who reads the informed consent form and any other written information supplied to the subject.

- **Investigator, Clinical Investigator (CI), Principal Investigator (PI)**

An individual who actually conducts a clinical investigation person responsible for the conduct of the clinical trial at a trial site.

If a trial is conducted by a team of individuals at a trial site, the investigator is the responsible leader of the team and may be called the principal investigator.

- **Inclusion and Exclusion criteria**

The characteristics that must be present (inclusion) or absent (exclusion) in order for a subject to qualify for a clinical trial, as per the protocol for the trial.

- **Institutional Review Board (IRB)**

Any board, committee, or group formally designated to review biomedical research involving humans as subjects, to approve the initiation of and conduct periodic review of such research.

- **Investigator Brochure**

A compilation of all information known to date about the test product, including chemistry and formulation information and preclinical and clinical data.

It is updated at least annually.

Once the product is marketed, it is replaced by the labelling (packaging insert) for the product.

- **Investigational New Drug**

A **new drug or biologic** that is used in a clinical investigation.

- **Investigational New Drug Application (IND)**

The **application to start clinical testing** of a new drug, biologic or device in humans

- **IRB Approval**

The determination of the IRB that the clinical investigation **has been reviewed** and **may be conducted within the constraints** set by the IRB and applicable regulations

- **ICH**

The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use is an initiative that brings together regulatory authorities and pharmaceutical industry to discuss scientific and technical aspects of pharmaceutical product development and registration.

- **Legally Authorized Representative**

An **individual or judicial or other body authorized** under applicable law **to consent on behalf of potential subject to the subject's participation** in research.

- **Medical Monitor (Sponsor Medical Monitor)**

A Medical Monitor essentially provides **medical expertise, oversight, and consultation** in a clinical trial..

- **Monitor**

Person employed by the sponsor or CRO who is responsible for determining that a trial is being conducted in accordance with the protocol.

- **Multi-cent trial**

A clinical trial conducted according to a single protocol but at more than one site, and therefore, carried out by more than one investigator.

- **New Drug Application (NDA)**

An application to FDA for a license to market a new drug.

- **Nuremberg Code**

Code of ethics for conducting human medical research set forth in 1947.

- **Open study**

A trial in which subjects and investigators know which product each subject is receiving; opposite of double-blind study.

- **Pharmacovigilance**

Term used for adverse event monitoring and reporting in some countries.

- **Prospective study**

Investigation in which a group of subjects is recruited and monitored in accordance with criteria described in a protocol

- **Randomization**

The **process of assigning trial subjects to treatment or control groups using an element of chance** to determine the assignments in order to reduce bias.

- **Source documents**

Original documents, data, and records (e.g., hospital records, clinical and office charts, laboratory notes, memoranda, subjects' diaries or evaluation checklists, pharmacy dispensing records, recorded data from automated instruments, copies or transcriptions certified after verification as being accurate copies, microfiches, photographic negatives, microfilm or magnetic media, x-rays, subject files, and records kept at the pharmacy, at the laboratories and at medico-technical departments involved in the clinical trial).

- **Sub investigator**

Any individual member of the clinical trial team designated and supervised by the investigator at a trial site to perform critical trial-related procedures and/or to make important trial related decisions (e.g., associates, residents, research fellows).

- **Serious Adverse Event (SAE)**

Any untoward medical occurrence at any dose that results in death, is life-threatening , requires hospitalization (or a prolongation of hospitalization in a patient who is already hospitalized), results in persistent or significant disability or incapacity, or is a congenital anomaly/ birth defect.

- **Site Management Organization (SMO)**

A group of investigational sites that have banded together and organized centrally to carry out clinical studies.

- **Standard Operating Procedures (SOPs)**

Official written instructions for the management and conduct of clinical trial processes.

SOPs ensure that processes are carried out in a consistent and efficient manner.

- **Subject/trial subject**

An individual who participates in a clinical trial, either as recipient of the investigational product(s) or as a control.

- **Study Coordinator**

The person at an investigational site who manages the daily operations of a clinical investigation and who reports to the investigator.

- **Subject identification code**

A unique identifier assigned by the investigator to each trial subject to protect the subject's identity and used in place of subject's name when the investigator reports adverse events and/or other trial- related data.

- **Unanticipated event**

Problem involving risks to human subjects or others participating in a clinical research study e.g., breach of confidentiality, incarceration of subject, suicide attempt, incorrect labelling of study drug.

This information needs to be collected and reported as part of the clinical investigation.

- **CFR**

The Code of Federal Regulations (CFR) is a codification of the general and permanent rules published in the Federal Register by the Executive departments and agencies of the Federal Government.

Title 21 of the CFR is reserved for rules of the Food and Drug Administration.

- **Good clinical practice**

The Good Clinical Practice (GCP) course is designed to prepare research staff in the conduct of clinical trials with human participants.

- **Good manufacturing practice (GMP)**

Is a system for ensuring that products are consistently produced and controlled according to quality standards. It is designed to minimize the risks involved in any pharmaceutical production that cannot be eliminated through testing the final product.

- **Good Laboratory Practice (GLP)**

Are a managerial quality control system covering the organisational process and the conditions under which non-clinical health and environmental studies are planned, performed, monitored, recorded, reported and retained (or archived).

eClinical Trials

- **eClinical trial**

A clinical trial in which primarily electronic processes are used to plan, collect (acquire), access, exchange, and archive data required for conduct, management, analysis and reporting of a trial.

- **American national standards institute (ANSI)**

Founded in 1981, ANSI itself does not develop standards ANSI's.

ANSI's roles include serving as a coordinator for US voluntary standard efforts, acting as the approval body to recognize documents developed by other national organizations as American National Standards, acting as the US representative in international and regional standards efforts, and serving as a clearing house for national and international standards development information.

- **Application**

Software designed to fill specific needs of a user, for example, software for navigation, payroll, or process control. i.e. software application

- **Application service provider (ASP)**

Third party entity that manages and distributes software-based services and solutions to customers across a wide area network from a central data center.

- **Audit Trail**

A secure time stamped record that allows reconstruction of the course of events relating to the creation, modifications and deletions of an electronic study record.

- **eCRF**

An auditable electronic record designed to record information required by the clinical trial protocol to be reported to the sponsor on each trial subject (FDA Guidelines on Computerized systems Used in Clinical Trials) a CRF in which related data items and their associated comments, notes, signatures are linked digitally

eCRFs may include special display elements, electronic edit checks and other display properties or functions and are used for both capture and display of linked data

- **Clinical Data Interchange Standards Consortium (CDISC)**

An open multidisciplinary, non-profit organization committed to the development of worldwide industry standards to support the electronic acquisition, exchange, submission and archiving of clinical trials and metadata for medical and biopharmaceutical product development.

- **Clinical Data**

Data pertaining to the medical characteristics or status of a patient or subject

- **Clinical Efficacy**

A relative concept referring to the ability of a therapy to elicit a beneficial clinical effect

- **Clinical information**

Referred to data contained in the patient record

The data may include problem lists, lab results, current medications, family history, etc.

- **eClinical trial record**

Any data collected in support of a clinical trial, including but not limited to, the eCRF data (e.g., medical history/patient data, patient contact information, IVRS data, electronic patient diary data, electronic health data relevant to clinical trials, clinical laboratory data)

- **Data element**

Data element entries are annotated with symbols to indicate generally what information is required to be submitted (and under which circumstances). The responsible party must ensure that the information provided complies with any applicable laws, regulations, or policies.

- **Data Entry**

Data entry may take a form of direct computer entry by a person transferring data from paper-based CRFs into a computer database, optical mark reading (scanning) or optical character recognition (DATAFAX) for scannable/faxable forms.

Input of data into a structured, computerised format using a human-computer-interface.

- **Data integrity**

The degree to which a collection of data is complete, consistent, and accurate.

Integrity of data may be assessed in terms of whether they are attributable, legible, contemporaneous(existing at or occurring in the same period of time), original and accurate (derived from FDA-GL, FDA Guidance IEEE)

- **Data interchange**

The transfer of information between two or more parties.

Data interchange is distinguished from data transfer in that the integrity of the contents of the data is maintained.

- **Data item**

A named component of a data element, usually the smallest component (ANSI)

- **Data model**

The packaged clinical data model captures information from electronic health records (EHR) and the problems, goals, and tasks associated with a care plan

- **Data safety monitoring board (DSMB)**

Is a committee of experts responsible for reviewing clinical trial data on an ongoing basis to ensure the safety of study subjects and validity and integrity of the data.

Members should be independent, with no vested interest in a specific treatment.

- **Data Validation**

Is the process of testing the validity of data in accordance with the protocol specifications.

Edit check programs are written to identify the discrepancies in the entered data, which are embedded in the database, to ensure data validity.

- **Source Data Verification (SDV)**

Is a method that clinical research organizations (CROs) can use to ensure the data they're collecting is accurate.

It involves comparing a study's source data against its final report, allowing for the identification of any errors before they become problematic or get published

- **An Electronic Record**

Is defined as any combination of text, graphics, data, audio, pictorial, or other information representation in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system

- **Electronic health record (EHR)**

EHRs are electronic platforms that contain health-related data collected during medical care in practices, clinics and other medical settings from various sources, connected to form a network of patient clinical data.

Declaration of Helsinki

- The Declaration of Helsinki was created in 1964 by the World Medical Association (WMA). The WMA was established seventeen years before in 1947 and was created to handle the growing concern of unethical medical practice which became more apparent during and after World War II.
- The WMA is an international representative organization for members of the medical community including doctors, researchers, dentists, and more. While the WMA is a well-respected body which is cited by health organizations around the world, it does not have any legal authority
- The Declaration of Helsinki is an **ethical guideline**, rather than a document with international legal implications.
- The Declaration **set the standard for ethical human experimentation** conducted by researchers.

Nuremberg Code

- The Nuremberg Military Tribunal's decision in the case of the United States v Karl Brandt et al. includes what is now called the Nuremberg Code, **a ten-point statement delimiting permissible medical experimentation on human subjects.**
- **What was the importance of Nuremberg Code?**
- The Nuremberg Code was designed to **protect the autonomy and rights of human subjects** in medical research, as compared with the Hippocratic Oath applied in the therapeutic, paternalistic patient-physician relationship.
- **What is Nuremberg Code in research?**
- The Nuremberg Code is a 10-point set of rules for the conduct of human experiments articulated in 1947 in the trials of Nazi doctors and bureaucrats convicted of crimes against humanity for their roles in concentration camp experiments.
- **Why were the Nuremberg Principles created?**
- The Nuremberg principles are a **set of guidelines for determining what constitutes a war crime**. The document was created by the International Law Commission of the United Nations to codify the legal principles underlying the Nuremberg Trials of Nazi party members following World War II.

Difference between Declaration of Helsinki and the Nuremberg Code

- The Declaration was designed with a focus on clinical research.
- The Nuremberg Code is broader in scope, involving any and all experimentation involving human subjects.
- The Nuremberg Code also states that "the voluntary consent of a human subject is absolutely essential," meaning that consent is a requirement. If that requirement is not met, then research cannot proceed.
- The Declaration is much more lax in this regard. It states that if an individual can give consent, then it must be given before research can proceed. However, if that individual is not capable of giving consent, then the physical or mental condition that prevents them from giving consent is a necessary characteristic of the research.
- The Declaration also outlines proxy consent, in which legal guardians can give consent for those unable to do so.

**THANK
YOU**



The background is a composite image. On the left, a portion of the periodic table is visible, showing elements like Nb, Mo, Tc, Ru, Rh, Pd, Ag, Au, Hg, Pt, Ir, Os, Re, W, Ta, and Hs. In the center, a glass flask is shown with a bright orange liquid inside. Above the flask, a dropper is dispensing a single drop of the same orange liquid. To the right of the flask, there are several chemical structures, including a complex polycyclic molecule and a linear chain with various functional groups and charges.

HISTORY OF CLINICAL RESEARCH

Ms. Sonali Correa

Ph.D., M.Sc.

605 BC BOOK OF DANIEL

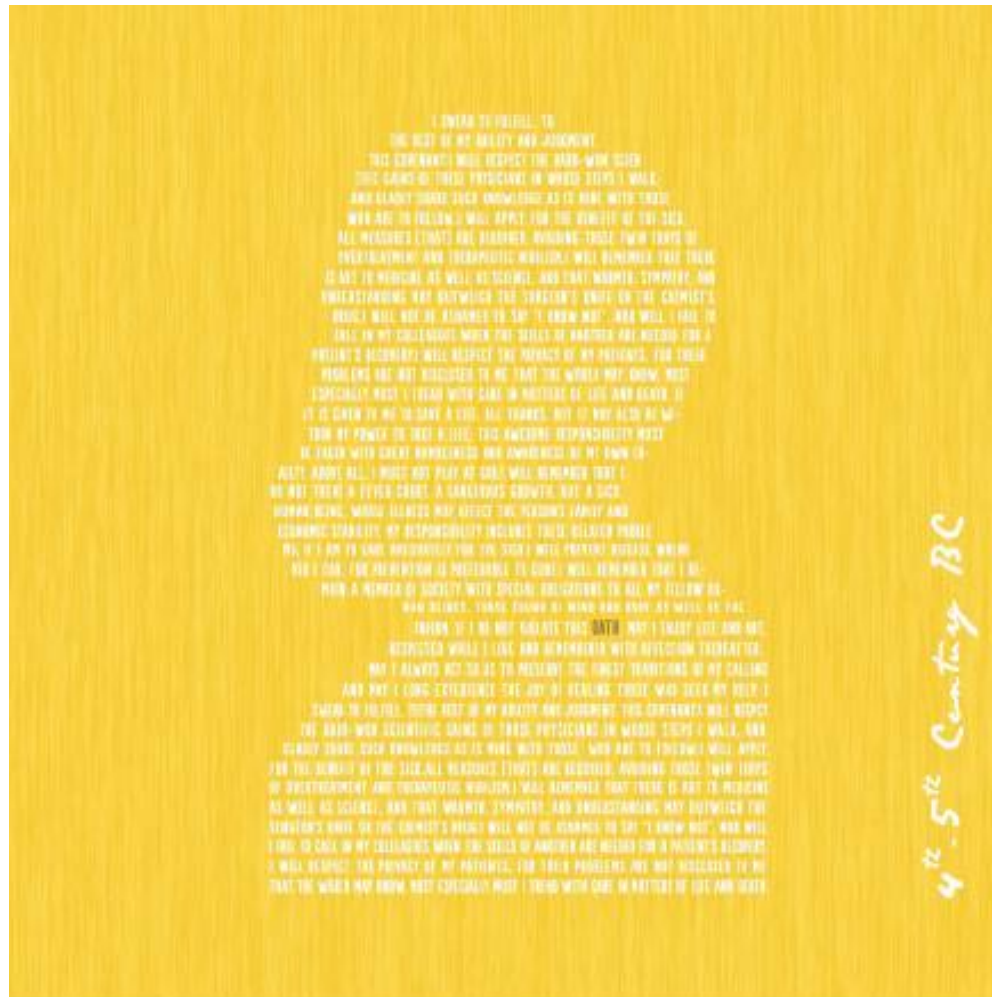
- In one of the earliest recorded examples of classical experimental design in history, Daniel sought to test the effects and benefits of a vegetarian diet. For ten days he ate only vegetables while the other subjects enjoyed the King's meat and wine. After the ten days, they assessed who was healthier – in this case Daniel.



500 BC

THE HIPPOCRATIC OATH

- Hippocrates, “the father of Western Medicine,” is believed to have created the oath that bears his name over 2,400 years ago. By taking it, physicians and other healthcare professionals swear to practice medicine justly and ethically. The responsibility of practicing medicine ethically carries over to clinical research, where the protection of human subjects is paramount.



1747

JAMES LIND'S SCURVY EXPERIMENT



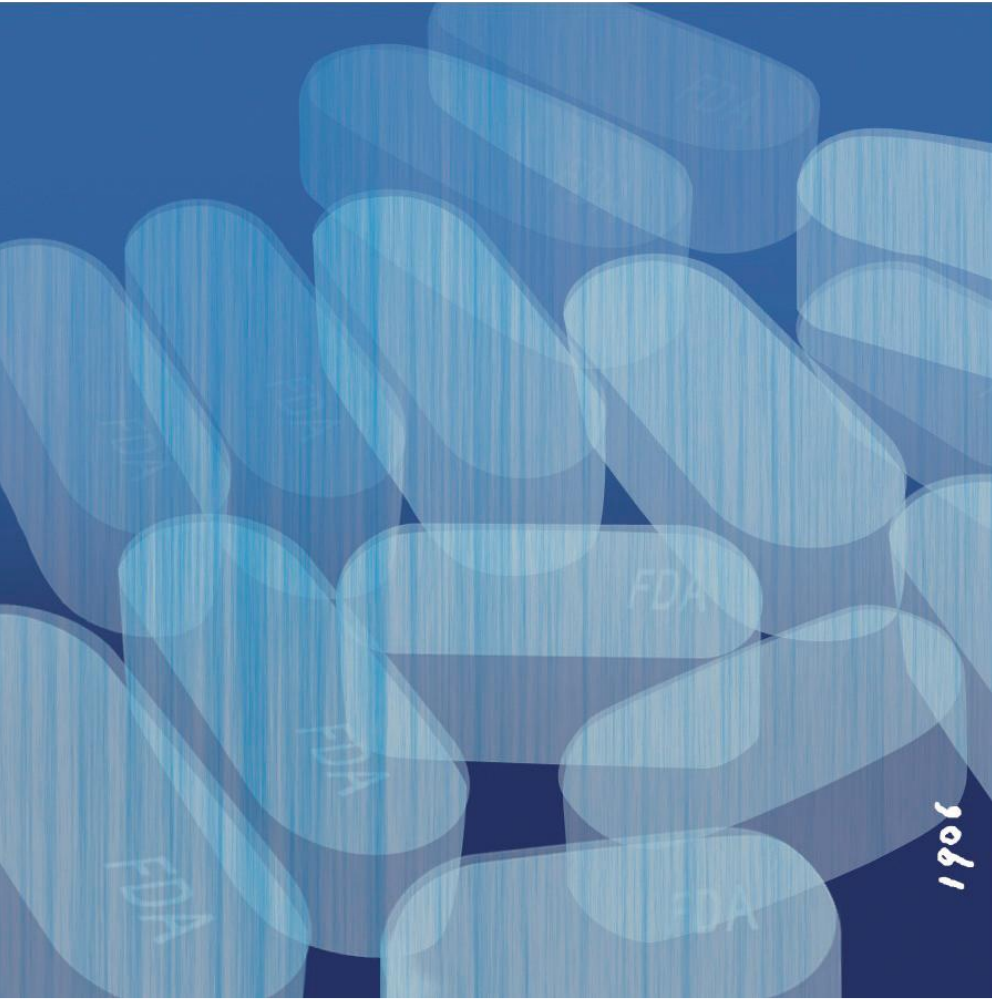
- In one of the first clinical experiments in the history of medicine, Scottish physician in the British Navy James Lind tested his theory that citric acids could prevent and cure scurvy, a huge problem among sailors at the time. As a result of scurvy, sailors suffered poor wound healing, skin changes, and loosening of teeth. With no way to treat scurvy, many sailors died at sea. During Lind's 'trial' twelve affected soldiers were divided into six groups, with each group receiving a different supplement in addition to their regular diet. Cider, sulfuric acid, vinegar, seawater, oranges and lemons, and a spicy paste with barley water were all tested. The group receiving the citrus fruit saw positive effects and began recovering immediately

1887

NATIONAL INSTITUTE OF HEALTH (NIH) FOUNDED

- Originally named the Hygienic Laboratory, the National Institute of Health was founded by Joseph J. Kinyoun. Today the NIH is the largest source of funding for medical research in the world, investing \$30.9 billion annually to support scientific discovery.





1906

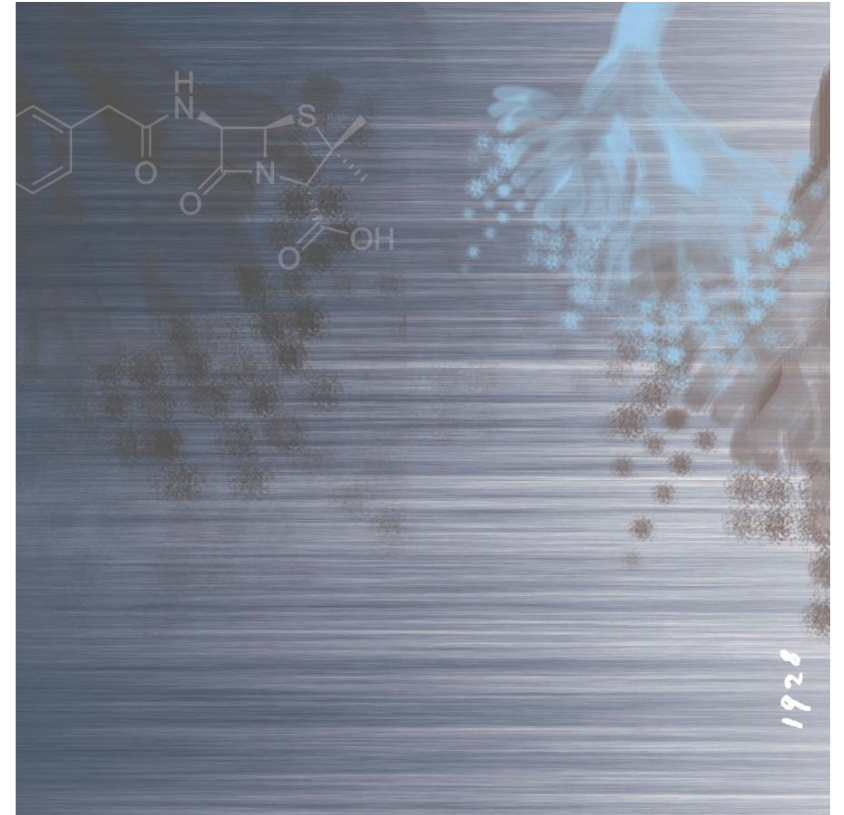
FDA PURE FOOD AND DRUG ACT

- President Theodore Roosevelt signed the FDA Pure Food and Drug Act into law, which regulated that products could not be sold for indications outside the labelling and prohibited interstate transportation of unlawful food and drugs.

1928

SIR ALEXANDER FLEMING DISCOVERS PENICILLIN

- Alexander Fleming's cluttered and untidy lab yielded one of the most important discoveries in the history of medicine as penicillin was identified on mold growing on a stack of staphylococci cultures. Soon thereafter, penicillin was recognized as one of the most efficacious life-saving drugs in the world, forever changing the treatment of bacterial infections.





1932-1972

THE TUSKEGEE SYPHILIS STUDY

- For 40 years, impoverished syphilis-positive African Americans were lied to and misled in the name of science. The US Public Health Service conducted studies on 600 (including 201 control) subjects to study the effects of syphilis. Subjects did not provide informed consent and were denied access to Penicillin, a proven treatment for syphilis. Many died as a result, infected others with the disease, and passed congenital syphilis to their children.

1937

ELIXIR SULFANILAMIDE DISASTER

- S. E. Massengill Co.'s work to market Sulfanilamide towards children became one of the deadliest mass poisonings of the 20th century. Chief chemist, Harold Cole Watkins, liquefied the drug by dissolving it in the toxic compound diethylene glycol, but failed to test the compound for toxicity since this was not a requirement at the time. Over 100 patients died after consuming Elixir Sulfanilamide. Watkins died while awaiting trial and it is believed that he committed suicide. S.E. Massengill paid a minimal fine for mislabelling the compound as an elixir even though it contained no alcohol. This was the only penalty the company was subject to under the 1906 Federal Food and Drugs Act



1938

FEDERAL FOOD, DRUG AND COSMETIC ACT



- Following the Elixir Sulfanilamide disaster, the U.S. Congress passed the 1938 Federal Food, Drug and Cosmetic Act, which required proof of safety before the release of a new drug. The 1938 law changed the drug focus of the FDA from that of a policing agency primarily concerned with the confiscation of adulterated drugs, to a regulatory agency increasingly involved with overseeing the evaluation of new drugs.
- This law, though extensively amended in subsequent years, remains the central foundation of FDA regulatory authority to this day. This act mandated that safety be demonstrated prior to market approval, and required that drugs be labeled with adequate directions for safe use.

1939-1945

WORLD WAR II EXPERIMENTS



- A series of grotesque medical experiments were conducted by the German Nazi Party on large numbers of concentration camp prisoners during World War II as part of the Holocaust. Prisoners were subjected to hazardous experiments with the goal of developing new weapons, to aid in treating injured German soldiers, and to advance their eugenic racial ideologies. These experiments often resulted in death, disfigurement, or permanent disability

1944

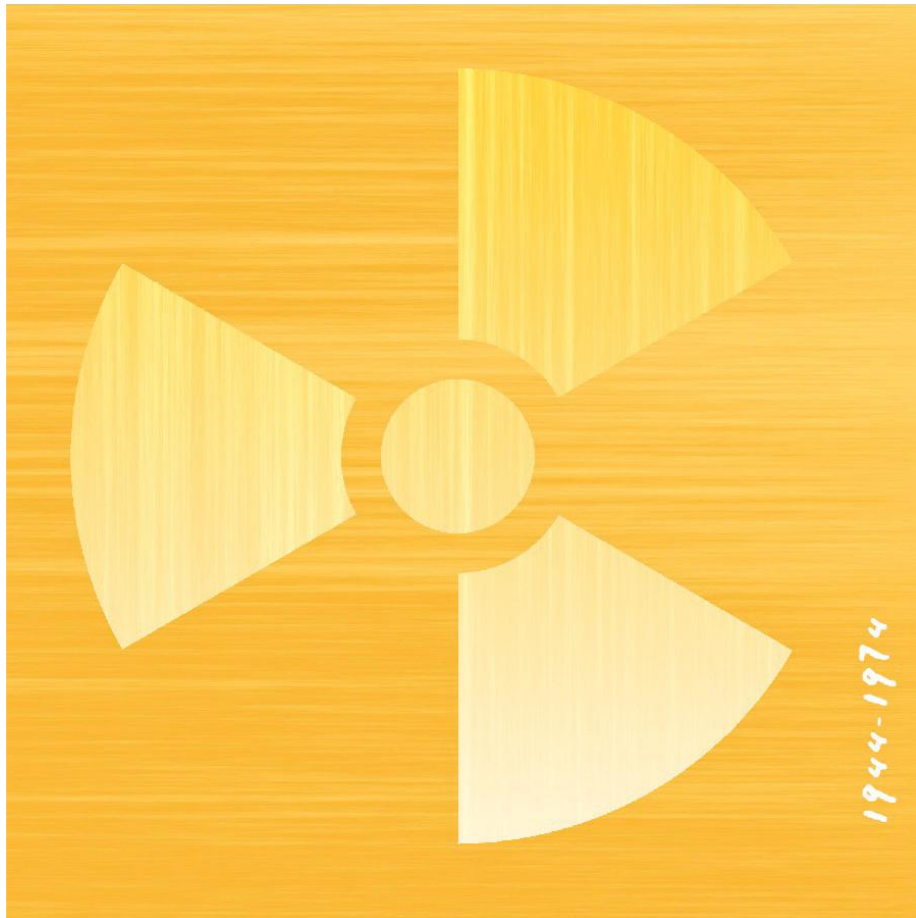
MULTICENTER STUDIES

- Conducting large trials at multiple sites represented a significant change in how studies were conducted. For the first time, trials were conducted at different sites using the same protocol, with all the centers' results assessed together. This finally allowed for larger numbers of participants and a wider range of population groups to be studied, which strengthens research trial designs and analyses.
- Today most large trials are conducted at multiple clinical research centers



1944-1974

HUMAN RADIATION EXPERIMENTS



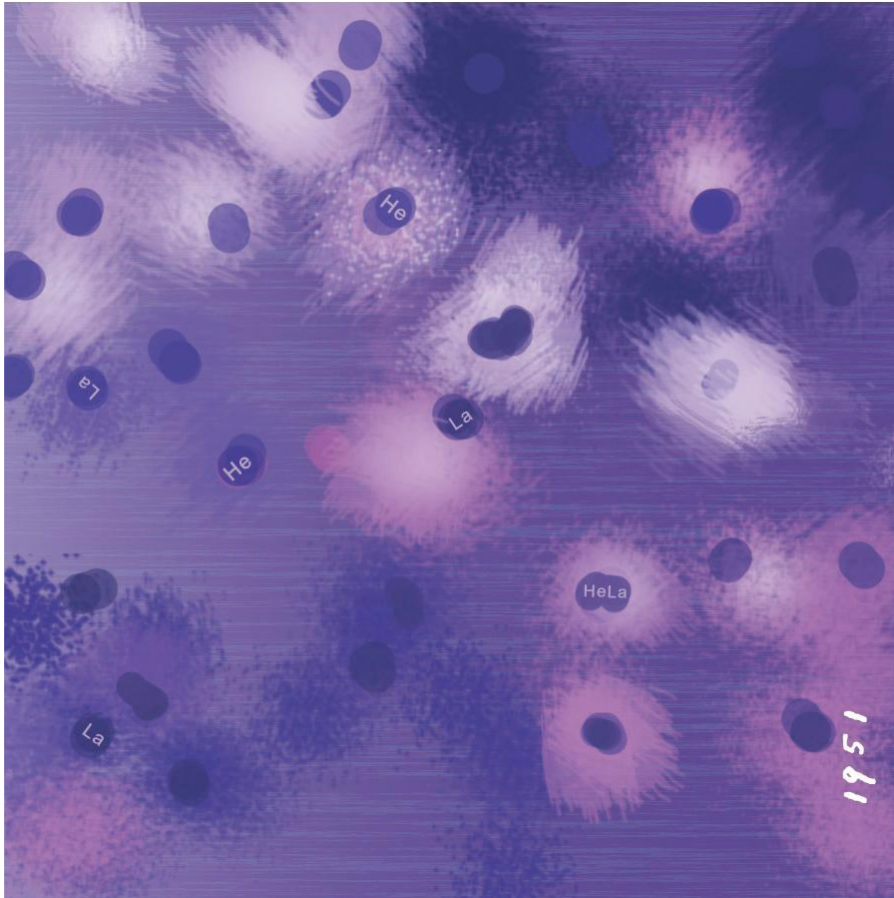
- During the Cold War, thousands of U.S. citizens became the innocent and unknowing victims of over 4,000 secret experiments sponsored by the U.S. government aimed at determining the effects of atomic radiation on the human body, often using doses likely to harm the subjects. The majority of victims included vulnerable groups such as prisoners, pregnant women, children, and impoverished or disabled patients.

NUREMBERG CODE

- [illegible]

1951

HENRIETTA LACKS



- Henrietta Lacks' contribution to medical science is enormous, yet she never knew or intended it to happen. During treatment for cervical cancer, cells were taken from her tumor without her knowledge. Researcher George Otto Gey took these cells and noticed that, remarkably, the cells could be kept alive in culture.
- Her cells would eventually become the HeLa (He for Henrietta and La for Lacks) immortal cell line, commonly used in biomedical research. Since the extraction of Henrietta's cells, some 20 tons of cells have been grown from the HeLa line and distributed to researchers across the world. These cells were used to test the first polio vaccine in the 1950's and have since been used for AIDS and cancer research, as well as many other scientific pursuits.

1960'S

HARVARD PSILOCYBIN EXPERIMENTS

- In the early 1960's, as Harvard was formulating its own policies regarding human research, two of its professors, Timothy Leary and Richard Alpert gave the psychedelic drug psilocybin to students and prisoners as part of a hypothesis that the then-legal drug could alter behavior in dramatic and beneficial ways. Due to the controversy surrounding an article published in the Crimson in 1962, Leary and Alpert were dismissed by the university.



1962

KEFAUVER-HARRIS DRUG AMENDMENT



- In addition to demonstrating safety, manufacturers were now required to provide proof of effectiveness of their drugs prior to approval and to disclose accurate information about their side effects. This prohibits cheaper generic drugs from being marketed as expensive new “breakthrough” medications.

1964

DECLARATION OF HELSINKI

- The Declaration expands on the ten principles necessary for ethical human experimentation set forth in the Nuremberg Code. Developed by the World Medical Association, the Declaration of Helsinki is generally regarded as the cornerstone document of human research ethics. Since its inception in June 1964, it has undergone six revisions, the most recent coming in 2008. The focus of the Declaration is on respect for the subject, the subject's right to self determination and their right to make informed decisions regarding research participation. It stands as a clear and powerful statement that the rights of the human subject shall never be compromised for the sake of science.



1974

THE NATIONAL RESEARCH ACT

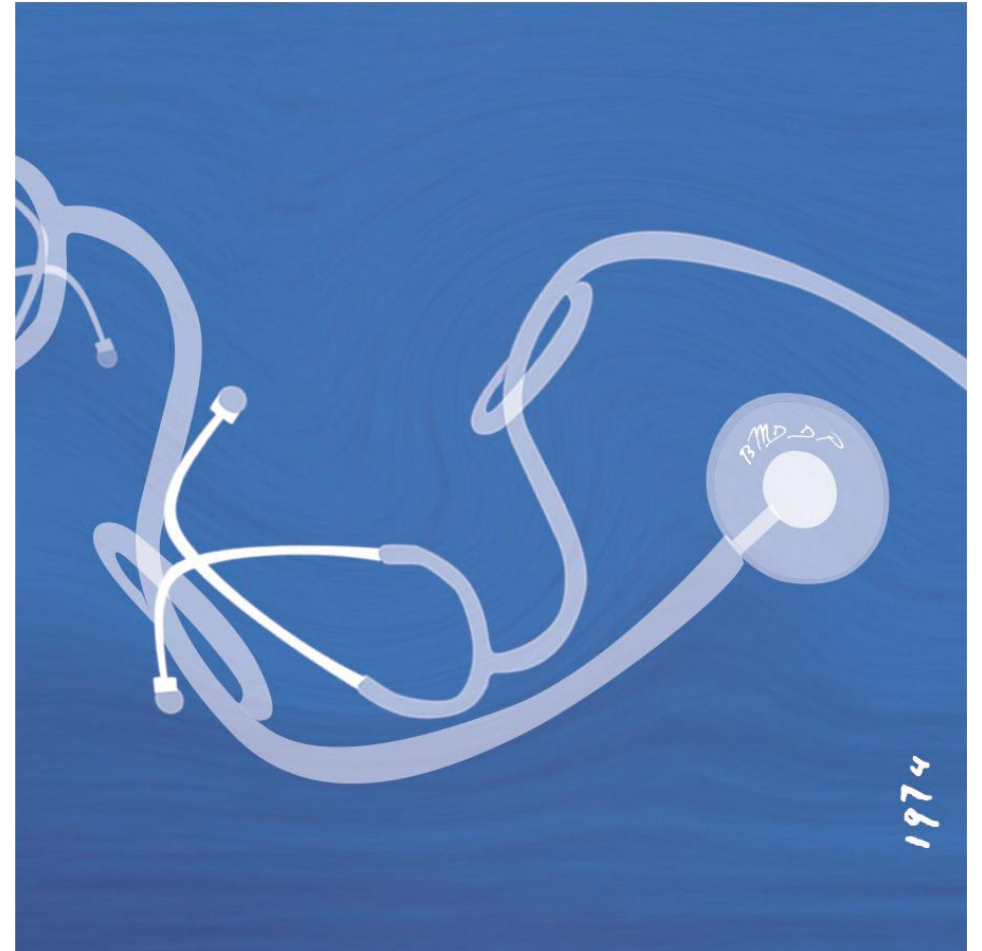


- President Richard Nixon signed the National Research Act into law in response primarily to the Tuskegee Syphilis Study. The act requires that all research using human subjects be reviewed by an Institutional Review Board as another step to ensure protection of human subjects.

1974

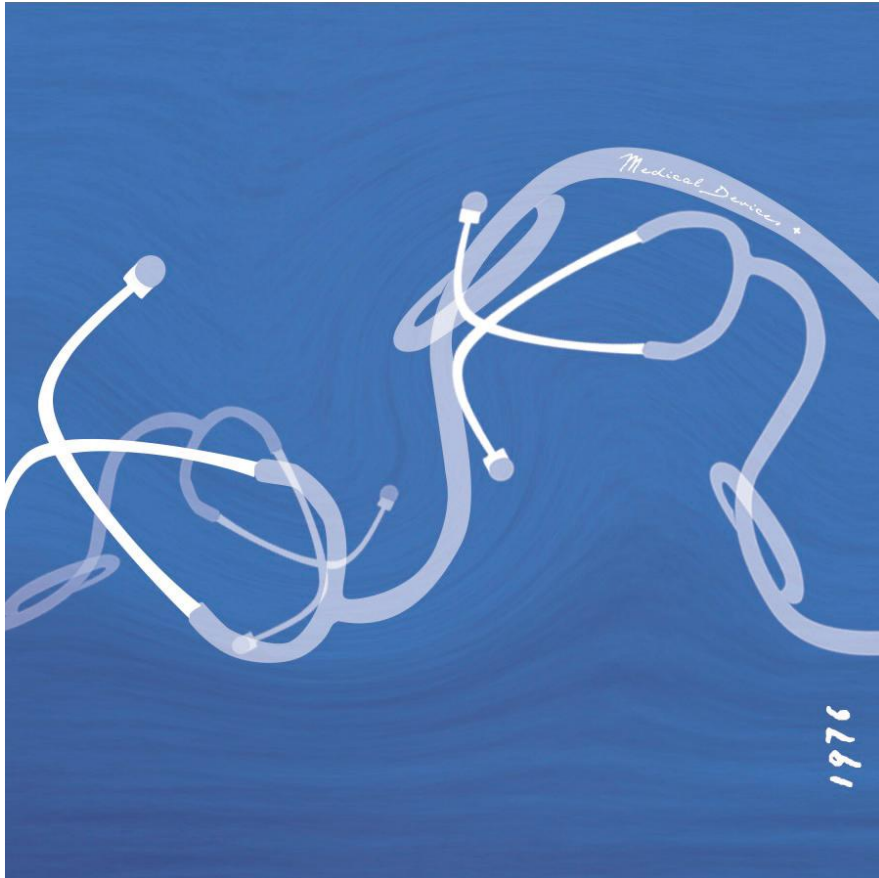
FDA BUREAU OF MEDICAL DEVICES & DIAGNOSTIC PRODUCTS

- Advances in technology led to a significant growth in the medical device field, yet the industry was still operating based on device regulations from the Federal Food, Drug and Cosmetic Act of 1938. Clearly, updated regulations were needed. In February 1974, the FDA created the Bureau of Medical Devices and Diagnostic Products to meet this need.



1976

MEDICAL DEVICE AMENDMENTS



- President Gerald Ford signed this amendment into law to increase FDA's authority over the production of medical devices. This created the current classification of devices with 3 classes based on risk, each having different regulatory pathways, and required devices to be proven safe and effective before they can be marketed.

1979

THE BELMONT REPORT

- Developed primarily in response to the Tuskegee Syphilis Study, The Belmont Report explains the three unifying ethical principles to guide human research:
- Respect for persons, Beneficence, and Justice.
- The Belmont Report has become the primary ethical framework for protecting human research subjects in the United States.



FDA REGULATIONS TITLE 21

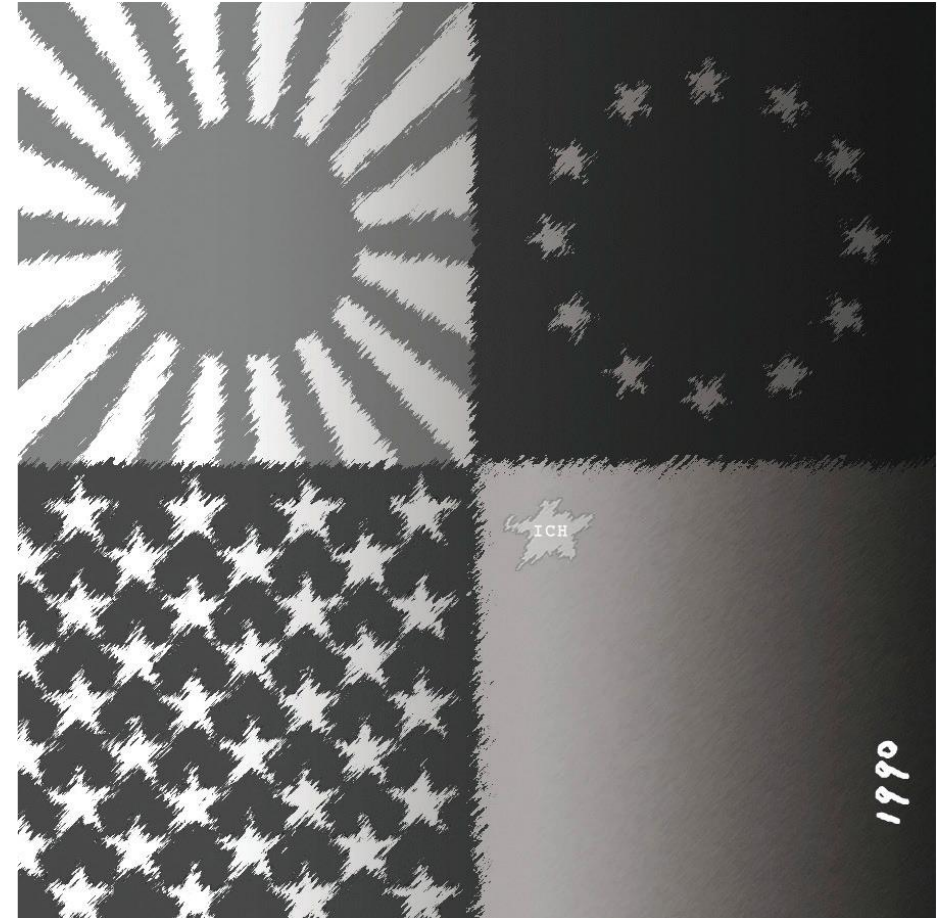
1981

- In follow-up to the Belmont Report, FDA and the Department of Health and Human Services formally revised regulations for human subject protections by creating
 - Title 21, which includes regulations for the Protection of Human Subjects (Part 50),
 - Financial Disclosure (Part 54),
 - IRBs (Part 56), Investigational New Drug Applications (Part 312),
 - Investigational Device Exemptions (Part 812),
 - Electronic Records (Part 11).

1990

INTERNATIONAL CONFERENCE ON HARMONIZATION GUIDELINES

- Many European nations along with Japan and the United States began creating plans for global harmonization of regulatory requirements to reduce duplicate, time-consuming, and expensive procedures needed to market products internationally, while maintaining safeguards on quality, safety, and efficacy. This need led to the creation of the ICH guidelines in April of 1990.



1990

THE SAFE MEDICAL DEVICES ACT

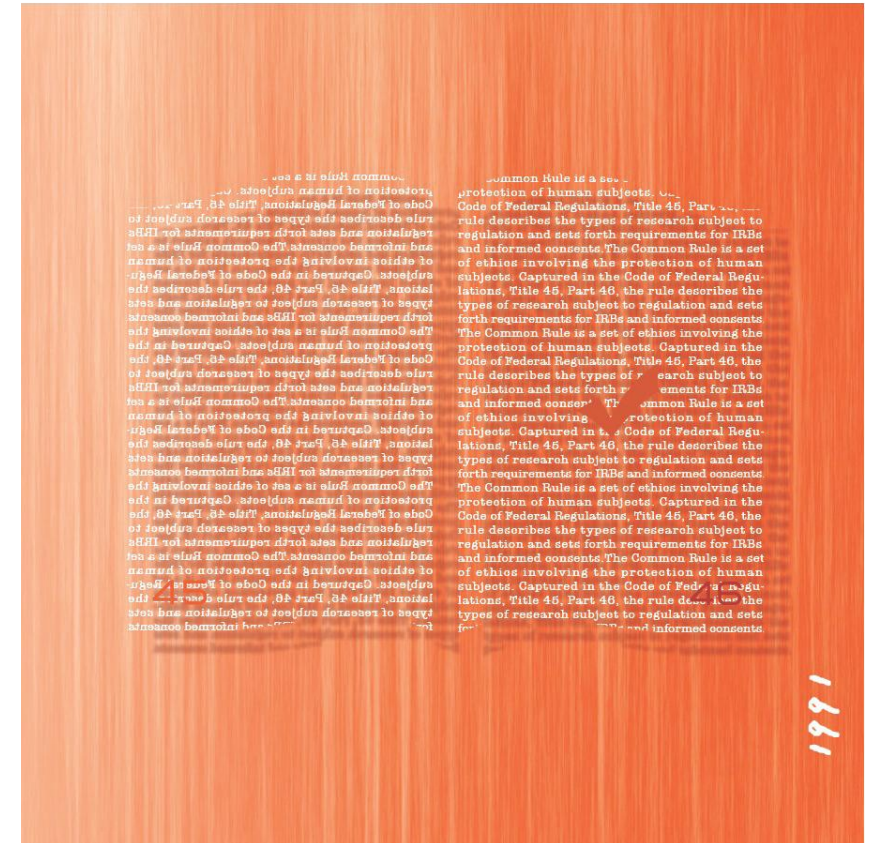


- An extension of the Medical Device Reporting Legislation, the Safe Medical Devices Act brought about requirements for hospitals and health professionals to report incidents to the FDA and manufacturers when devices cause serious injury or death.
- The act authorized the FDA to order device product recalls and take other actions. This reinforce the importance of protecting patients' health from defective or risky devices.

1991

THE COMMON RULE

- The Common Rule is yet another set of ethics involving the protection of human subjects.
- Captured in the Code of Federal Regulations, Title 45, Part 46, the rule describes the types of research subject to regulation.
- The rule also sets forth requirements for the Institutional Review Board's oversight of clinical trials and details protection of human subjects regarding informed consent.



1993

MEDWATCH



- MedWatch was launched by the FDA as a system designed to collect health professionals' reports of adverse events involving medical products. When safety hazards are detected, the FDA issues medical product safety alerts or orders product recalls, withdrawals, or labelling changes to protect the public health.

<https://www.fda.gov/safety/medwatch-fda-safety-information-and-adverse-event-reporting-program>

1996

HEALTH INSURANCE PORTABILITY & ACCOUNTABILITY ACT

- The Health Insurance Portability and Accountability Act requires that patients be informed of how their protected health information will be stored and kept private when they participate in a research trial. President Bill Clinton signed this act into law in 1992, but the final HIPAA rule did not go into effect until 1996.



1996

NATIONAL BIOETHICS ADVISORY COMMISSION



- Created by President Clinton, the National Bioethics Advisory Commission was created to explore ethical issues in science and medicine and advise the President on bioethical issues.
- The commission examined topics including cloning, human stem cell research, and research involving human subjects

1996

THE WORLD HEALTH ORGANIZATION GUIDELINES FOR GOOD CLINICAL PRACTICE

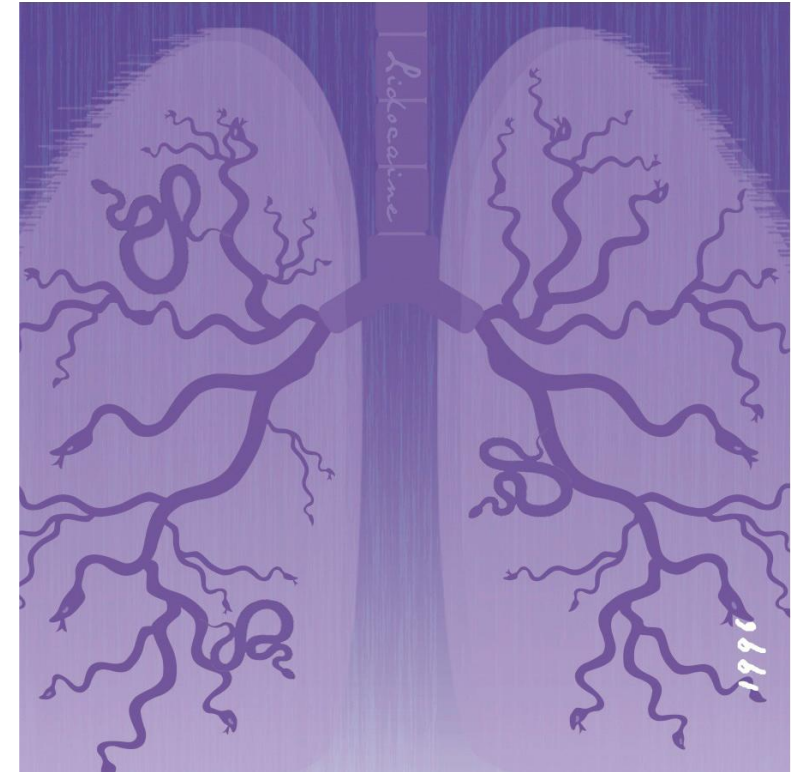
- The guideline addresses justifications for a trial and protocol; protection of subjects; responsibilities of investigators, sponsors and monitors; assurance of data integrity and product accountability; and roles of regulatory authorities



1996

HOIYAN (NICOLE) WAN

- Nineteen-year-old Nicole Wan participated in a minimal risk smoking and air pollution research study where she underwent a bronchoscopy to obtain some of her lung cells. Ms. Wan received four times the maximum tolerated dose of lidocaine administered by an inexperienced intern.
- She died two days later from complications. The attending physician was found guilty of failing to clearly state the maximum dosages in the study's protocol, violating the stated guidelines, and failing to properly monitor the young woman's condition after the procedure.



1999

IMARC RESEARCH, INC.



- Sandra Maddock founded IMARC Research, Inc. in 1999 to deliver the highest-quality clinical research monitoring, auditing, training, development and consulting services for companies seeking to correctly follow the guidelines for legal and ethical trials.

1999

JESSE GELSINGER

- Jesse Gelsinger suffered from a rare x-linked genetic disease of the liver but did not suffer effects at the time he enrolled in a gene therapy study at a University to help others.
- The eighteen year-old died four days after the research procedure.
- The University failed to report patients had experienced serious side effects, the informed consent did not disclose known deaths during animal trials, and the co-investigator and University had a financial interest in the trial.
- Additionally, Jesse's high ammonia levels should have excluded him from participating in the study in the first place.



2004-2007

KETEK



- The antibiotic Ketek won FDA approval in 2004; since then, Ketek has been linked to dozens of cases of severe liver injury, including a link to four deaths. It has been the subject of a series of increasingly urgent safety warnings and sparked two Congressional investigations of the FDA's acceptance of fraudulent safety data and inappropriate trial methods when it reviewed the drug for approval.

CONCLUSION

- Ensuring accurate and credible data from clinical trials and protection of human subjects are the roles of every individual on the research team.
- The price for compromise is high; history provides an array of haunting reminders of our failure to uphold our obligation to ethical excellence in our work.
- Even more sobering is the realization that even today, research professionals sometimes fail to protect innocent people from harm and research misconduct.
- Advances in the field of medicine are dependent on the quality of research that is conducted.
- It thus important to be armed with the knowledge that on the other side of those investigational products are real people; patients who are taking a risk to help the research and medical community and protecting them is our biggest responsibility.

Clinical Research: Types, Phases & Design

Dr. Ms. Sonali Correa

Clinical Research

All scientific approaches to evaluate medical disease in terms

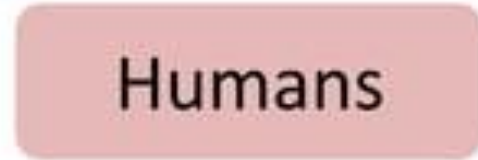
Prevention

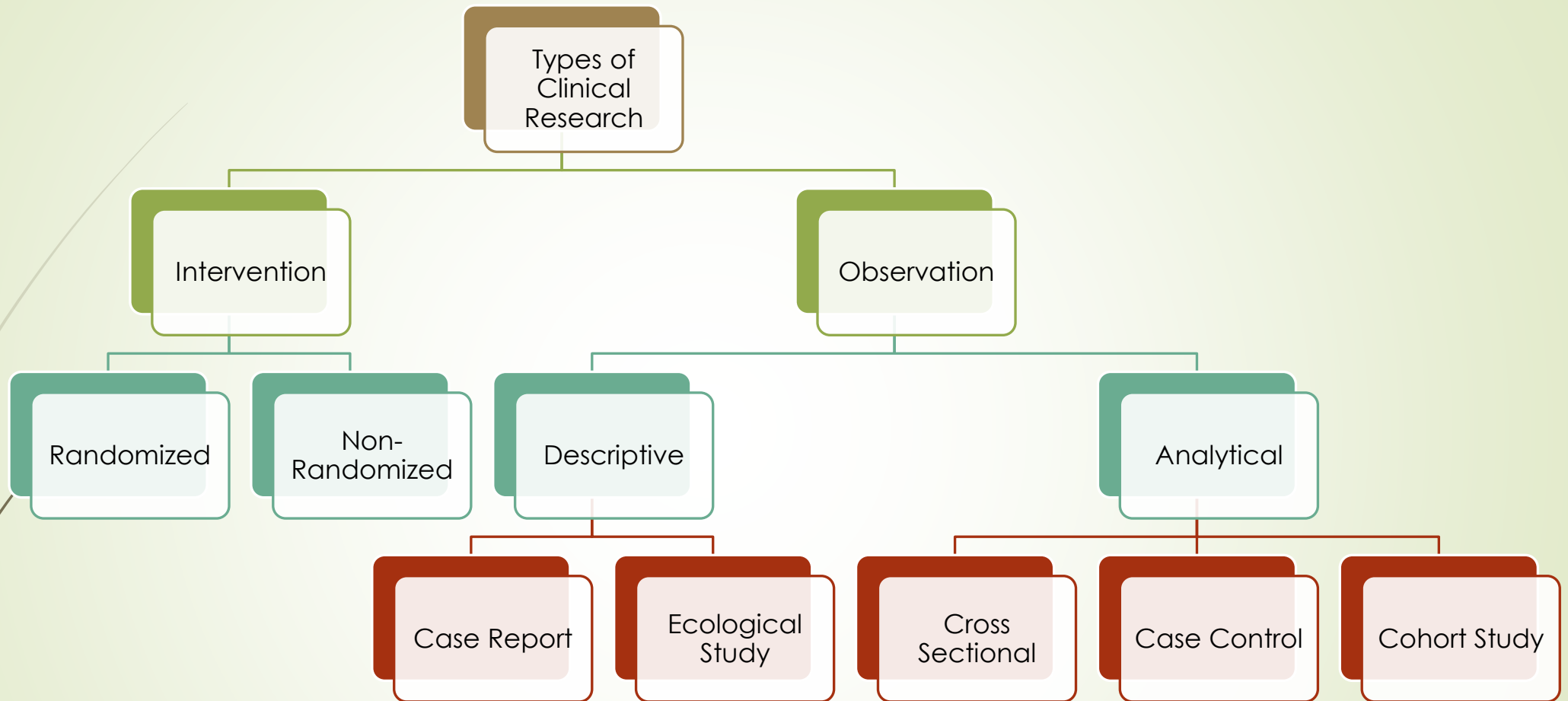
Diagnosis

Treatment



Humans







An **interventional study tests** (or tries out) an intervention — a potential drug, medical device, activity, or procedure -- in people. It is also commonly referred to as a clinical trial.

- A new potential drug, such as the compounds on the drug development pipeline
- A medical device, such as a vibration vest
- An activity, such as exercise or therapy

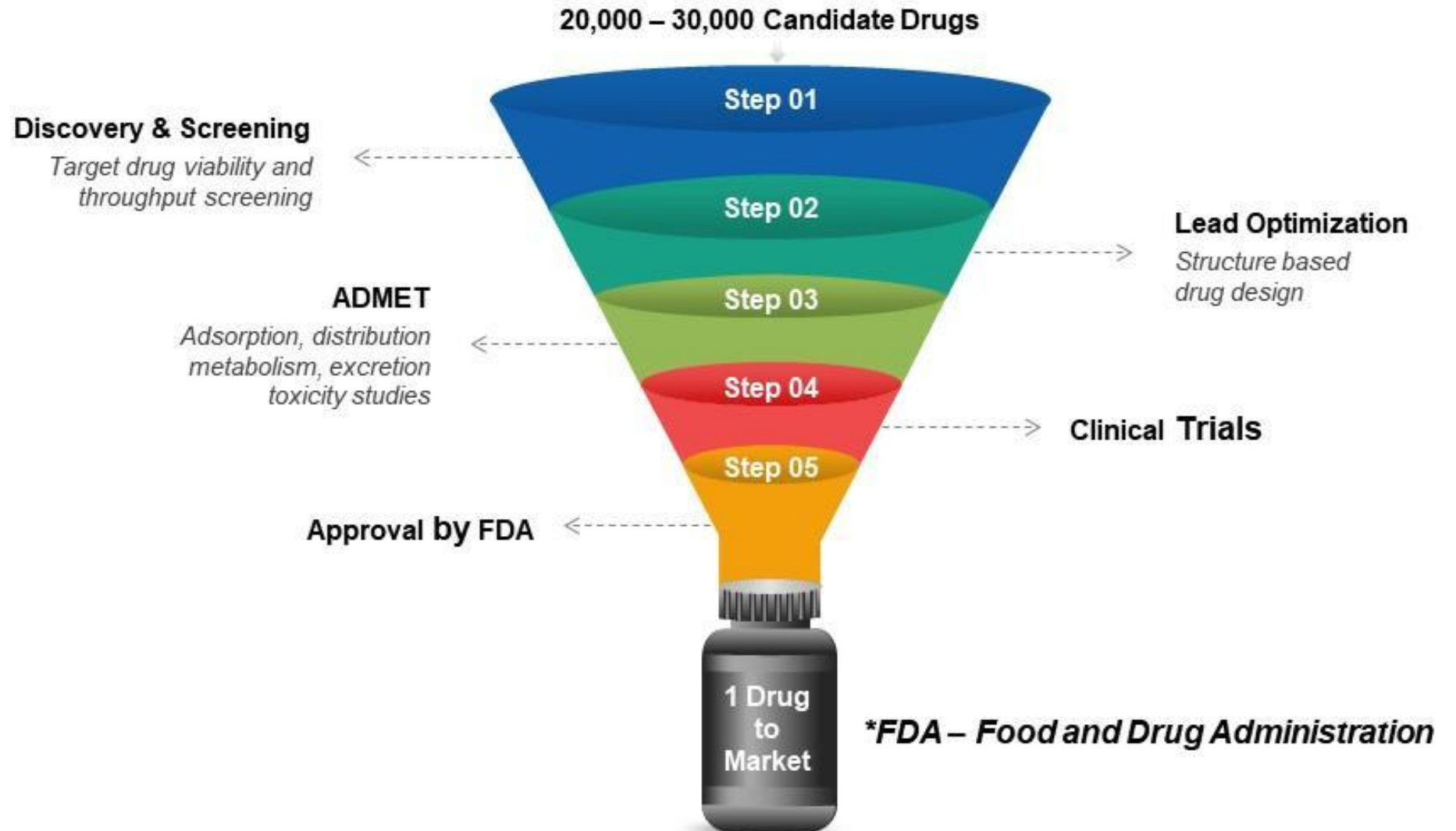
New potential drugs need to pass through **three phases** of interventional testing to show that they are **safe and efficacious** before receiving approval

Observational studies don't test potential treatments. Instead, participants stay on their normal treatment plan.

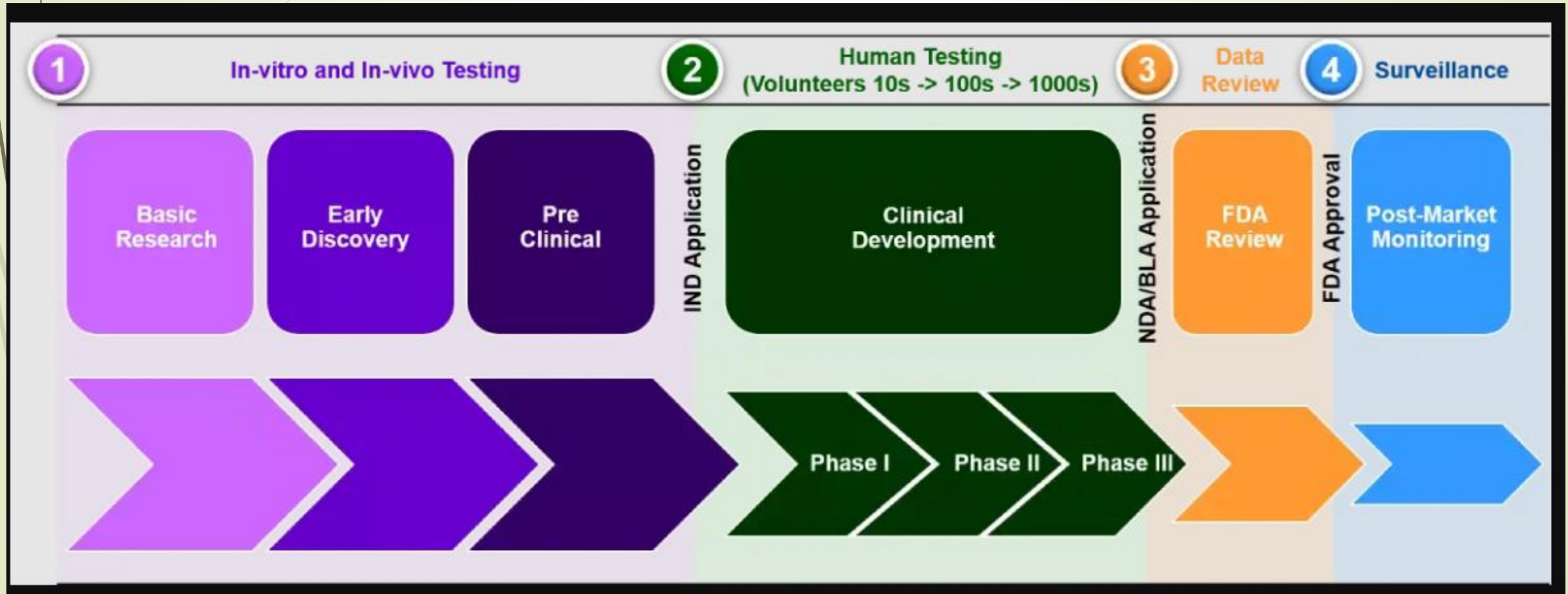
- Researchers observe participants and track health outcomes over time.
- These studies help us understand trends in survival and complications. They can also help us understand how effective a treatment is in a day-to-day or “real-world” setting.
- For example, after a new drug is approved, the FDA often requires that the pharmaceutical company continue to monitor the effectiveness and safety of the drug in what is known as a Phase 4 study.

Observational studies often take less time to participate. Participation could be as simple as a single blood draw or filling out a questionnaire. Unlike interventional studies, you can be in more than one observational study at the same time because the studies do not involve any changes to your regular treatment plan.

Drug Discovery and Development Process Funnel



Pathway to research

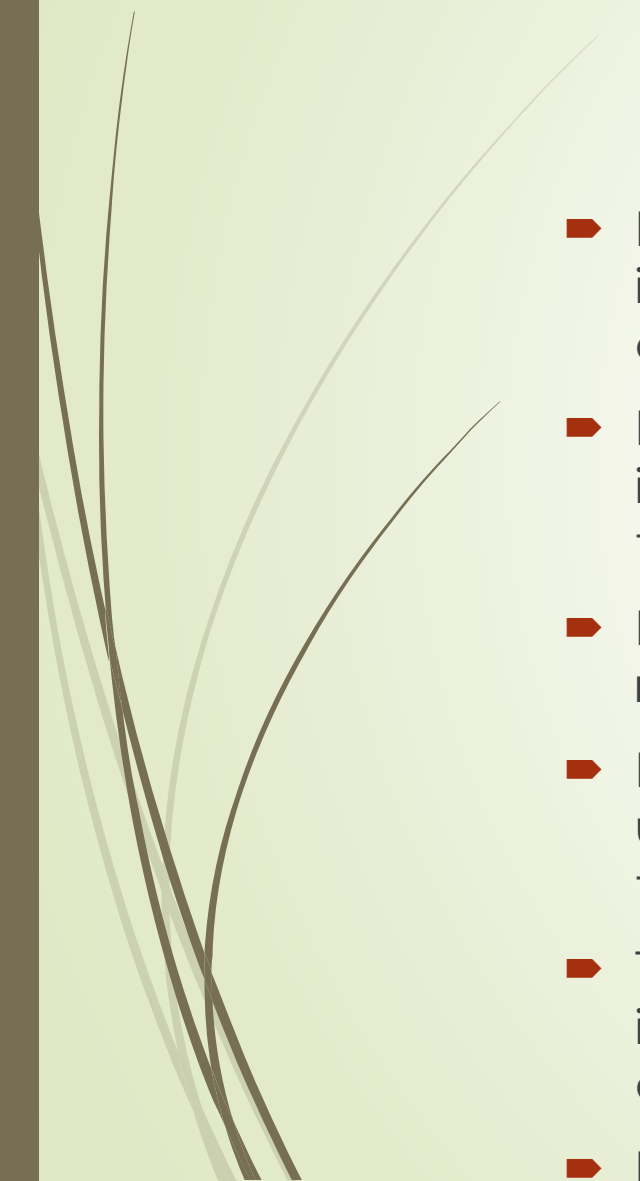


Step 1: Discovery & Development

Phase	Target Discovery	Target Validation	Lead Generation & Refinement	Preclinical Development	
Goal	Find All Targets	Eliminate Wrong Targets	Generate Molecules	Eliminate Molecules	Advance Molecules
					



Progress timeline



- Historically, drug discovery, design, and development mostly started with identifying active ingredients from traditional medicines or purely by chance.
- Later, classical pharmacology was used to investigate chemical libraries including small molecules, natural products, or plant extracts, and find those with therapeutic effects.
- Post human DNA was sequencing, reverse pharmacology has found remedies to existing diseases through modern testing
- Disease processes, molecular compound tests, existing treatments with unanticipated effects, and new technologies spur drug discovery timeline through the cycle
- Today the steps in drug discovery and development involve screening hits, iterative medicinal chemistry, and optimization of hits to reduce potential drug side effects (increasing affinity and selectivity).
- **Efficacy or potency, metabolic stability (half-life), and oral bioavailability**



► Target Identification And Validation

- Target identification **finds a gene or protein** (therapeutic agent) that plays a significant role in disease
- **record** the target's therapeutic characteristics
- **Criteria are:** efficacious, safe, usable, and capable of meeting clinical and commercial requirements
- **Validation of targets**, using modern tools and techniques such as disease association, bioactive molecules, cell-based models, protein interactions, signalling pathways analysis, functional analysis of genes, in vitro genetic manipulation, antibodies, and chemical genomics
- **EXAMPLE:** the Sanger Whole Genome CRISPER library and Duolink PLA are excellent sources for drug discovery targets
- https://www.sanger.ac.uk/news_item/2015-01-29-crispr-collaboration-aims-to-speed-up-drug-discovery

➤ **Hit Discovery Process**

Following target validation, compound screening assays are developed.

➤ **Assay Development And Screening**

- Assay development in drug discovery is a crucial component of drug discovery workflow. Assays are test systems that evaluate the effects of the new drug candidate at the cellular, molecular, and biochemical levels.

➤ **High Throughput Screening**

- High Throughput Screening (HTS) uses robotics, data processing/control software, liquid handling devices, and sensitive detectors to rapidly conduct millions of pharmacological, chemical, and genetic tests, eliminating hours of painstaking testing by scientists. HTS identifies active compounds, genes, or antibodies that affect human molecules.

➤ **Hit To Lead**

- In the Hit to Lead (H2L) process, small molecule hits from an HTS are evaluated and optimized in a limited way into lead compounds. These compounds then move on to the lead optimization process.

➤ Lead Optimization

In the lead optimization (LO) process, the lead compounds discovered in the H2L process are synthesized and modified to improve potency and reduce side effects. Lead optimization conducts experimental testing using animal efficacy models and ADMET tools, designing the drug candidate.

➤ Active Pharmaceutical Ingredients

- Active pharmaceutical ingredients (APIs) are biologically active ingredients in a drug candidate that produce desired effects.
- All drugs are made up of the API or APIs and excipients. Excipients are inactive substances that deliver the drug into the human system.
- High Potency Active Pharmaceutical Ingredients (HP APIs) are molecules that are effective at much smaller dosage levels than standard APIs.
- They are classified based on toxicity, pharmacological potency, and occupational exposure limits (OELs), and used in complex drug development involving more than ten steps.

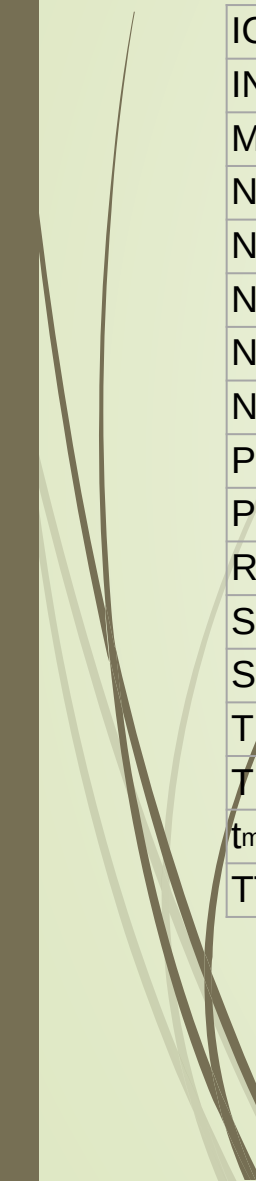
➤ **The drug discovery process gets narrowed when one lead compound is found for a drug candidate, and the process of drug development starts.**



Pre-Clinical Research

ACRONYMS

ADME	Absorption, distribution, metabolism, and excretion
API	Active pharmaceutical ingredient: any component intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment or prevention of disease.
CFR	Code of Federal Regulations
CGMP, GMP	(Current) good manufacturing practice
CMC	Chemistry manufacturing and controls
CoA	Certificate of analysis
CRO	Contract research organization
CTM	Clinical trial material
C _{max}	Maximum plasma concentration
DP	Drug product: finished dosage form (for example, tablet, capsule, solution) that contains an active drug ingredient, generally in association with inactive ingredients
DS	Drug substance: any substance that is represented for use in a drug and that, when used in manufacturing, processing, or packaging of a drug, becomes an active ingredient or a finished drug form
FDA	US Food and Drug Administration
FIH	First in human
FRS	Foreign related substances
GLP	Good laboratory practice
HPLC	High performance liquid chromatography

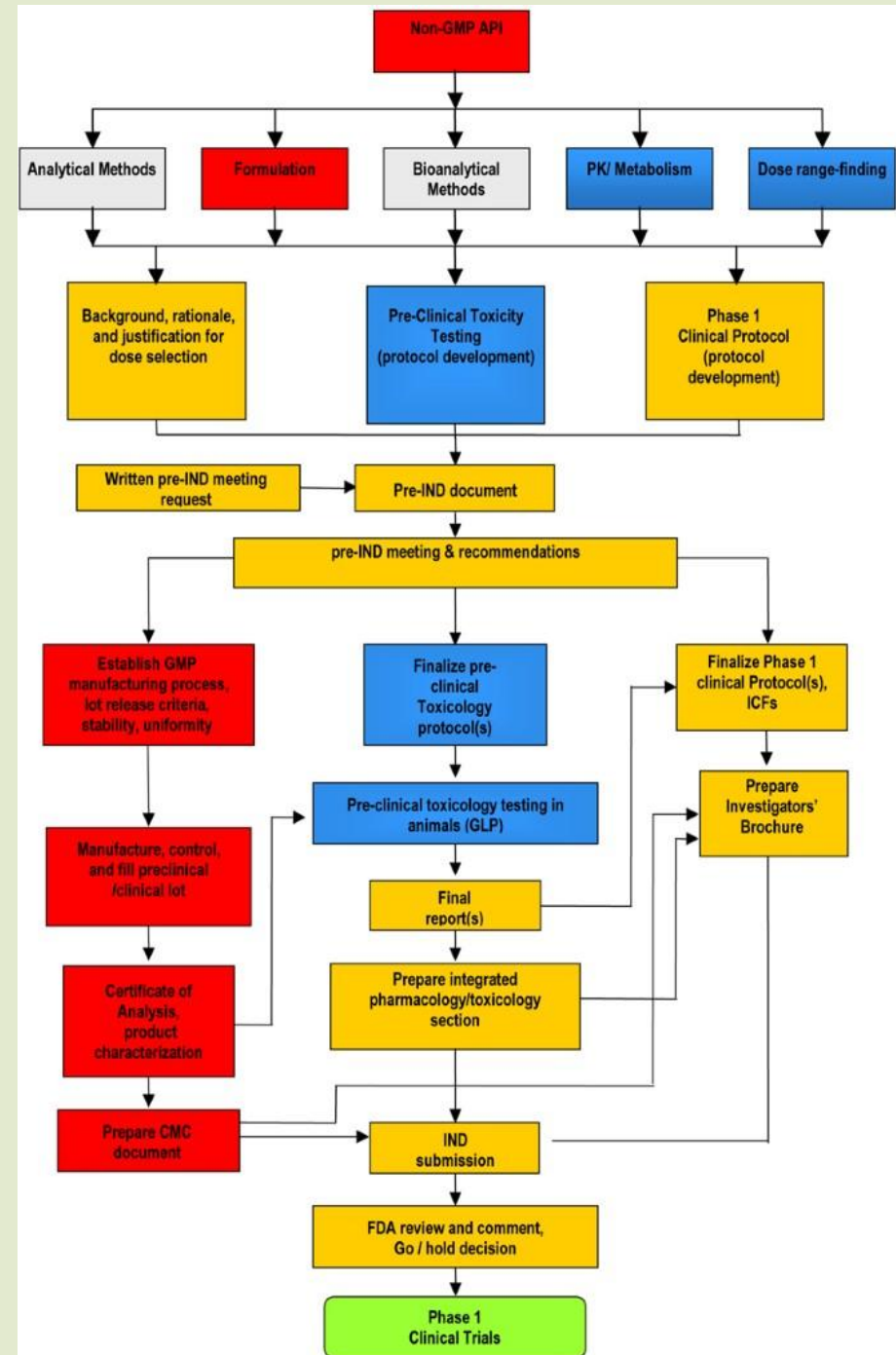


ICH	International Conference on Harmonisation
IND	Investigational New Drug application
MTD	Maximum tolerated dose
NCE	New chemical entity
NDA	New drug application
NCI	National Cancer Institute
NIA	National Institute on Aging
NOAEL	No observed adverse effect level
PIB	Powder in bottle
PK	Pharmacokinetics
RAID	Rapid Access to Intervention Development (preclinical program)
SBIR	Small Business Innovative Research (grant)
STTR	Small Business Technology Transfer (grant)
TI	Therapeutic Index
TK	Toxicokinetic
t _{max}	Time of maximum plasma concentration after dose administration
TTP	Target product profile

Preclinical flow diagram.

The parallel and inter-related activities contributing to preclinical development are summarized with color coding to denote related components: manufacturing (red), analytical (grey), documentation (orange), safety (blue), clinical (green).

API, active pharmaceutical ingredient; CMC, chemistry, manufacturing, and controls; FDA, US Food and Drug Administration; GLP, good laboratory practice; GMP, good manufacturing practice; ICF, informed consent form; IND, Investigational New Drug; PK, pharmacokinetics.





Preclinical development components

Target product profile

- The goal in drug development is FDA approval of a new drug application (NDA) and prescribed use in the clinic.
- Many project development teams find it helpful to develop a target product profile (TPP) to guide preclinical development.
- The TPP is a useful tool for delineating the required as well as desired features of the new drug product, critical milestones, and metrics to success.
- The TPP provides a framework to ensure that the preclinical development program supports the intended clinical trial design and therapeutic use



The contents of a TPP may vary from compound to compound and from team to team, but each profile generally includes

- i. therapeutic indication(s);
 - ii. market size, competition, and differentiators;
 - iii. expected clinical use, including key trial endpoints;
 - iv. drug target and mechanism of action;
 - v. patient age range; dose route, form, and frequency of administration;
 - vi. bioavailability and duration of action;
 - vii. safety, precautions, and contraindications;
 - viii. chemistry, manufacturing, and controls (CMC) profile, including solubility, manufacturing process, formulation, storage conditions, and stability;
 - ix. patent status and any modifiers of exclusivity (for example, orphan drug status).
- The TPP document continues to evolve as the drug development program progresses, and it should be regularly reviewed by the team to assess whether required goals are being met

- 
- Any results indicating potential safety concerns, or the inability of a given new chemical entity to fulfil stated TPP criteria should be evaluated by the team for their potential impact on project success.
 - **Implementation of a TPP helps to keep the focus of the project on key product criteria, thereby increasing the likelihood that 'Go/No-Go' decisions will be made in a timely manner and reducing the overall project risk.**



Pharmacology and experimental toxicology

- ▶ The prelude to IND-enabling preclinical development generally includes **efficacy, pharmacology, and experimental toxicology studies to define the dose, route, and frequency required for subsequent studies.**
- ▶ Using one or more pharmacological animal models of the disease, the initial efficacy studies demonstrate that treatment with drug candidates has the desired therapeutic effect.
- ▶ **Efficacy studies also help to identify the best drug candidates for further development.**
- ▶ **A number of studies are used to address the ADME characteristics of the drug.**
- ▶ **Bioavailability studies are generally conducted in vivo on candidates intended to be administered by a non-intravenous route.**



Bioavailability results provide information on the percentage of drug that is absorbed by the body as defined by quantity in plasma.

- Pharmacokinetics (PK) studies provide information on the maximum attainable plasma concentration (c_{max}), the time after dose administration to c_{max} (t_{max}), the mean residence time in the plasma, clearance, and other information used to characterize the body's effect on the drug.
- Initial dose range-finding and toxicity studies include single- and multiple-dose administration protocols with varied observation times.
- These earliest studies are intended to determine the maximum tolerated dose (MTD), identify observable signs of toxicity, and provide a rationale for setting dose levels in more complex definitive studies.
- Regulatory requirements almost always call for definitive studies in at least two laboratory animal species, one rodent (rat or mouse) and one nonrodent (rabbit, dog, nonhuman primate, or other suitable species).

- 
- 
- Preliminary toxicity, bioavailability, and PK studies should also include one or more rodent and nonrodent species, including the species to be used in the definitive studies.
 - The group sizes for the early range-finding studies may consist of only a few animals and one sex (one animal per dose level).
 - Once a suitable dose range is identified, group sizes are increased to at least three per sex per dose level to allow for statistical comparisons.
 - Only a few endpoints are collected in these range-finding studies unless there are particular toxicity concerns; a larger number of required endpoints will be evaluated in the eventual definitive studies described below.
 - Although scientific and reporting integrity is expected for any study, quality assurance audit and review are rarely required for bioavailability, PK, and range-finding studies.



Active pharmaceutical ingredient

- ▶ API and DS are different terms for the same type of chemical entity used in the FDA Guidance for Industry CGMP for Phase 1 Investigational Drugs (2008) and ICH document Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients (2000) and, therefore, these terms are used interchangeably.
- ▶ According to the 2008 FDA guidance
- ▶ *“any substance or mixture of substances intended to be used in the manufacture of a drug (medicinal) product and that, when used in the production of a drug, becomes an active ingredient of the drug product.”*
- ▶ APIs include substances manufactured by processes such as **chemical synthesis, fermentation, recombinant DNA or other biotechnology methods, isolation/recovery from natural sources, or any combination of these processes.**



API must be well characterized in terms of **structure identity (crystalline or polymorphic), counter ions (salts) and co-crystals, impurities, stability, chirality and enantiomer(s), appearance, solubility, and other chemical and physical properties.**

- These properties will continue to be referenced throughout API scale-up process chemistry and GMP manufacturing.
- All preclinical drug development programs require an adequate drug supply.
- As development progresses, increasing quantities of higher quality APIs are required for small non-good laboratory practice (non-GLP) efficacy studies, early formulation activities, in vivo dose range-finding studies, and finally rigorous IND-enabling GLP toxicology studies.
- For small molecule drugs, milligram quantities of research grade material are usually suitable for early stage in vivo efficacy and ADME studies, as well as small PK and dose range-finding studies (the latter may require gram quantities).



When a molecular entity is selected, pre-formulation activities commence to determine its physical and chemical properties, including counter ion salt or polymorphic form, solubility, and stability.

- The outcome of this stage is a recommended form, and the API portion of the project will transition to issues surrounding reaction efficiency, cost of goods, purity and control of impurities, and batch-to-batch consistency
- synthetic steps, along with control documentation, will be written into the master batch record and included in the IND package in the CMC section.
- the investigator may opt to transfer the synthetic process (along with appropriate legal intellectual property documentation) to a specialized contract research organization (CRO) that will produce required batches along with a Certificate of Analysis or GMP release for each batch.
- Once an API batch is released, it is ready to be used in GLP safety toxicology studies or prepared/formulated for clinical use.



Formulation

- Drug formulation, the mixing of API with other chemical ingredients to create the drug product (DP), is often a major hurdle in drug development.
- At this stage, the route of administration intended for the clinic should be clearly identified i.e. enterally (oral, buccal, and rectal) or parenterally (not through the alimentary canal) including by injectable routes (intravenous, intramuscular, and subcutaneous), topically, and by inhalation
- Drugs intended to be administered orally may be formulated as a solution, suspension, capsule, or tablet.
- Any formulation intended for human use is subject to rigorous quality control manufacturing and safety testing.
- clinical application, the specific physicochemical properties of a DS will influence formulation options



Parameters considered when creating the DP formulation.

- i. The components of any formulation must have physical and chemical compatibility with the DS.
 - ii. Patient factors to consider include tablet/capsule size, taste, stability and shelf life, and ease of use.
 - iii. Formulations may incorporate components such as dissociation enhancers that are found to improve bioavailability of the active DS.
 - iv. Solid formulations, particularly tablets, may be coated to improve swallowing, mask an unpleasant taste, protect ingredients during storage, improve appearance, and control drug release over time or target it to specific regions of the gastrointestinal tract.
- In most instances, the clinical drug formulation is not fully optimized before submission of the IND and the initial first-in-human (FIH) clinical studies thus a simple formulation to be presented as the DP in the IND and used in phase1 studies to deliver the drug to human subjects.



Analytical methods

- Reliable analytical methods are required to test and qualify in-coming materials, in-process methods, equipment, formulations, DSs, and DPs.
- These methods are critical for analyzing the various formulations that may be investigated for a final dosage form and are integral to quality control in GLP and GMP settings.
- FDA and ICH guidelines require stability testing on each lot of DS and DP.
- analytical methods may need to be developed for a variety of materials and circumstances, each with a different intended purpose
- the analytical method required for formulation development may not require the same performance characteristics as those required for a stability-indicating method for DS or DP.

- 
- Analytical work consists of two components: a research and development component and a GLP/GMP component
 - The research and development component includes analytical method development and analytical support for preformulation and formulation.
 - The rest of the analytical work is conducted according to GLP and GMP guidelines. The quality control unit is responsible for the oversight of GMP analytical work.
 - It is essential that the validated methodology used to test the DS be used in clinical manufacturing.
 - The developed method must satisfy two requirements:
 - i. it must be accurate, requiring high specificity, good precision, and good reproducibility;
 - ii. it should be practical, with the necessary ruggedness, sensitivity, and linearity.



Assay methods are verified under the ICH guidelines for reproducibility, specificity, selectivity, accuracy, linearity, precision, applicable concentration range, limit of detection, limit of quantification, ruggedness, and robustness.

- The specifications for DS typically include a physiochemical characterization program that generally requires determination of the composition, physical properties, and primary structure of the desired product.
- The suitability of a final compound for pharmaceutical use requires establishment of its identity and purity, as well as knowledge of its chemical and physical properties.
- Formulation analysis verifies the active and inactive components and dosage, assesses potency, determines shelf-life stability, confirms dissolution properties, and determines whether decomposition has occurred or impurities have been introduced during the formulation process.



Bioanalytical methods

- Physiologic fluids such as blood, plasma, and urine are analyzed to determine the fate and disposition of a DS administered to a test animal or patient by sampling aliquots of blood over time to determine therapeutic drug concentration ranges. Often the goal is to assess the overall ADME characteristics of the DS.
- The concentration of the drug in the biological matrix changes with time, typically over a broad range, and, therefore, bioanalytical quantification limits are at concentrations much lower than those required for formulated or bulk drugs.
- bioanalytical method is required to detect drug at these low levels, as well as linearly over an appropriate range.
- Matrix effects and stability issues can also make accurate analysis of the analyte difficult; these include, among many others, endogenous materials extracted from the biological matrix that may interfere in the analysis, enzymes in the biological fluid that are capable of metabolizing the analyte, plasma proteins that the analyte can bind to, and concomitant drugs that might interfere in the analysis.



Pharmacokinetics, toxicokinetics, and metabolism

- PK, or the study of the time course of a drug in the body incorporating the processes of ADME, is a key determinant in the selection of a viable drug candidate.
- many potential drugs are eliminated from further development because of poor PK properties, drug discovery programs now incorporate early ADME screens for desirable 'drug-like' properties to optimize the selection of successful candidates.
- These predictive ADME approaches include in silico models, physiochemical parameters, and in vitro assays of permeation and drug metabolism to better evaluate the properties of potential pharmaceuticals and to concentrate additional efforts on only the most promising compounds.
- these are screened for data and evaluated in concert with efficacy results, compounds that are predicted to have favorable PK properties are studied further using in vivo animal models.



PK parameters are extrapolated from measurements of drug concentration in the plasma, blood, or other relevant biological matrix over a selected time period (usually several time points concluding at 24 or 48 hours post dose).

- PK provides information that can guide future animal and clinical studies for the selection of the dose levels and frequency of administration.
 - The IND package requires PK data generated in two species (one rodent, one nonrodent), preferably using the same two species used for the safety studies.
 - These studies usually include multiple dose levels so that PK dose dependency can be evaluated. Oral and intravenous administrations are compared to determine the oral bioavailability of the drug if an oral route is anticipated for the clinic.
 - However, it is not necessary to delineate all ADME characteristics of a molecule at the time of IND submission, since the first clinical study generally focuses on the pharmacokinetics and metabolism of the drug in humans.
- 



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- 



Toxicokinetic (TK)

- ▶ Toxicokinetic (TK) endpoints are similar to PK except that the samples are collected from animals during the toxicology and safety studies (described below) and usually do not cover as many time points.
 - ▶ They help the investigator understand the drug's behavior at the maximal dose levels used in toxicology studies as well as steady state, accumulation, and trough levels after repeated administration.
 - ▶ The same bioanalytical methods as developed for in vivo PK evaluations are usually suitable for measuring drug concentration in the relevant biological matrix (for example, plasma); however, the method will need to be validated if it is applied to GLP study samples.
- 



Metabolism studies

- Metabolism studies are also recommended by the FDA.
- These studies can be useful to evaluate the potential for drug-drug interactions and cytochrome P450 inhibition and to generate drug metabolite profiles for different species, including humans.
- Metabolism studies are typically conducted using in vitro methods for exposing hepatic microsomes, cytosolic fractions, hepatocyte cultures, or other applicable test systems to the drug.
- Using a bioanalytical method, any metabolites are described as a fractional percent of parent-drug peak. Any metabolite occurring at a significant level (for example, accounting for 10% or more of parent drug level, particularly when using test systems derived from primary human tissues) likely represents a significant metabolic pathway for the drug and may require further investigation, including elucidation of metabolite identity and biological activity.



In vitro metabolism studies are frequently the first glimpse into the effects of human metabolic pathways on a drug and may reveal considerable metabolic differences between humans and preclinical species.

- In the event of significant interspecies metabolite profile differences, investigators may compare these with the human metabolite profile to identify which species, if any, most closely matches the humans, and then select this species for the pivotal safety and toxicology studies.
- The importance of these interspecies results is for understanding the potential disposition of the drug in humans, many development plans begin these studies about the same time as the first animal PK studies.

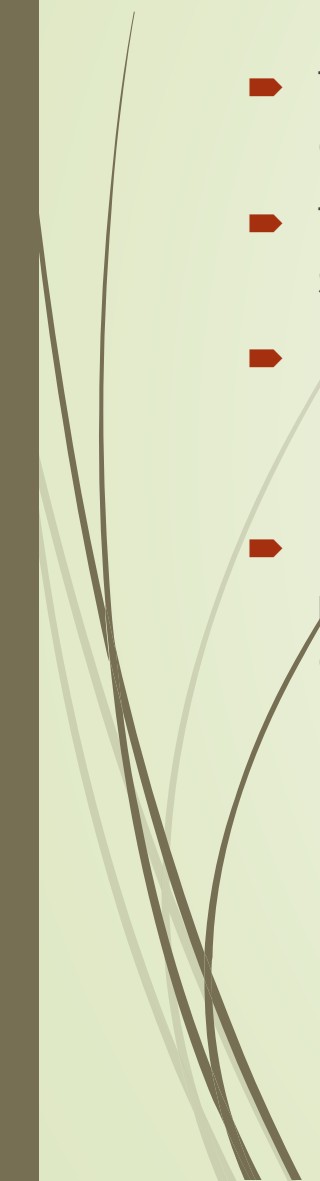


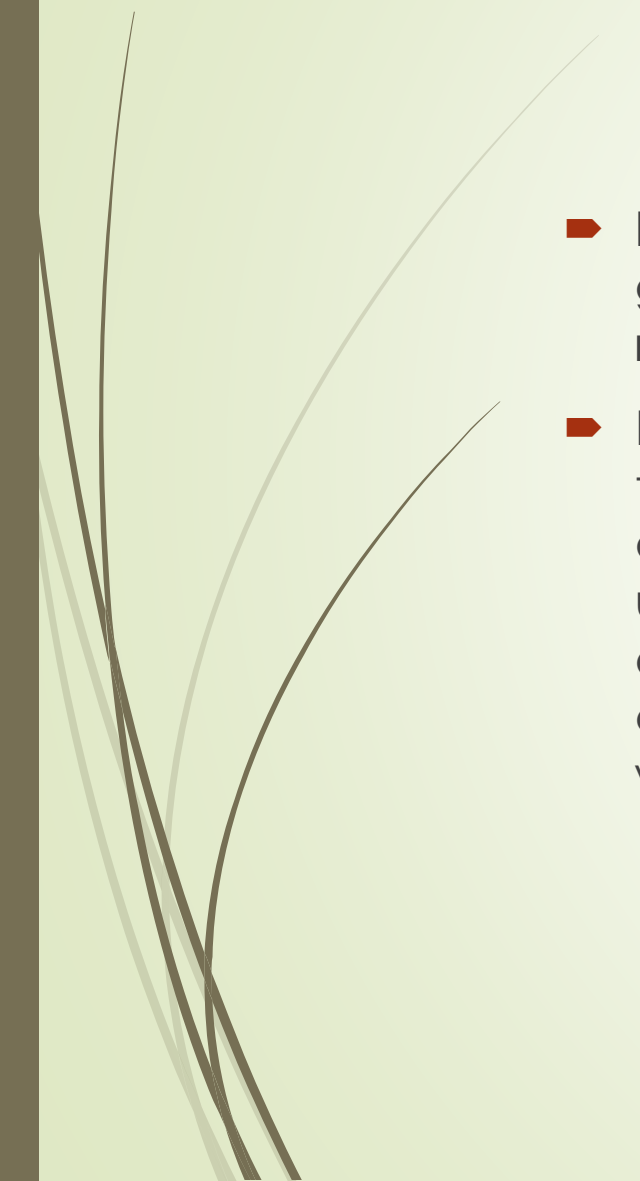
Pivotal toxicology and safety

- Despite numerous technical advances in the science of toxicology and attempts to develop in silico screening, the primary methods used to assess safety remain single- and repeat-dose toxicology studies conducted in rodent and nonrodent species.
- Definitive animal studies establish the safety characteristics, including the **no observable adverse effect level** (NOAEL), of the candidate drug
- These studies are rigorously documented and conducted under regulatory guidelines –FDA GLP.
- The highest dose levels tested in the definitive animal studies are almost always based on the MTD determined from the range-finding studies rather than either the expected dose level to be used in the clinical trial or an expected plasma concentration.
- By the time a drug reaches definitive animal safety testing, a human trials clinician is included in the drug development team to provide study details on the proposed FIH clinical trial



In general, the route and frequency of drug administration, vehicle, and dose formulation (if applicable) to be used in initial human trials is reflected in the definitive animal studies.

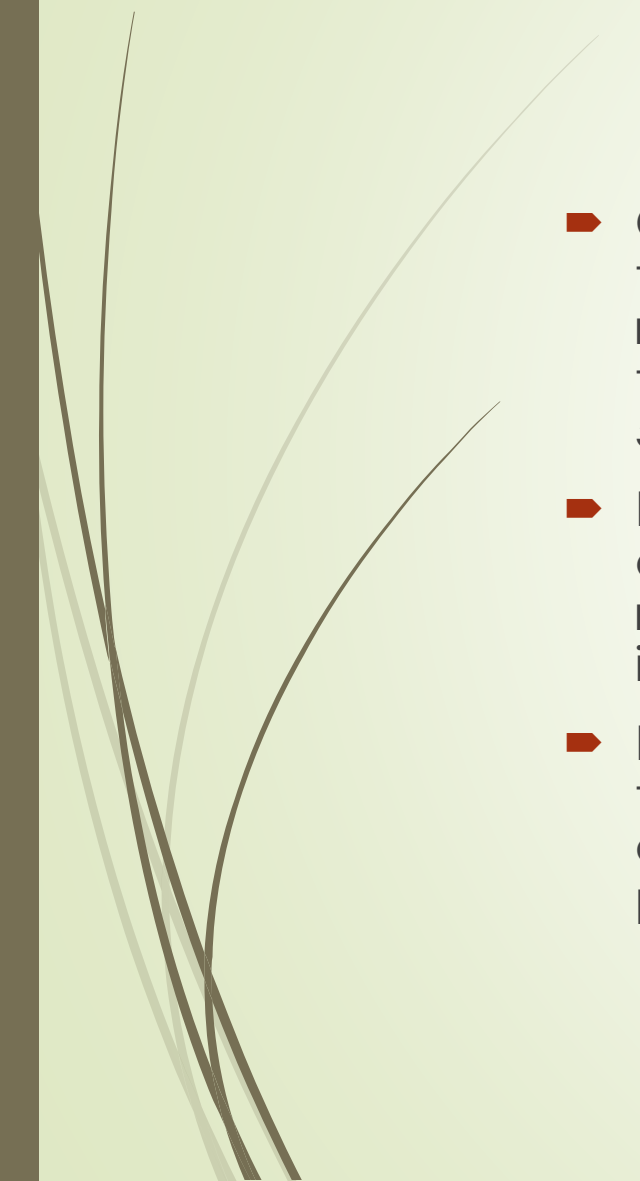
- To account for differences between humans and laboratory species, a safety margin is established based on the NOAEL in the 'most sensitive' of the tested species.
 - Toxicology studies are commonly conducted in rats and dogs, though other large animal species may be appropriate for specific products or therapeutic applications.
 - For example, rabbits are frequently the species of choice for safety testing of vaccines. Preliminary toxicity studies are often performed as part of the lead compound selection process.
 - For IND-directed safety studies, two complete GLP-compliant safety studies are generally required. The route of administration in these studies must be the same as the proposed clinical route.
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- If the proposed route is oral, drug is administered by gavage to rats and by gavage or capsule to dogs. The duration of administration and dose regimen must, at a minimum, conform to the proposed clinical protocol.
 - For example, if 14 days of continuous drug administration is proposed for the phase 1 clinical trial, then animal toxicity studies of at least 14 to 28 days are typically required to support a clinical study of this length. Although usually occurring after phase 1 dosing, longer term animal studies (for example, 60 days and longer) will be needed to support later stage human clinical trials. The frequency of dosing (for example, three times a week for 4 weeks) in the animal studies should also mimic the clinical dosing schedule.



Pivotal safety studies

- ▶ Pivotal safety studies are performed with drug manufactured under GMP conditions whenever possible, although the FDA does not state this requirement.
- ▶ In the event that a drug is not manufactured under GMP conditions, the investigator is required to demonstrate that the clinical drug is essentially the same as that used in animal safety studies.
- ▶ If significant differences (for example, different counterion salts) are observed between GMP materials scheduled for the clinic and the materials used in a pivotal preclinical safety study, the regulating agency may deem the safety study to be invalid and request that bioequivalency studies be conducted or the pivotal study be repeated using the correct materials.

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- ▶ One strategy used by some investigators to pre-emptively qualify some of the possible manufacturing impurities, such as enantiomers or side reactions, that may be present in batch production is to utilize a batch in the animal studies that contains at least 95% purity of the drug and up to 5% impurities.
 - ▶ Because most repeat-dose toxicity studies of therapeutics reveal some adverse effects at higher dose levels, group assignment also includes a recovery group to evaluate whether adverse effects are transient or irreversible after repeat dosing.
 - ▶ For dose level selection, allometric body surface area scaling may be used to convert from many preclinical species to human dose levels, with additional multiples added to account for interspecies differences and potential safety factors.



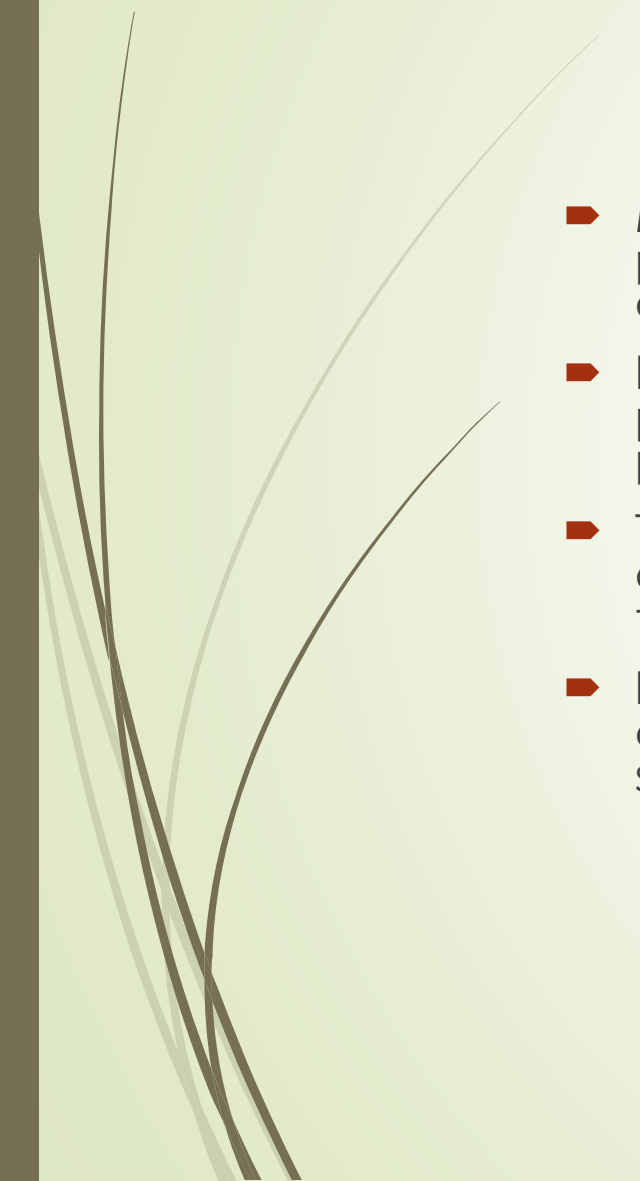
To the pivotal safety studies, required ancillary studies, specified in ICH guidelines, complete most regulatory packages.

- Safety pharmacology studies are required regulatory elements outlined in ICH guideline S7A (This document provides a definition, general principles and recommendations for safety pharmacology studies. This Guideline generally applies to new chemical entities and biotechnology-derived products for human use)
- Standard procedures are established for the safety pharmacology core battery, which includes assessments for the central nervous system, cardiovascular system, and respiratory system. Genetic toxicology studies, outlined in ICH guidelines S2(R1) are required for most small molecules
- The FDA requires submission of data from three genetic toxicology assays for most IND applications:
 - i. gene mutation in bacteria,
 - ii. including four strains of *Salmonella typhimurium* and one strain of *Escherichia coli* (Ames test);
 - iii. an in vitro mammalian cell assay (either evaluation of chromosomal damage in Chinese hamster ovary cells or the mouse lymphoma mutagenesis assay);
 - iv. in vivo chromosomal damage in rat or mouse hematopoietic cells (such as the rodent bone marrow micronucleus assay).

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- If the candidate drug may have immunosuppressive effects identified in earlier toxicity or safety studies, ICH guideline S8 provides an overview of follow-up programs.
 - Other studies that are frequently required in later stage drug submissions (after phase 1 clinical studies) include carcinogenicity and reproductive toxicology studies.



Pre-IND meeting



- Meeting for the drug submitter to proposed preclinical and clinical trial protocols, as well as the proposed manufacturing and control testing of the DS and DP, to the appropriate FDA division representative(s).
- begins a dialog with the FDA to address specific questions concerning the proposed preclinical drug development program and its relation to the proposed early stage clinical trial(s).
- These discussions should occur prior to the initiation of complex preclinical activities (for example, GLP multiple-dose studies) to ensure that the design of the preclinical program addresses any potential concerns posed by the FDA.
- Incorporating this step into the process greatly improves the likelihood that the clinical trial will proceed within the planned timeframe following the IND submission.



Phases in Clinical Trials



A new treatment goes through several phases. Each phase has a different purpose:

Phase I trials test if a new treatment is safe and look for the best way to give the treatment. Doctors also look for signs that cancer responds to the new treatment.

Phase II trials test if one type of cancer responds to the new treatment.

Phase III trials test if a new treatment is better than a standard treatment.

Phase IV trials find more information about long-term benefits and side effects.

How does a clinical trial work?

Clinical trials occur in four phases, and each phase has a different purpose.

Phase I



Focus on **safety** and the proper dose.

15 to 50 patients

Phase II



Focus on **effectiveness** and side effects.

Less than 100 patients

Phase III



Compares the **new treatment** to existing treatment.

Hundreds of people

Phase IV



Treatment is **approved and available**. Long-term effects are observed.

Thousands of people

Phase I

- Phase I trials are concerned primarily with establishing a new drug's safety and dose range in about 20-100 healthy volunteers.
- Pk studies ADMET
- This is determined through frequent blood draws (usually in an inpatient environment) to check for the level of drug in the blood plasma.
- Pharmacokinetic trials are usually considered Phase I trials regardless of when they are conducted during a drug's development.
- Dosage range of a new drug is determined by administering increasingly larger doses to one or more groups of subjects, who are closely monitored for harmful side effects.
- The goal is to learn the **maximum tolerated dose** that does not produce unacceptable side effects.

Phase I studies may involve risks even though an investigational drug has passed the Preclinical phase of testing.

Phase I studies typically offer little or no benefit to the volunteer subjects; therefore they typically are compensated for their time and effort.

Although usually conducted with healthy volunteers, Phase I trials are sometimes conducted with severely or terminally ill patients, for example those with AIDS or cancer.

A Phase I trial takes several months to complete. About 70 percent of experimental drugs pass this initial phase of testing.



Phase II

- Phase II studies determine the effectiveness of an experimental drug on a particular disease or condition in approximately 100 to 300 volunteers.
- This phase may last from several months to two years.

A Phase II trial answers the question, "Does Drug X improve Disease Y?"

A secondary objective for a Phase II trial is to ascertain therapeutic dose level and dosing frequency. This answers the questions, "What quantity of Drug X works better on Disease Y, (1 mg, 2 mg or 3 mg)?" and "Does Drug X work better on Disease Y taken once or twice a day?"

Most Phase II studies are randomized, which means that subjects are assigned randomly (by chance not by choice) to receive either the experimental drug, a standard treatment or a placebo (harmless, inactive substance). Those who receive the standard treatment or placebo are called a control group.

Phase II

Randomized Phase II studies are often double-blind, which means that both subject and physician don't know which treatment is being used. Blinding prevents any unscientific influence on the study results that could be caused by knowledge of the treatment. In a single-blind study, only the subject is unaware of the treatment used.

Since larger numbers of patients receive a treatment in Phase II studies than in phase I studies, there is a greater chance to observe and compile side effect information. Subjects in a Phase II trial may benefit from their participation if they receive an active treatment. Approximately 33 percent of experimental drugs which pass Phases I and II will go on to Phase III.



Phase III

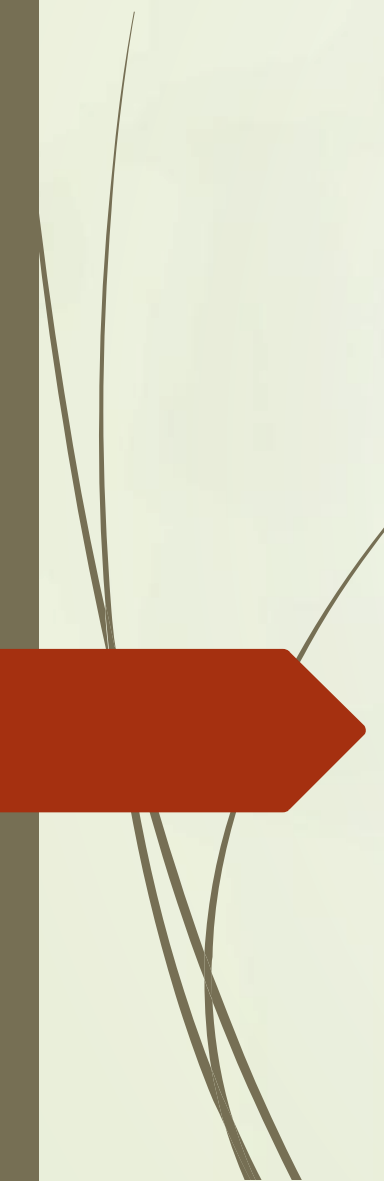
- Phase III studies are conducted at multiple centers with several hundred to several thousand patients for whom the drug is intended. Massive testing of a drug provides continued generation of data on a drug's safety and efficacy.
- As in Phase II, most Phase III studies are randomized and blinded.

Phase III trials provide the bulk of information needed for the package insert and labeling of a medicine, after it has been FDA approved.

A drug in this phase can be studied for several years and may be one of 25-30 percent that pass Phases I, II and III. Once a Phase III study is completed, a pharmaceutical company can request FDA approval to market the drug. This is called a New Drug Application (NDA). The NDA contains all the scientific data that the company has gathered throughout the phases in all trials.

The background of the slide features a faint, dark ECG (heart rate) line plotted on a grid of small orange dots. On the left side, there is a solid orange arrow pointing towards the right, partially overlapping the text.

Types of Clinical Trials



Treatment Trials

- Clinical trials are research studies that test the safety and effectiveness of new treatments, vaccines, or diagnostic procedures. They are the primary way researchers determine if a new treatment is safe and effective

Prevention Trials

- Prevention clinical trials are designed to prevent the onset of a disease, condition, or disorder. They can involve vaccines, vitamins, medicines, and lifestyle changes.

QoL Trials

- A diagnostic trial is a research study that evaluates disease detection methods. Diagnostic tests are used to help determine the presence or absence of a disease based on a person's signs and symptoms. Diagnostic tests can also be used to plan treatment, determine how well treatment is working, and make a prognosis.

Diagnostic Trials

- A diagnostic trial is a research study that evaluates disease detection methods. Diagnostic tests are used to help determine the presence or absence of a disease based on a person's signs and symptoms. Diagnostic tests can also be used to plan treatment, determine how well treatment is working, and make a prognosis.



Parallel Design

- Parallel arm design is the most commonly used study design. In this design, subjects are randomized to one or more study arms and each study arm will be allocated a different intervention. After randomization each participant will stay in their assigned treatment arm for the duration of the study

Cross-over Design

- In a crossover study design, two or more treatments (e.g., drugs, procedures) are provided to subjects at different time periods, and the sequence of treatments is randomized for each subject.

Factorial Design

- A factorial clinical trial design involves conducting two or more intervention comparisons at the same time. This allows researchers to obtain evidence about efficacy from fewer patients than would be needed if the comparisons were conducted in separate trials.

Randomized Design

- A randomized clinical trial (RCT) is a study where participants are randomly assigned to treatment or control groups. RCTs are considered the gold standard for evaluating the effectiveness of interventions.



Adaptive Design

- An adaptive design is defined as a design that allows modifications to the trial and/or statistical procedures of the trial after its initiation without undermining its validity and integrity. The purpose is to make clinical trials more flexible, efficient and fast.

Superiority Design

- A superiority trial is a clinical trial design that shows that one treatment is better than another. In a superiority trial, a new drug is tested against a placebo. The sample size calculation for a superiority trial is based on the idea that the relevant confidence limit for the difference between the two treatments excludes zero.

Non-Inferiority Design

- A non-inferiority trial is A non-inferiority trial is a clinical trial that aims to show that a new treatment is not worse than an active treatment by more than a pre-specified margin

Parallel Design

- A parallel clinical trial is a type of clinical study where two or more groups of participants are assigned to different treatment arms at the beginning of the trial.
- Participants remain in the same treatment arm throughout the trial.
- Most commonly used design



Controls in CT

Placebo

A placebo-controlled trial is a clinical trial that has two or more groups. One group receives the active treatment, while the other receives a placebo. The subjects in these trials are blinded as to whether they receive the treatment or a placebo

No Treatment

A no-treatment control group is a group of people who are not exposed to any experimental manipulation or intervention. They serve as a neutral comparison for study groups receiving the treatment under investigation.

Active

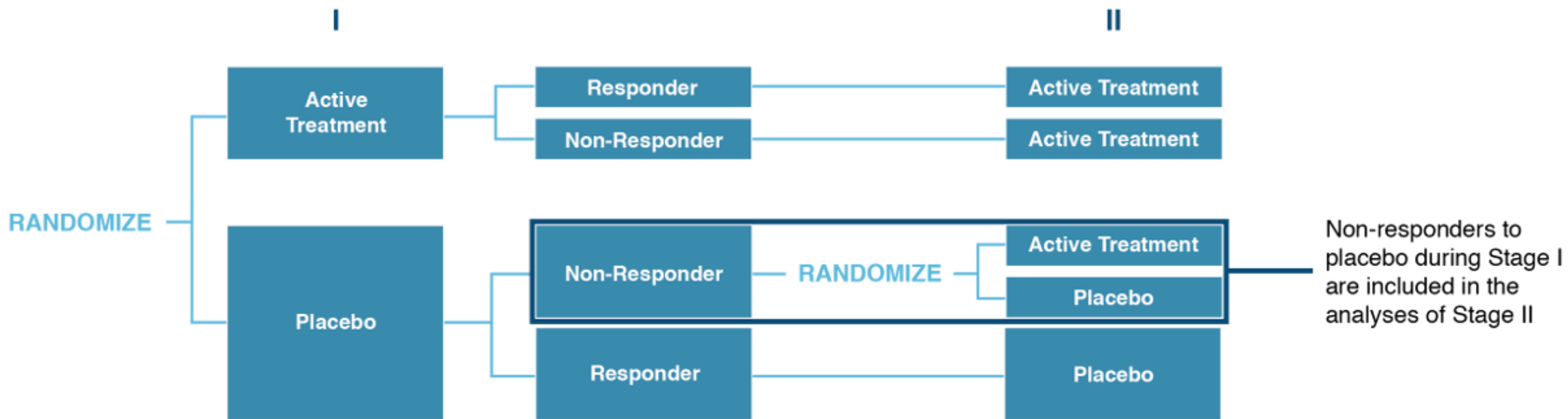
active control is when an experimental treatment is compared to an established treatment that has a known level of effectiveness. The goal is to either demonstrate that the experimental treatment is as good as or better than the active treatment. Active control trials are performed when it is considered unethical to include a placebo control group.

External Control

An external control in a clinical trial is a cohort of patients who are not part of the randomized study. The control group is made up of patients who are not receiving the new treatment being studied. The control group is compared to the group that receives the new treatment to see if the new treatment works

Dose Response Control

Dose-response (D/R) studies are generally conducted in phase II clinical trials. These studies help to provide proof of the concept, to define the dose for phase III clinical trial, and to establish a D/R relationship to manage adverse drug reactions





Placebo Control

- Inert by nature
 - The placebo is specifically designed to have no real effect.
 - In some cases, none of the participants know whether they are taking the active or inactive (placebo) substance
 - Used in **Efficacy testing**
 - **Double blinded studies**
 - The placebo effect makes it more difficult to evaluate new treatments.
 - For example, if you take a specific pill for headaches, you may begin to associate that pill with pain relief. If you receive a similar-looking placebo pill for a headache, you may still report decreased pain due to this association.
 - Placebo controls should not be used if there is evidence of the efficacy of another treatment or “good evidence” that the experimental therapy is effective.
-
- **Adv:** Minimized bias, demonstrating efficacy
 - **Disadv:** Ethical issues: Lack of treatment, Declaration of Helsinki- use of a standard
 - Patient issues: demotivated to participate, lack of enrollment and withdrawal



When is Placebo Control used:

- There **is no proven effective treatment** for the condition being studied
- Withholding treatment poses **minimal risks** to participants
- There is **no known or available alternative therapy** that can be tolerated by subjects
- The study question cannot be answered in an active-controlled trial design
- The risks of delaying or foregoing an efficacious vaccine are mitigated



No Treatment control

- A no-treatment control group is a group of people who are not exposed to any experimental manipulation or intervention.
- They serve as a neutral comparison for study groups receiving the treatment under investigation
- A no-treatment control can only be used when it is possible to ensure that:
 - The control group participants are not given the experimental treatment
 - They cannot obtain it from any other provider
 - They cannot obtain any other treatment for the same problem
- In a trial with no-treatment concurrent controls, a group receiving the experimental intervention is compared with a group not receiving the treatment or placebo.



Active control

- An active control is a treatment that all participants in the trial receive, instead of a placebo.
- The goal of an active control trial is to compare an experimental treatment with a known, effective treatment.
- The goal is to show that the experimental treatment is as good as, or better than, the known treatment
- Active control trials are performed when it's considered unethical to include a placebo control group. For example, for certain drug classes, such as analgesics, antidepressants, or anti-anxiety drugs, it's common for a study to fail to show superiority to a placebo.
- On the other hand, an active control trial is one in which an investigational drug is compared with an established treatment that has a known degree of effectiveness, with the aim of either demonstrating that the test treatment is as good as or is superior to the active treatment.



Dose response control

- Dose-response (D/R) studies are generally conducted in phase II clinical trials. These studies help to provide proof of the concept, to define the dose for phase III clinical trial, and to establish a D/R relationship to manage adverse drug reactions
- Both PK and PD differences in individuals could result in varied D/R relationships. If a previous study of PK and PD suggested the association of genetic variants, genotyping must be considered during the D/R studies.
- Genetic information should also be deemed while evaluating the blood levels of a drug.

External Response control

- An external control in a clinical trial is a cohort of patients who are not part of the randomized study.
- The control group is made up of patients who are not receiving the new treatment being studied.
- The control group is compared to the group that receives the new treatment to see if the new treatment works.
- One major limitation of externally controlled trials is the potential for confounding bias. This means that it can be difficult to establish comparability between the treatment and control groups



Uncontrolled

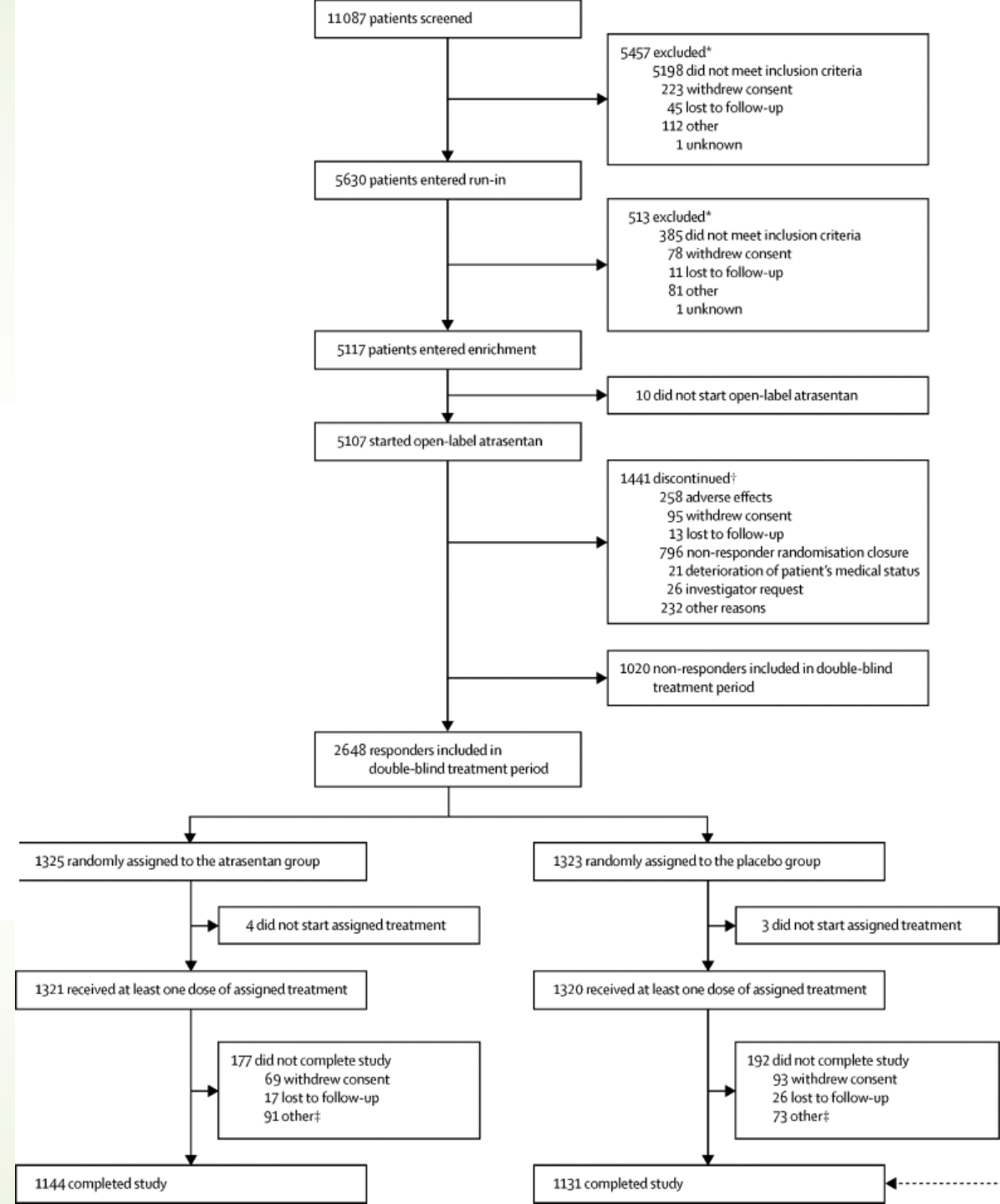
- An uncontrolled clinical trial is a clinical study that doesn't have a control group.
- Uncontrolled trials are often used to determine a new drug's pharmacokinetic properties, such as in Phase 1 trials
- Uncontrolled trials have one treatment arm, and all patients receive the same intervention.
- Uncontrolled trials are clinical studies that study new interventions without a control group.
- Uncontrolled trials are clinical studies that lack a comparison group.
- Uncontrolled trials are known to produce higher mean effect estimates than controlled trials, which can inflate expectations from the intervention.
- They also usually produce less information on the therapy than controlled trials.



Run-in Design (Placebo run-in design)

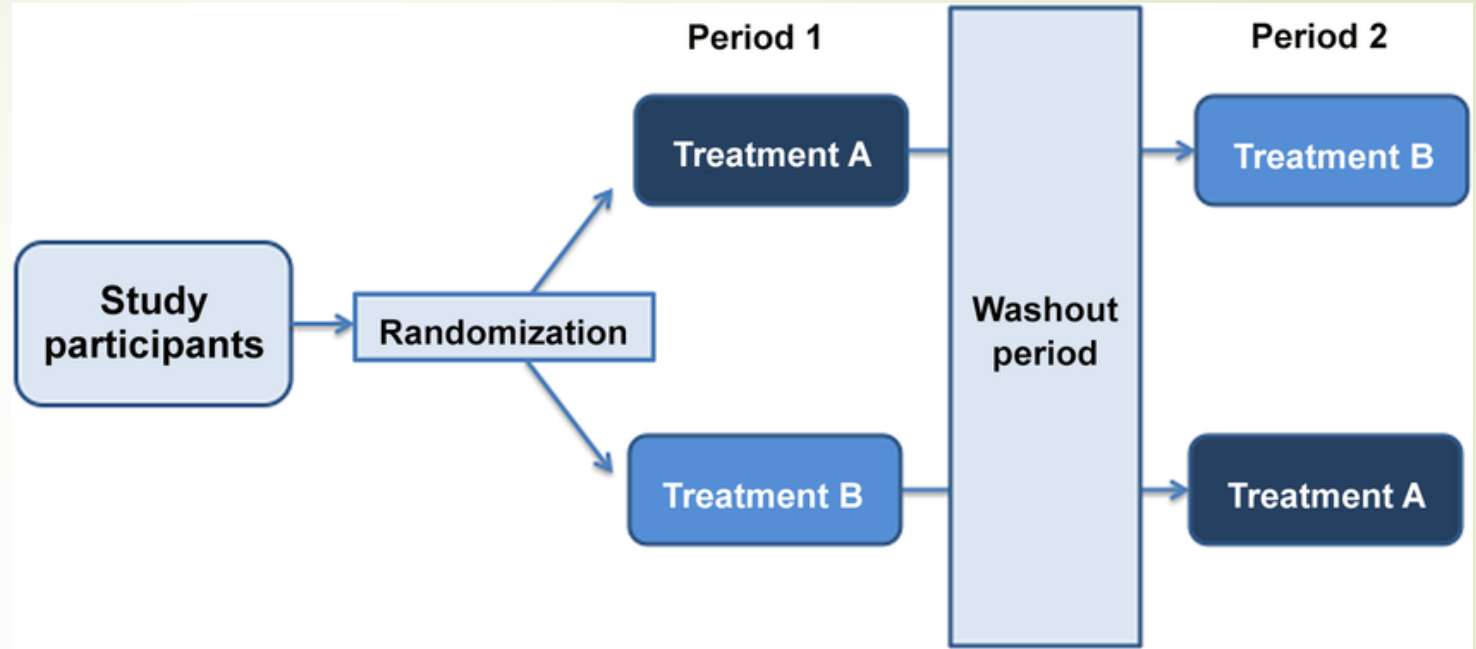
- Placebo run-in design – Placebo run-in period is a period before a clinical trial is commenced, when placebo is administered for all study subjects.
- A run-in period is a time period after inclusion, but before randomization, used to exclude certain patients.
- Other pre-randomization periods exist, for example, extended screening periods and washout periods. These different pre-randomization periods may overlap in purpose, design and terminology.
- Run-in periods are used to select or exclude patients in an increasing number of clinical trials. However, the implications of run-in periods for interpreting the results of clinical trials have not been systematically examined.

Run-in Design



Crossover Design

- A crossover design is a type of randomized trial where each subject receives each treatment in succession.
- A crossover study design randomly assigns subjects to study arms.
- Each arm consists of a sequence of two or more treatments given consecutively
- USE: BioEquivalence studies and Phase I



- Each patient serves as their own control
- Avoids between participation variation
- Requires small patient size
- Assumptions
 - The effects of the First Period do not carry over to the Second period
 - Internal & External factors are constant

Prerequisites of Crossover Design

- Disease- chronic
- Effects of the drug should develop within the trial period
- washout period: Sufficiently long enough for reversibility of the drug effect
- Wash-Out Period: FIVE HALF LIVES of the drug

Problems of Crossover Design

- Not useful acute disease
 - Carryout Effect
 - Allocation of ADR affects difficulty
 - Generally not used in Vaccine trials as it affects immune system completely
- 



Parallel

Groups assigned different treatments

Shorter duration

Sample size- large

No carryover effect

Acute cases

Crossover

Each patient receives both treatments

Longer

Smaller

Carryover effect

Not in acute,
Chronic,stable

Latin Square Design

		Period		
		I	II	III
Subjects	1	A	→ B	→ C
	2	B	→ C	→ A
	3	C	→ A	→ B

Intensive Design: Comparing 2 Tts.(A & B) each subject receives same Tt. several times.

Period	I	II	III	IV	V	VI
Treatment B		A	B	B	A	A

Greco-Latin Squares

Effect of treatment & effect of another factor (eg. ancillary Treatment, diet etc)

		I	II	III
		$A\alpha$	$B\beta$	$C\gamma$
Subjects	1	$A\alpha$	→ $B\beta$	→ $C\gamma$
	2	$B\gamma$	→ $C\alpha$	→ $A\beta$
	3	$C\beta$	→ $A\gamma$	→ $B\alpha$

- For short period of R_x /single dose
- For testing efficacy of new compound

Randomized Design

- Randomized design is an experimental design approach that involves randomly assigning experimental conditions.
- Three of the more widely used experimental designs are the **completely randomized design, the randomized block design, and the factorial design**
- **CRD :**
- **A completely randomized design (CRD) is one where the treatments are assigned completely at random so that each experimental unit has the same chance of receiving any one treatment.**
- For the CRD, any difference among experimental units receiving the same treatment is considered as experimental error.
- In the design of experiments, **completely randomized designs** are for studying the effects of **one primary factor** without the need to take other nuisance variables into account. The experiment **compares the values of a response variable based on the different levels of that primary factor.**

Randomized Design

- A **randomized block design (RBD)** is an experimental design that organizes experimental units into blocks and randomly assigns treatments to those units.
- A randomized block design is a restricted randomized design, in which experimental units are first organized into homogeneous blocks and then the treatments are assigned at random to these units within these blocks.
- The main advantage of this design is, if done properly, it provides more precise results.



Factorial Design

- A factorial design is a research method that allows researchers to examine multiple factors simultaneously.
- It's a common experimental design that combines multiple studies into one.
- A factorial design allows the effect of several factors and even interactions between them to be determined with the same number of trials as are necessary to determine any one of the effects by itself with the same degree of accuracy.

2x2 Factorial Design

- The 2 x 2 factorial design calls for randomizing each participant to treatment A or B to address one question and further assignment at random within each group to treatment C or D to examine a second issue, permitting the simultaneous test of two different hypotheses.



Factorial Design- Example

- A factorial design example is a 2×2 factorial design, which involves testing two factors, each with two levels. This results in four different conditions, representing all possible combinations of the levels of the two factors.
- Factorial designs are the basis for examining several factors simultaneously.
- For example, if k number of variables/factors are studied to determine/screen the important ones, the total number of treatment combinations for a k number of factors can be calculated.
- This screening technique is known as the 2^k design of experiments
- Fractional factorial designs are a proportion of a full factorial design. They allow the size of an experiment to be kept practical while still enabling the estimation of important effects.
- These methods are of particular interest when more than one change is implemented at the same time

2x2 factorial design

Both A and B	A only
B only	Neither A nor B

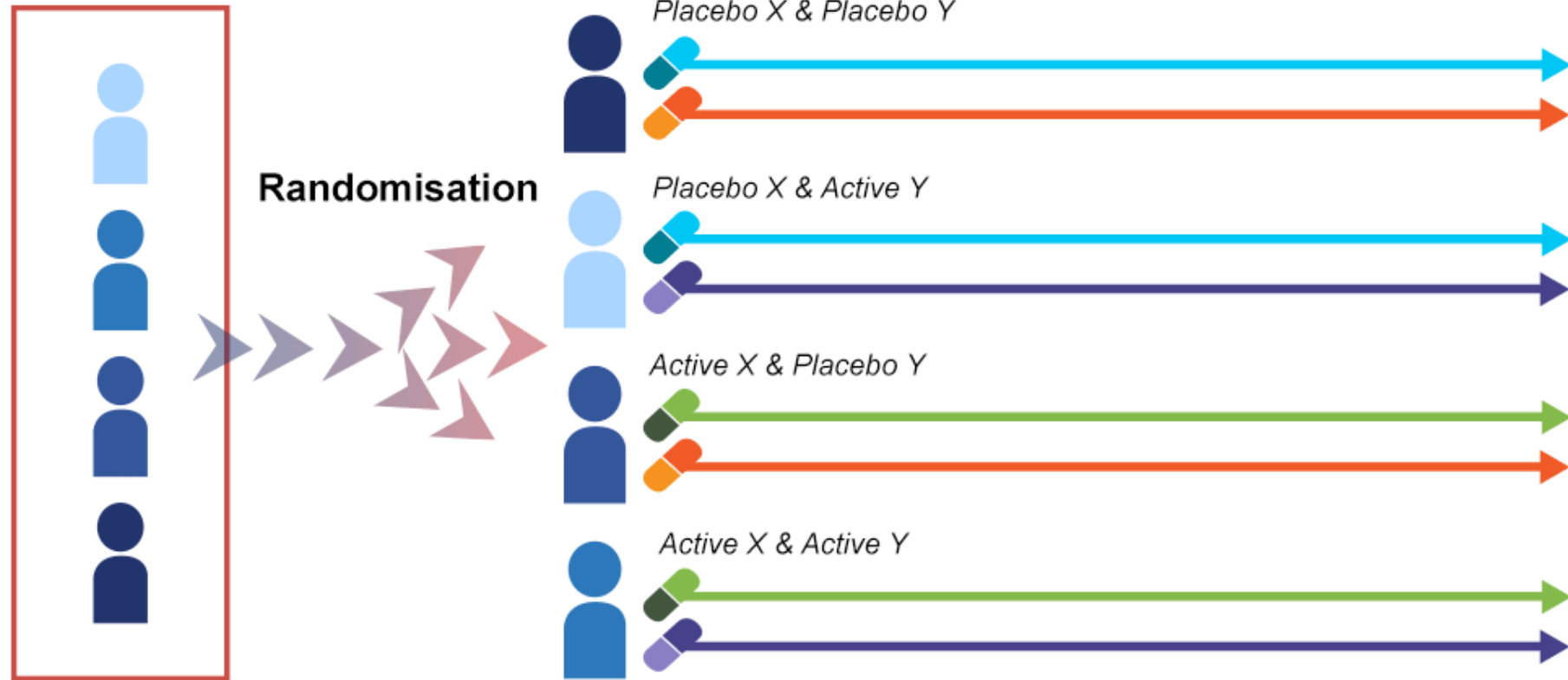
Advantages

- Two drugs studied at same time
- Discover interactions
- Test FDC

Disadvantages

1. Complexity of trial design
2. Complexity of statistical analysis

2x2 Factorial design



Active Treatment X



Placebo X



Active Treatment Y



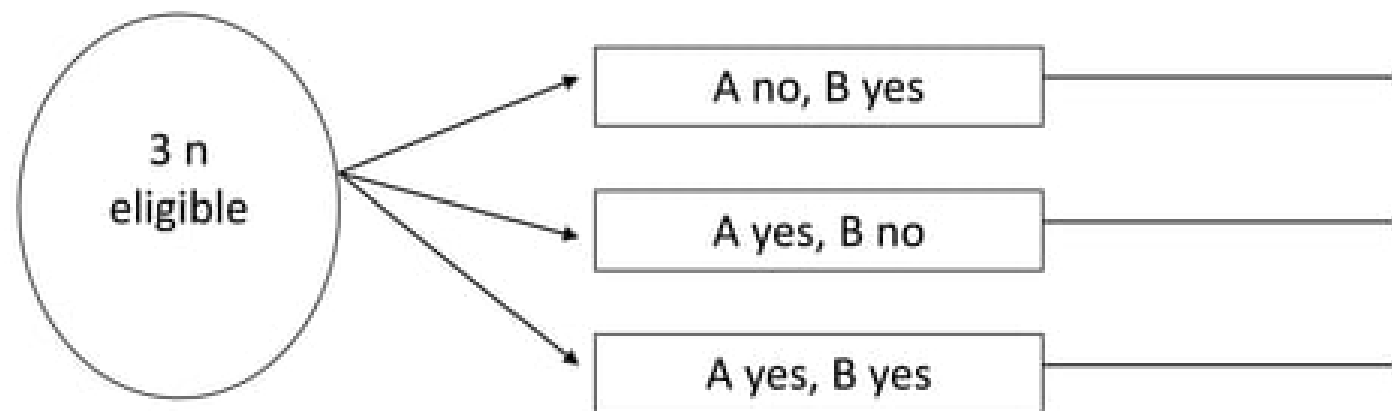
Placebo Y



Incomplete Factorial Design

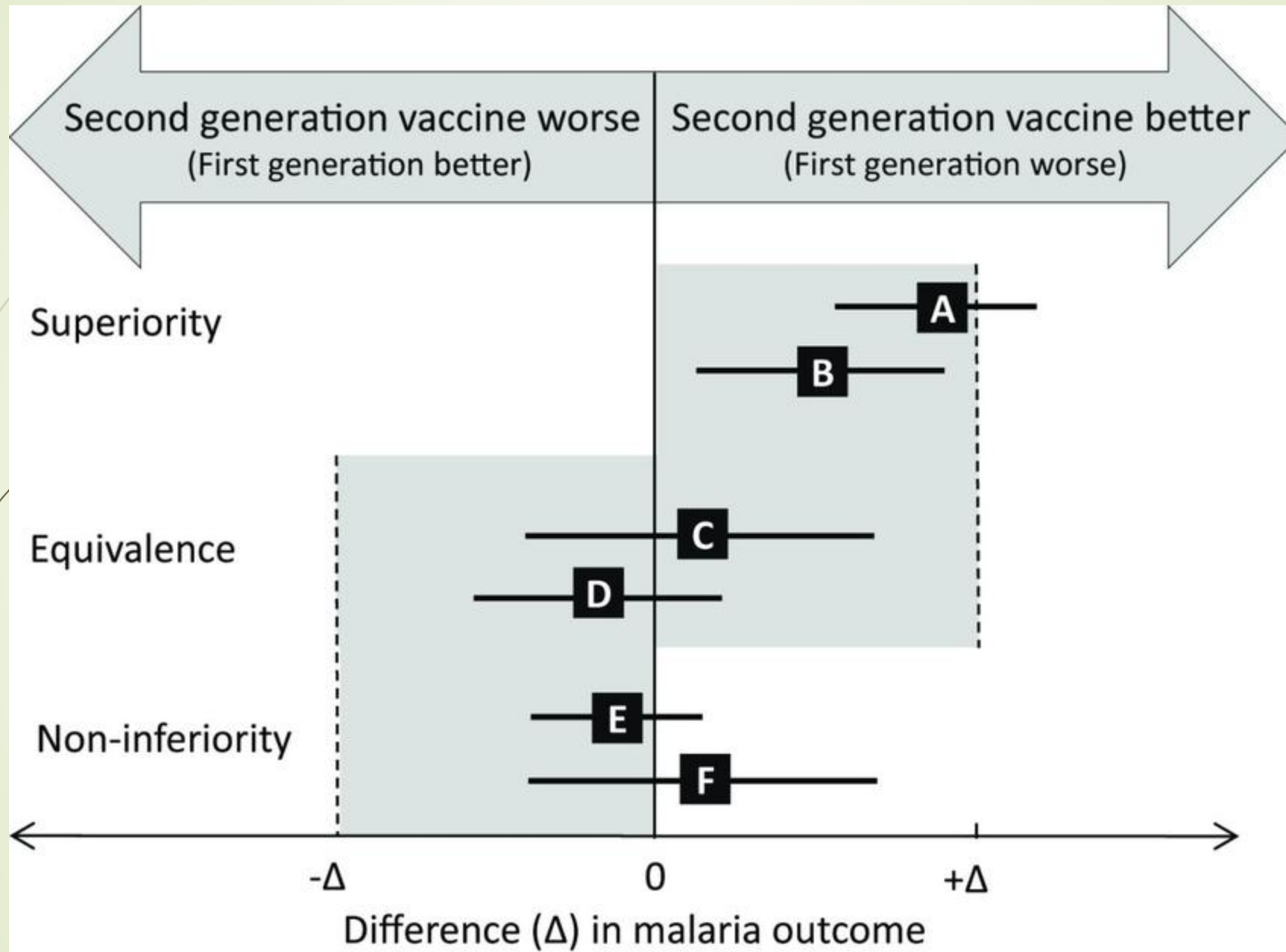
Eg. depression; if unethical to do nothing

- A – Desipramine
- B – Cognitive therapy
- C – Combination of A & B



Superiority Trials, Non-Inferiority

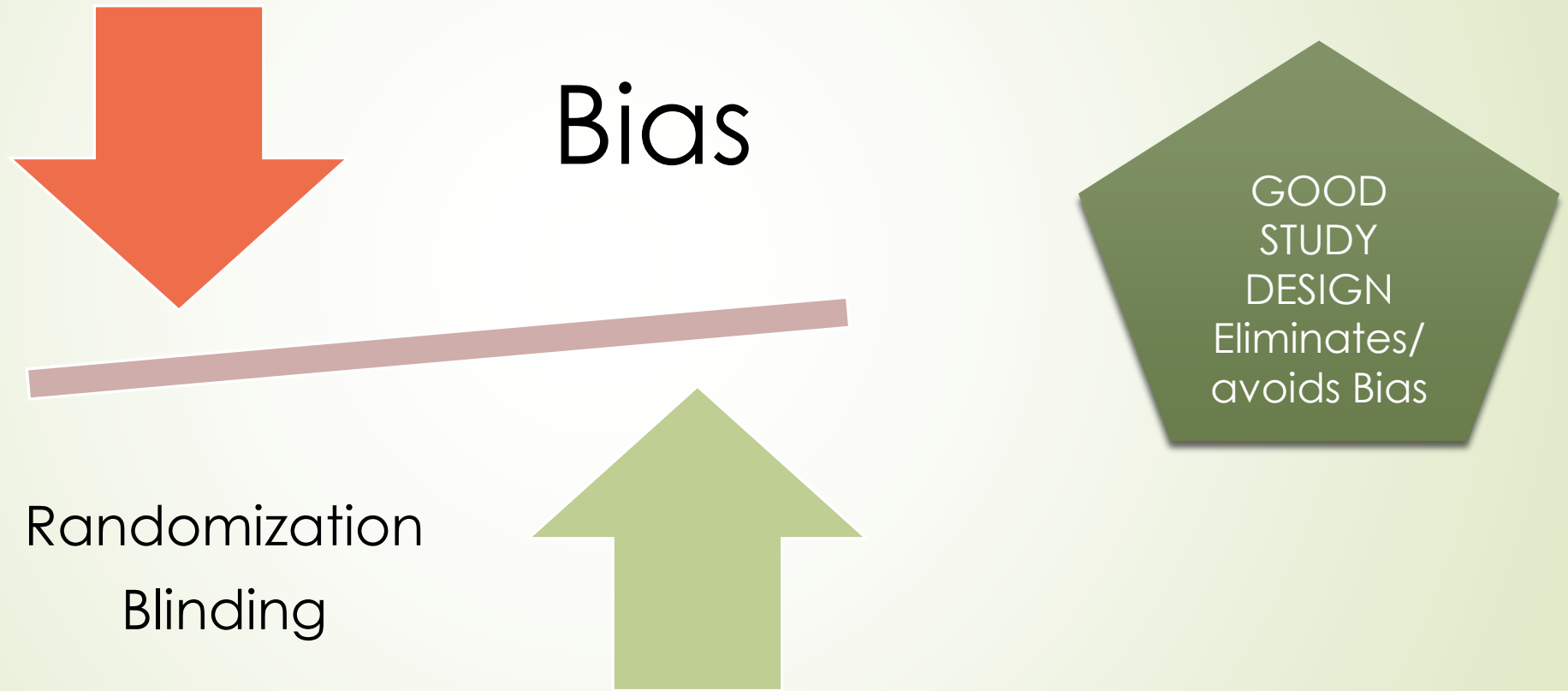
- A superiority trial is a type of clinical trial that compares a new treatment to a placebo or active control. The goal of a superiority trial is to show that one treatment is better than another
 - A non-inferiority design is a one-sided test that determines if a new intervention is no worse than a standard intervention. It's a major tool for evaluating drugs, devices, biologics, and other medical treatments
 - A non-inferiority design is often used when the new treatment may not have additional efficacy, but may have other benefits. For example, the new treatment may be less toxic, more convenient, or reduce the cost of supportive care
 - Non-inferiority trials are conducted by measuring an outcome for two groups of patients, each of which has been given one of the treatments. For example, if the outcome is survival time, and the two treatments are equally effective, then both treatments have a median survival time of 12 months.
 - If the difference between the new drug and the active comparator does not exceed this prespecified margin, non-inferiority can be concluded
- 



Multi-centric Trials

- A multicentric trial, or MCCT, is a clinical trial that takes place at more than one medical institution. It's a collaborative effort that involves more than one independent center in enrolling and following study participants.
- MCCTs are costly and require extensive resources and coordination. The planning phase of a MCCT can take years.
- Strict protocol compliance
- Central monitoring committees
- Phase III
- The goal of a multicentric trial is to:
 - Discover potential beneficial effects of a drug, biologic, or device
 - Determine the safety and efficacy of a drug, biologic, or device
 - Find better ways to prevent, diagnose, or treat cancer

Major Challenges in Design



Bias

- Bias in clinical trials is a systematic error that favors one outcome over another.
- It can occur at any stage of research, from study design to data collection and analysis, and even in the publication process.
- Bias can lead investigators to draw incorrect conclusions about the effects of interventions.
- Some types of bias in clinical trials include:
- **Reporting bias** When the results of a trial affect how and whether they are published. For example, analyses with statistically significant differences between intervention groups are more likely to be reported than non-significant differences.
- **Sampling bias** When the data used to train a machine learning model doesn't represent the whole population being modeled. For example, if a new drug trial only includes participants from a specific region or demographic, the drug may not work the same way in other groups.
- **Allocation bias** When the allocation sequence is not concealed. For example, if patients with the worst side effects from a new drug do not return for follow-up, the results would most likely be biased.

Randomization

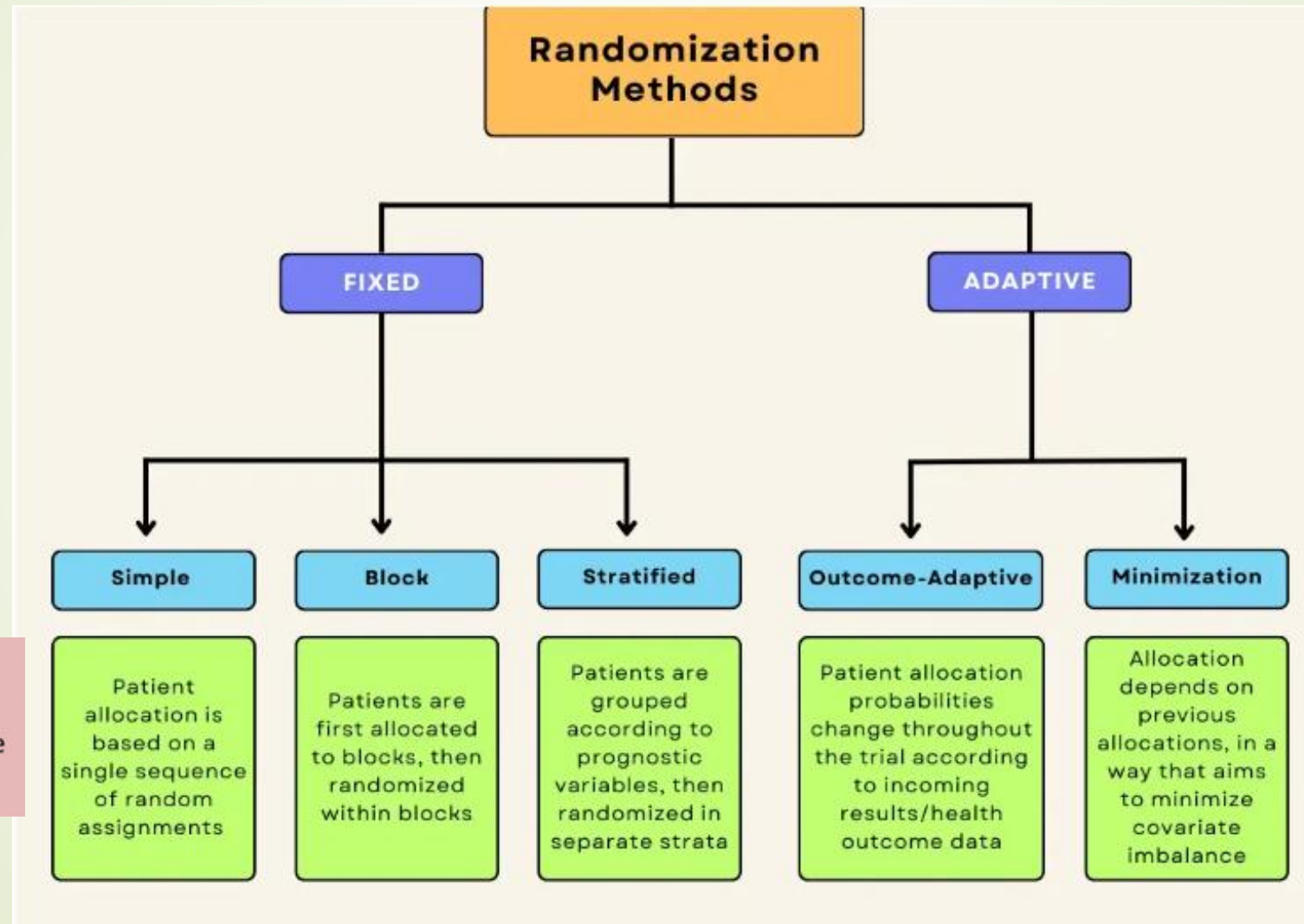
- A randomized clinical trial (RCT) is a study that randomly assigns participants to experimental or control groups.
- The random assignment ensures that neither the doctor nor the patient knows the treatment assignment in advance
- Randomization is a fundamental aspect of scientific research methodology.
- It's the process of assigning participants to treatment or control groups using an element of chance to reduce bias.
- The basic benefits of randomization are that it:
 - **Eliminates selection bias**
 - Balances the groups with respect to many known and unknown confounding or prognostic variables
 - Forms the basis for statistical tests
 - Assumes that each participant has an equal chance of being assigned to any group

Randomization Methods

- Randomization is a statistical process that uses a random mechanism to select a sample from a population or assign subjects to different groups. It is a fundamental aspect of scientific research methodology
- Randomization consists of three steps:
 - Generate a random allocation sequence
 - Conceal the allocation
 - Implement the random allocation sequence



Randomization Methods



Tossing coin
Dice
Random number table
Computer generated

Randomization Methods

- **Stratified randomization**

- A two-stage procedure where patients are grouped into strata according to clinical features that may influence outcome risk. Within each stratum, patients are then assigned to a treatment according to separate randomization schedules.

- **List randomization**

- Also known as the item count or unmatched count technique, this provides a simple way for respondents to report on sensitive behavior without allowing the researcher or surveyors to identify individual responses.

- **Permuted-block randomization or blocked randomization**

- A "block size" and "allocation ratio" (number of subjects in one group versus the other group) are specified, and subjects are allocated randomly within each block.



RCT

- Gold standard
- Minimizes Bias
- Costly, time consuming

Allocation Concealment

- Preventing next assignment in clinical trial from being known
- procedure for protecting randomization process so that treatment to be allocated is not known before the patient is entered into the study

Methods of Allocation concealment

- Sequentially numbered, opaque, sealed envelopes,
- Pharmacy-controlled allocations
- Coded identical containers or kits
- Central randomisation systems

Blinding

- Blinding refers to the concealment of group allocation from one or more individuals involved in a clinical research study, most commonly a randomized controlled trial (RCT).
- Blinding, or masking, is a process in clinical trials that prevents researchers, doctors, and patients from knowing which group a patient is in.
- The purpose of blinding is to protect against bias and isolate the effect of the intervention.
- In randomized controlled trials (RCTs), blinding is most commonly used. Here are some tips for blinding in surgical trials:
 - Blind as many individuals as possible
 - Do not inform patients of what group they are in
 - Conceal incisions and scars
 - Use independent outcome assessors
 - Alter digital radiographs or images
- In an **unblinded**, or “**open**”, **study**, information about the assigned interventions is available to all people and groups involved in the research

Purpose of Blinding

- Blinding is a cornerstone of many clinical trials.
- It's used to prevent bias in the design, execution, conduct, and interpretation of a clinical trial.
- **Observer bias** Most researchers have some expectations regarding the effectiveness of an intervention. Blinding of observers provides a strategy to minimize this form of bias.
- **Selection bias** If participants know whether they are in a control or treatment group, they may adjust their behavior in ways that affect the outcome that researchers are trying to measure.
- There are different **types of blinding**:
- **Single-blind study**: Only the participants are blinded.
- **Double-blind study**: Both participants and experimenters are blinded.
- **Triple-blind study**: The assignment is hidden not only from participants and experimenters, but also from the researchers analyzing the data.
- **Complete blinded study**: An extension of the double-blinded study that expands the scope of blinding to include all personnel who evaluate the patient or analyze data

Types of blinding

- **Open-label trial** is a clinical study where both the researchers and participants know which treatment is being given to participants.
- Open-label trials are not completely blinded, and they do not have to involve randomization or include a control treatment.
- Some advantages of open-label trials include:
 - Compassionate use studies
 - Patients with serious diseases that have few effective treatments may want access to experimental interventions.
 - Open-label extension studies
 - Subjects can remain on the study intervention for an extended period of time.
- Open-label trials lack the rigor of blinded studies, so they should only be used when blinding is not feasible or ethical.

Types of blinding

- **A Prospective Randomized Open Blinded End-point (PROBE)** study is a design for intervention trials.
- It's a cost-effective, alternative to double-blind, prospective study designs
- In a PROBE study, patients are allocated to different treatment regimens using a strict randomization procedure. The end-points are evaluated by a blinded end-point committee.
- Benefits of PROBE studies include:
 - Lower cost
 - Greater similarity to standard clinical practice
 - Results are more easily applicable in routine medical care
 - According to a meta-analysis, PROBE trials yield the same results as double-blind, placebo-controlled trials

Phase I (Human Pharmacology)

Aims:

- To find safe dose range
- Pharmacokinetics of drug
- Drug food interaction
- First in man study

Sample:

- Healthy volunteers
- Drug -too toxic (cancer, HIV): patients

Sample size- 20-50

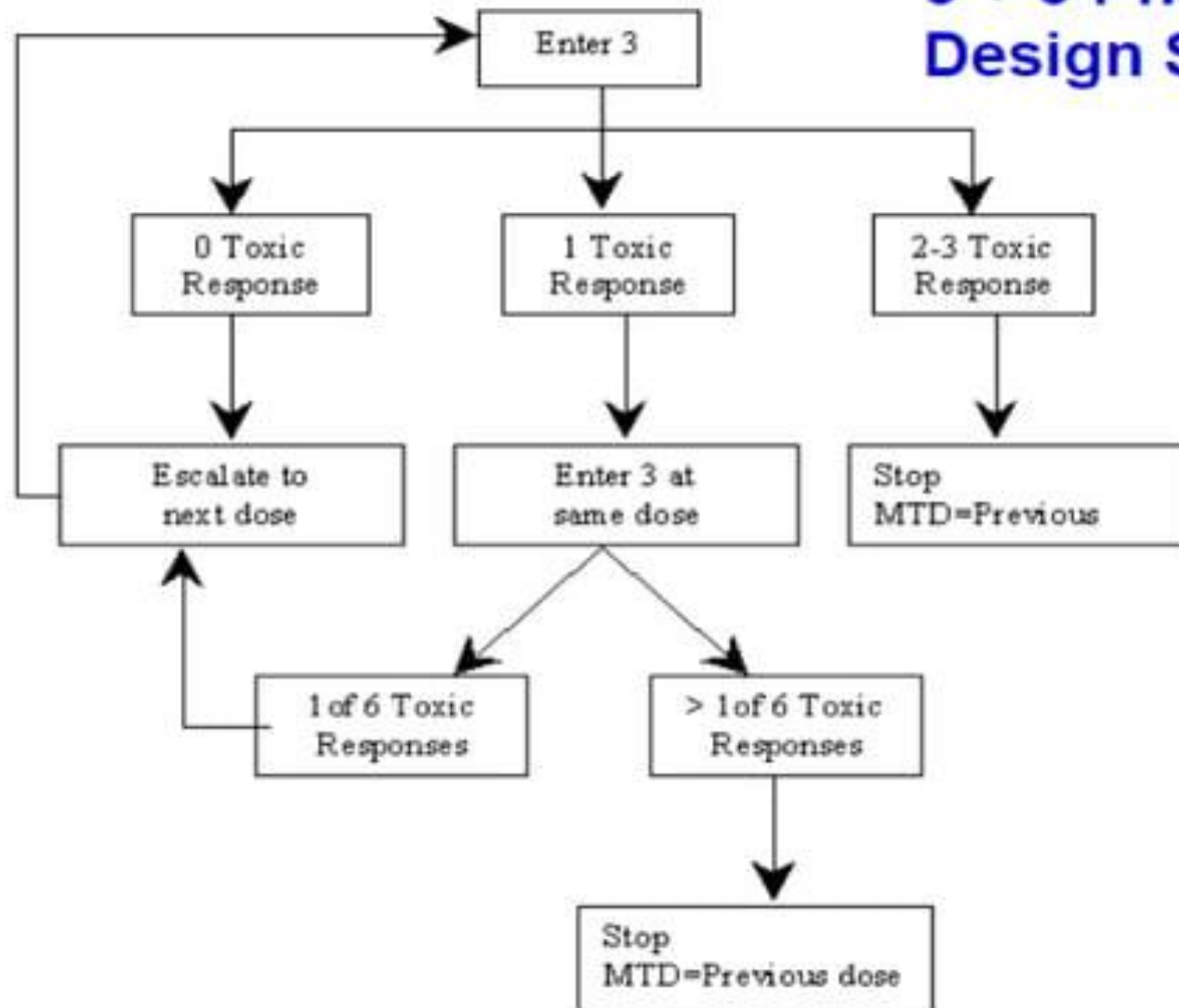
Phase I contd

Design

- Open label
- Non-randomized
- Dose escalation
- Uncontrolled

Randomized 2 way crossover study of one dose level of drug under fasting and fed conditions

3 + 3 Phase 1 Study Design Schematic



Phase II (Therapeutic exploratory)



II a- Proof of concept

Sample – Patients

Sample size- 40-100 subjects

Placebo control preferred

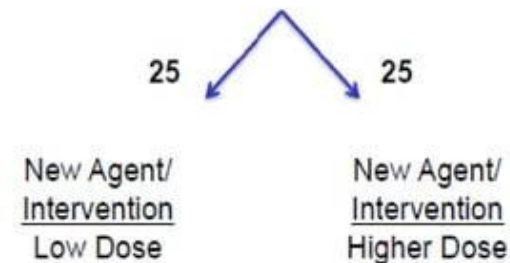
not- multi centered

Phase II b- Dose range finding

- To find optimal dose response range
- 300-400 patients
- Placebo/ active control
- Multicentric

Low dose vs. higher dose

Randomization (n=50)



Placebo (Inactive) vs agent

Randomization (n=50)



Phase III

Confirmatory trials- confirm drug is safe and effective

- Sample- 1000-3000 patients
- Active controlled
- Randomized
- Double blinded
- Parallel
- Non-inferiority
- Multicentric

Phase IV

Post- marketing surveillance

- Detect long term ,rare side effects
- Pharmacoeconomics
- New indication

New drug status-
4 years

Uncontrolled
Observational

Bioequivalence studies

- For generic drug submission
- ANDA- Abbreviated new drug application
- RLD- Reference listed drug
- Parallel-Group Design
 - Even number of subjects in two groups
 - Each receive a different formulation

PHARMACEUTICAL INDUSTRY- GLOBAL & INDIAN PERSPECTIVE

Ms. Sonali Correa

Ph.D., M.Sc.



INTRODUCTION

The pharmaceuticals market covers remedies used to cure, treat, prevent, and diagnose diseases.

These small molecule drugs or biologic medical products may be called drugs, medications, or medicines.

Prescription drugs and over-the-counter (OTC) drugs are both included.

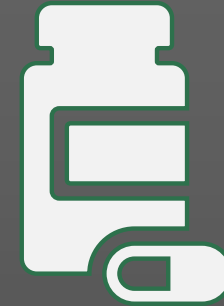
Depending on the regulations in each country, these products are provided by pharmacies, hospitals, physicians, and/or drug stores, and/or they can be bought elsewhere.

They may be administered in the form of pills, capsules, gels, ointments, injections, infusions, lozenges, or aerosols/sprays.

The Pharmaceuticals market is organized in segments that cover drugs used in specific medical fields, i.e., medicines for a specific disease or a certain type of medicine.

They include original brands, biosimilars, generics, and orphan drugs.

PHARMACEUTICAL INDUSTRY

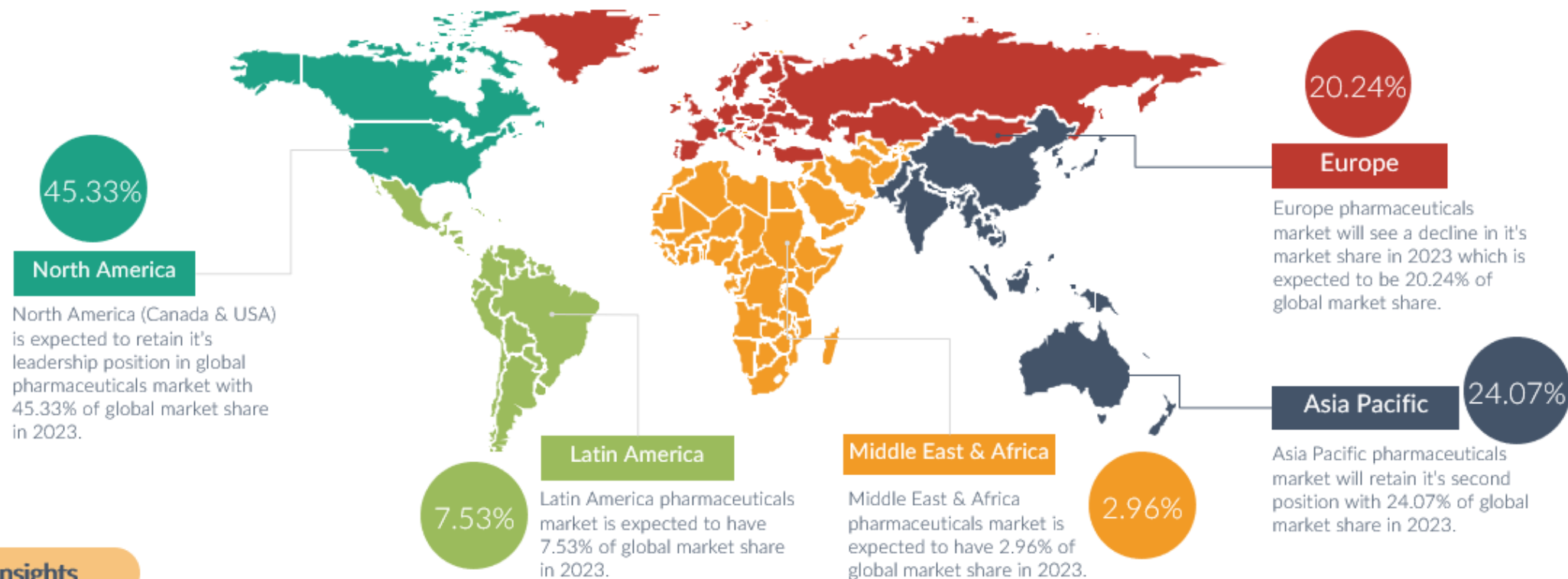


The pharmaceuticals market covers remedies used to cure, treat, prevent, and diagnose diseases.



Global Pharmaceuticals Market 2023

Geography-wise Market Share of Global Pharmaceuticals Market in 2023

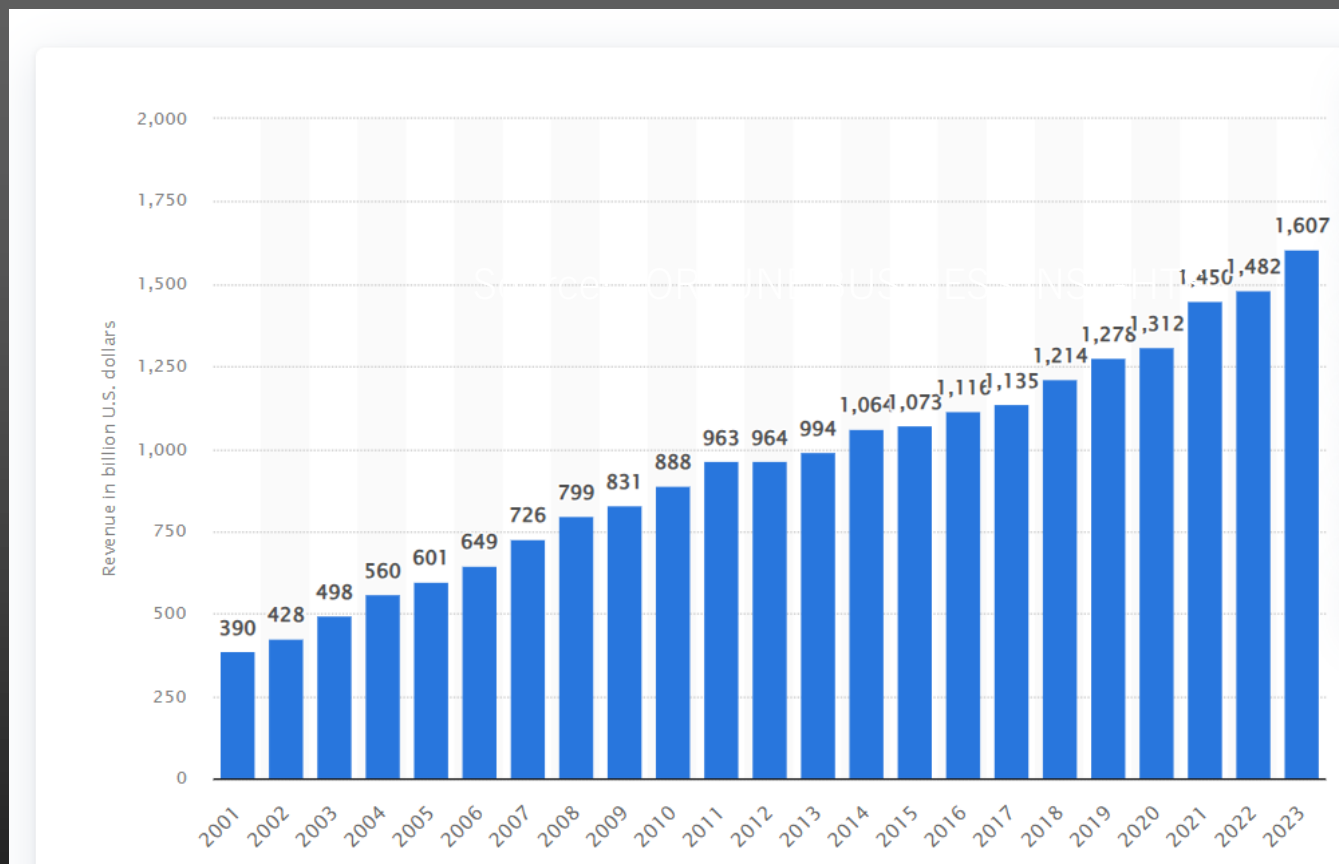


Insights

North America is expected to its leading position in Global Pharmaceutical market while Europe is likely to see a decline in 2023.

Source: NAVADHI Intelligence Rediscovered
Deloitte Data

GLOBAL MARKET SIZE



Year	Value (USD Trillion)
2022	1.4
2023	1.6
2024 Q3	2.3 est.

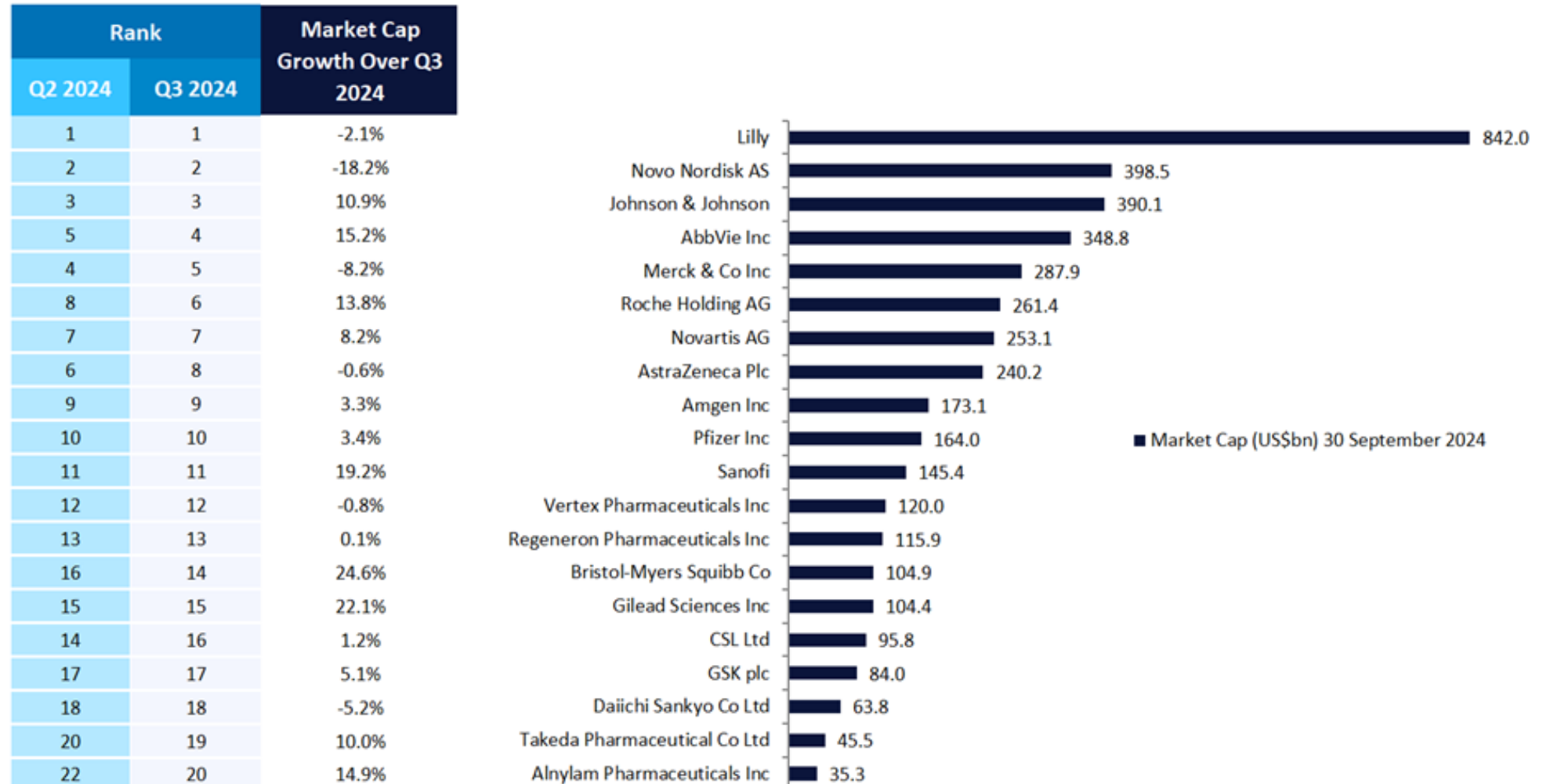
- The CAGR of pharmaceutical is 6.4% during the analysis period of 2023 to 2032.

GLOBAL MARKET VALUE

- The global pharmaceutical market value in the third quarter of 2024 was \$2.3 trillion, a 2% increase from the previous quarter.
- Global Pharmaceutical Market revenue is expected to increase by USD 2.8 Trillion by 2032, with a 6.4% CAGR from 2023 to 2032
- North America region led with more than 38% of Pharmaceutical Market share in 2022
- Asia-Pacific Pharmaceutical Market growth will record a CAGR of more than 7.1% from 2023 to 2032
- By therapeutic category, the anti-diabetic segment is anticipated to grow at a CAGR of 6.7% during the forecast period
- By prescription type, the OTC drugs segment is projected to expand at the fastest CAGR over the projected period
- Rising geriatric population and increasing burden of chronic diseases, drives the Pharmaceutical Market value

MAJOR PLAYERS GLOBALLY

Figure 1: Top 20 global biopharmaceutical companies, by market cap, as of September 30, 2024



MAJOR PLAYERS IN GLOBAL MARKET

- Johnson & Johnson: A global pharmaceutical company with headquarters in New Brunswick, New Jersey. In 2023, they generated around \$85 billion in revenue.
- Novartis: A Swiss pharmaceutical company that operates in many therapeutic areas, including oncology, immunology, neuroscience, and ophthalmology.
- Bristol-Myers Squibb: A New Jersey-based biopharma company that develops therapies for conditions like HIV and hepatitis.
- Pfizer: A major pharmaceutical company.
- Merck & Co: A major pharmaceutical company.
- Roche: A major pharmaceutical company.
- Sanofi: A major pharmaceutical company.
- AstraZeneca: A major pharmaceutical company.
- GSK: A pharmaceutical company that specializes in pharmaceuticals, biologics, and vaccines.

FUTURE

- North America held the dominating position in pharmaceutical industry during the analysis period of 2023 to 2032.
- Asia-Pacific region exhibited fastest growing CAGR for market of pharmaceutical during the analysis period of 2023 to 2032.

TRENDS IN PHARMACEUTICAL INDUSTRY

SHAPING THE INDUSTRY IN 2023

- As of 2023 and looking into 2024, several key trends are shaping the global pharmaceutical market. These trends reflect ongoing shifts in healthcare needs, technological advancements, regulatory changes, and market dynamics.
- Increased Investment in Biologics and Biosimilars: There is a growing focus on biologic therapies, including monoclonal antibodies and gene therapies. The market for biosimilars is also expanding as patents for various biologics expire, leading to increased competition and reduced healthcare costs.
- Digital Health and Telemedicine: The adoption of digital health solutions, including telemedicine, wearables, and mobile health applications, is on the rise. This trend is driven by demand for remote healthcare services, which saw a significant boost during the COVID-19 pandemic. Pharmaceutical companies are increasingly incorporating digital tools into their offerings to enhance patient engagement and adherence.

Personalized Medicine: There is an ongoing shift towards personalized or precision medicine, where treatments are tailored to the genetic makeup of individual patients. Advances in genomics and biotechnology are facilitating this move, which is particularly relevant in oncology and rare diseases.

Artificial Intelligence and Data Analytics: Pharma companies are leveraging AI and big data for drug discovery, clinical trials, and market analytics. These technologies can streamline the R&D process, improve clinical trial design, and enhance patient recruitment and retention.

Regulatory Changes and Market Access: Regulatory bodies are increasingly focusing on expedited pathways for drug approvals, especially for innovative treatments addressing unmet medical needs. Additionally, market access strategies are evolving to include real-world evidence and pay-for-performance models.

Sustainability and Ethical Practices: There is growing pressure on pharmaceutical companies to adopt sustainable practices and address environmental concerns. This includes responsible sourcing of materials, reducing carbon footprints, and improving supply chain transparency.

Emergence of Rare Disease Treatments: As understanding of rare diseases improves, there is increased investment in therapies for these conditions. The orphan drug market is expanding, driven by incentives for developers and growing patient advocacy.

Global Supply Chain Resilience: The COVID-19 pandemic highlighted vulnerabilities in pharmaceutical supply chains. Companies are now focusing on building more resilient supply chains, including diversification of suppliers and localizing production to mitigate risks.

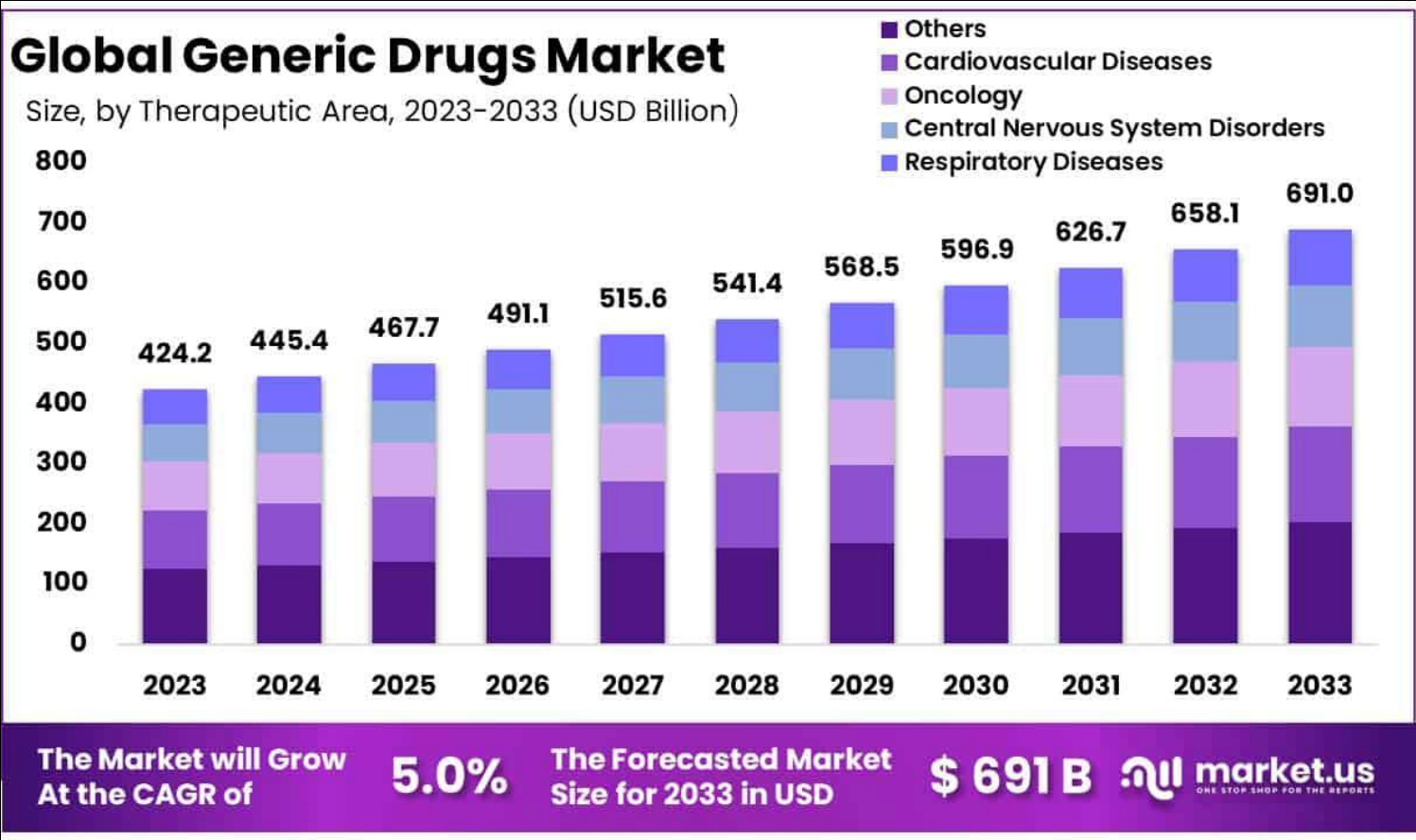
Global Health Challenges and Access: Issues related to global health inequities are prompting pharmaceutical companies to explore innovative pricing strategies and partnerships to improve access to essential medications, particularly in low- and middle-income countries.

Mental Health Treatments: There is an increasing recognition of mental health issues, leading to greater investment in treatments and therapies for conditions like depression, anxiety, and post-traumatic stress disorder (PTSD). This trend includes both traditional pharmaceuticals and innovative approaches like psychedelic-assisted therapies.

According to recent data, the global pharmaceutical market in 2023-2024 is seeing a trend towards slower growth in developed regions like North America and Europe, with emerging markets in Asia, Latin America, and other regions driving the majority of growth, fueled by increased access to medicine and rising healthcare needs in those areas; key focus areas include oncology treatments, personalized medicine, and a continued emphasis on R&D for novel therapies, particularly in areas like CNS disorders and obesity treatments

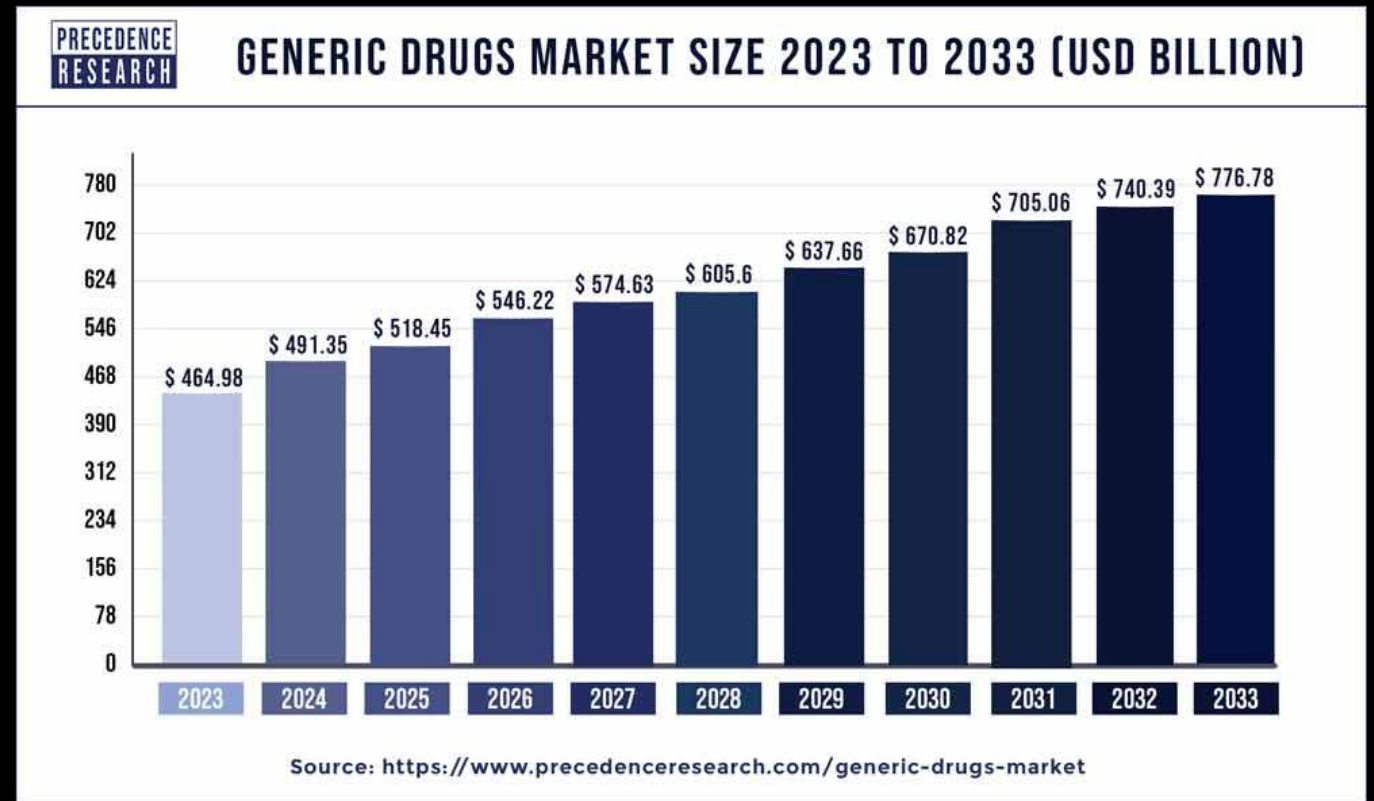
<https://www.zs.com/insights/trends-shaping-pharmaceutical-landscape-2024-and-beyond>

MARKET SHARE



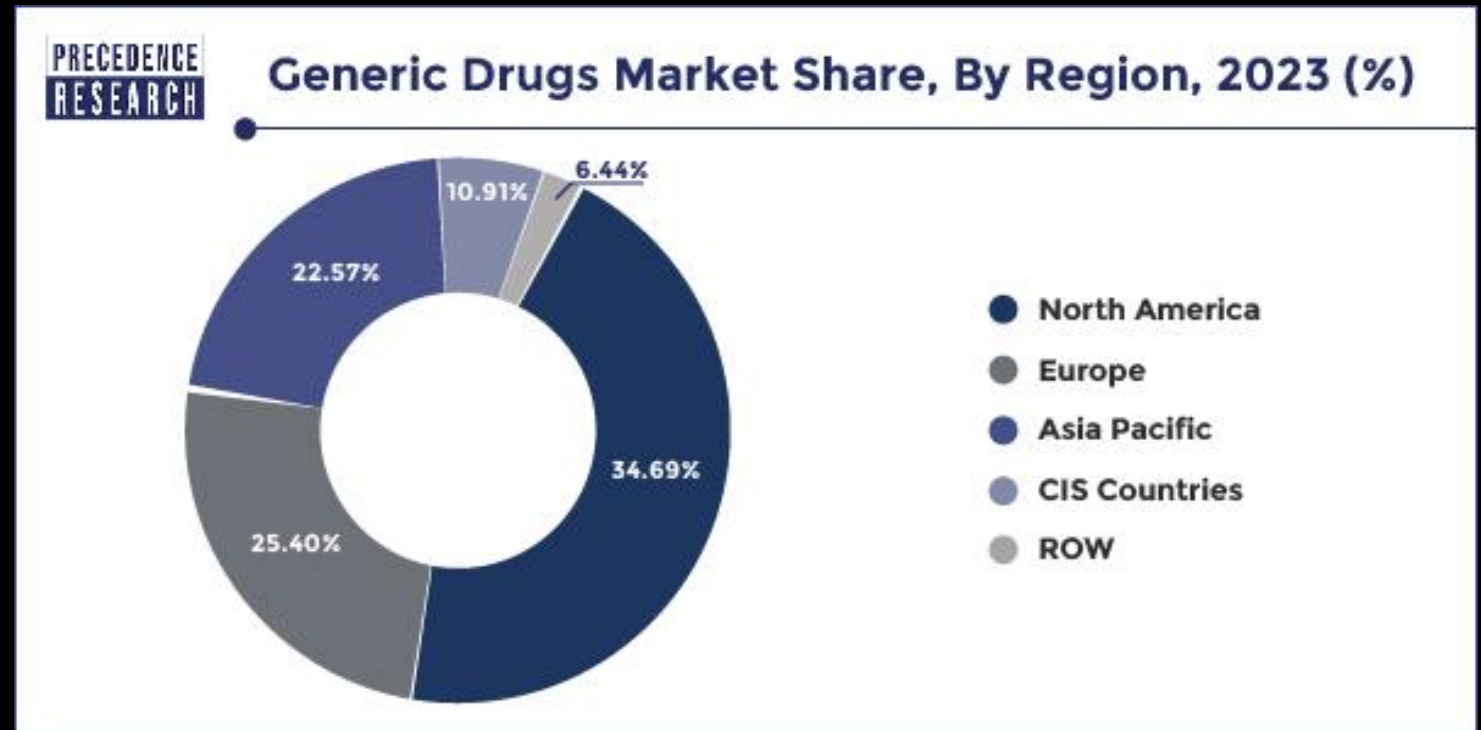
GENERIC DRUGS MARKET SHARE

- The global generic drugs market size was valued at USD 464.98 billion in 2023
- projected to hit around USD 776.78 billion by 2033
- growing at a CAGR of 5.2% over the forecast period from 2024 to 2033
- North America led the global market with the highest market share of 34.69% in 2023.
- By brand, the pure generics segment has held the largest market share of 52.57% in 2023.



- By route of administration, the oral segment captured the biggest revenue share of 65.33% in 2023.
- By drug type, the simple generic segment is estimated to hold the highest market share of 60.74% in 2023.
- By therapeutic application, the oncology segment is expected to grow at a remarkable CAGR of 6.6% during the forecast period.

- The growth is attributed to speedy increase in the prevalence of chronic disease in the countries of APAC owing to changing lifestyle which is creating demand for the generic drugs.
- Additionally, growing demand for the generic drugs for the end users in LATAM and the countries of Middle East and Africa is anticipated to propel industry growth in the next 10 years.



- North America contributed revenue of around USD 161.29 billion in 2023 and registered a CAGR of 3.7% from 2024 to 2033.
- The Asia Pacific was valued at USD 104.96 billion in 2023 and growing at a CAGR of 8% over the forecast period 2024 to 2033.
- Europe generated revenue of USD 118.09 billion in 2023 and registered a CAGR of 4.5% between 2024 and 2033.

INDIAN PHARMACEUTICAL MARKET

India is the largest supplier of generic drugs globally, responsible for approximately 20% of global generic drug exports, particularly in the United States and Europe.

India is ranked among the top 10 pharmaceutical markets globally, with the industry valued at over \$42 billion in 2021 and expected to reach around \$65 billion by 2024, representing significant growth.

India's pharmaceutical exports have been substantial, accounting for a significant portion of global pharmaceutical exports. The country exports to over 200 countries, including highly regulated markets like the U.S., the European Union, and other developed regions.

India has the most number of pharmaceutical manufacturing facilities that are in compliance with the US Food and Drug Administration (USFDA) and has 500 API producers that make for around 8% of the worldwide API market.

Indian pharmaceutical sector supplies over 50% of global demand for various vaccines, 40% of generic demand in the US and 25% of all medicine in the UK

The domestic pharmaceutical industry includes a network of 3,000 drug companies and ~10,500 manufacturing units.

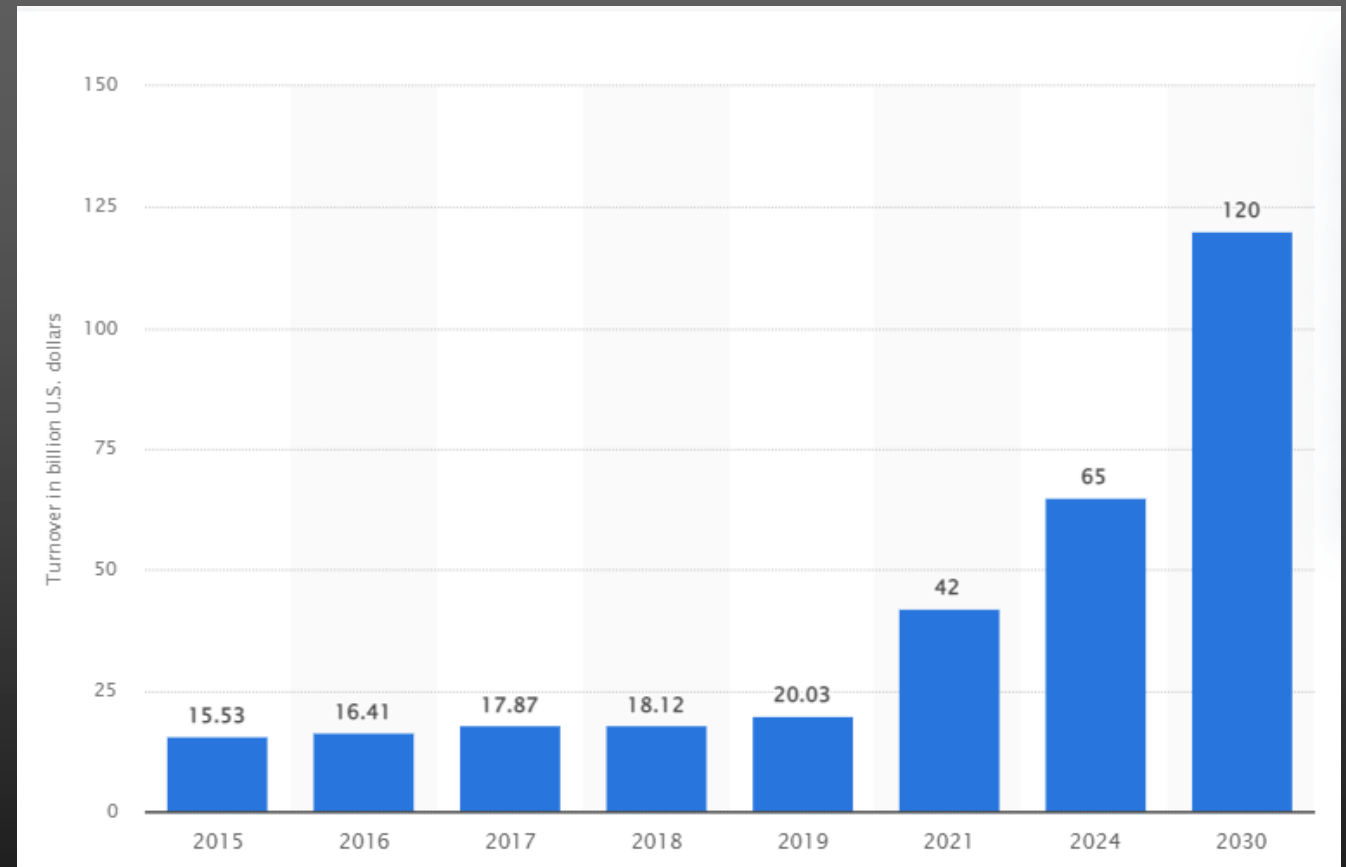
The Indian pharmaceutical industry is governed by the Central Drugs Standard Control Organization (CDSCO), which ensures the quality and safety of pharmaceuticals. Regulatory reforms over the years have increased compliance with international standards and enhanced the global competitiveness of Indian manufacturers.

Presently, over 80% of the antiretroviral drugs used globally to combat AIDS (Acquired Immune Deficiency Syndrome) are supplied by Indian pharmaceutical firms.

India is rightfully known as the "**pharmacy of the world**" due to the low cost and high quality of its medicines.

INDIAN PERSPECTIVE

- According to the Indian Economic Survey 2021, the domestic market is expected to grow 3x in the next decade.
- India's domestic pharmaceutical industry's annual turnover reached over **42 billion U.S. dollars in 2021**.
- In 2018, the India pharmaceutical market was valued at **18 billion U.S. dollars**.
- There was a 9.8 percent year-on-year growth rate recorded in the market in 2019 compared to 2018
- is likely to reach **US\$ 65 billion by 2024** and further expand to reach **US\$ 120-130 billion by 2030**.



- In the global pharmaceuticals sector, India is a significant and rising player. India is the world's largest supplier of generic medications, accounting for 20% of the worldwide supply by volume and supplying about 60% of the global vaccination demand. The Indian pharmaceutical sector is worth US\$ 42 billion worldwide

INDIAN PHARMACEUTICAL MARKET QUARTERLY INSIGHTS – Q2 2024

- The Pharma Industry registered 9% growth in Q2 2024. Some of the key trends of Pharma Industry for Q2 2024 are as below –
- Chronic therapies (11%) reflected better growth than Acute therapies (8%).
- Cardiac, Neuro/CNS and Respiratory led in terms of growth amongst Chronic therapies. Gastro and Derma led in terms of growth amongst Acute therapies.
- Indian companies (9%) grew somewhat faster than MNCs (8%).

TRENDS IN THE IPM

- **Increased Generic Drug Production:** India is a leading producer of generic drugs, and this trend continues to grow. Pharmaceutical companies are focusing on developing affordable generic medications, especially as patent expirations open up opportunities in various therapeutic areas.
- **Rise of Biologics and Biosimilars:** The market is witnessing a shift towards biologics and biosimilars. Companies are investing in research and development to produce biosimilars, particularly in immunology and oncology, as these areas offer significant growth potential.
- **Digital Transformation:** The integration of digital technologies in pharmaceutical operations, including telemedicine, e-pharmacies, and digital marketing, is on the rise. Companies are investing in digital infrastructure to enhance efficiency, improve patient engagement, and streamline supply chains.

TRENDS IN THE IPM

- Focus on Research and Development (R&D): There is an increasing emphasis on R&D to innovate and develop new drugs, particularly in areas lacking effective treatment options. This is driven by both global competition and the need to address local healthcare challenges.
- Regulatory Changes: The Indian government is implementing various reforms to improve the ease of doing business in the pharmaceutical sector. This includes streamlining drug approval processes and enhancing regulatory compliance, which is expected to boost foreign investment and encourage domestic innovation.
- Public Health Initiatives: The government's focus on improving public health infrastructure and access to medicines continues, which is likely to boost the pharmaceutical market. Initiatives related to vaccines, antiretrovirals, and treatments for non-communicable diseases are particularly emphasized.

TRENDS IN THE IPM

- **Increased Exports:** India remains a significant player in global pharmaceutical exports, with a strong push toward expanding its presence in regulated markets like the US and Europe. Companies are complying with international standards to cater to these markets.
- **Sustainability and Green Chemistry:** There is a growing awareness and commitment to sustainable practices within the pharmaceutical industry. Companies are working on reducing waste and energy consumption, and developing greener processes and packaging.
- **Personalized Medicine:** Advances in genomics and biotechnology are paving the way for personalized medicine. The focus on tailored therapies for individual patients is becoming more prevalent, with companies investing in research to support this trend.
- **Collaboration and Partnerships:** There is an increase in strategic alliances and collaborations between domestic companies and multinational corporations to enhance innovation, expand product portfolios, and penetrate new markets.

ADVANTAGE INDIA

COST EFFICIENCY

- * Low cost of production and R&D boosts efficiency of Indian pharma companies, leading to competitive exports.
- * Indian drug & pharmaceutical exports stood at US\$ 24.60 billion in FY22 and US\$ 24.44 billion in FY21.



ECONOMIC DRIVERS

- * High economic growth along with increasing penetration of health insurance to push expenditure on healthcare and medicine in India.



POLICY SUPPORT

- * In June 2021, Finance Minister Ms. Nirmala Sitharaman announced an additional outlay of Rs. 197,000 crore (US\$ 26,578.3 million) that will be utilised over five years for the pharmaceutical PLI scheme in 13 key sectors such as active pharmaceutical ingredients, drug intermediaries and key starting materials.



INCREASING INVESTMENT

- * The FDI inflows in the Indian drugs and pharmaceuticals sector reached US\$ 19.90 billion between April 2000-June 2022.



EXPORTS

India is the 12th largest exporter of medical goods in the world.

Indian drugs are exported to more than 200 countries in the world, with US being the key market.

Generic drugs account for 20% of the global export in terms of volume, making the country the largest provider of generic medicines globally.

Indian drug & pharmaceutical exports stood at US\$ 24.60 billion in FY22 and US\$ 24.44 billion in FY21.

Indian drug & pharmaceutical exports stood at US\$ 2,196.32 million in September 2022.

GOVERNMENT INITIATIVES

- **Pharmaceuticals Vision 2020:** This initiative aimed to make India a global leader in the production of affordable and quality medicines. It includes strategies for both domestic and export markets.
- **Production-Linked Incentive (PLI) Scheme:** This scheme was launched to boost domestic manufacturing in key sectors, including pharmaceuticals. The government offered financial incentives to firms to increase their scale of production and develop high-value products.
- **National Health Mission (NHM):** The NHM emphasizes access to affordable healthcare services and includes programs that promote the availability of essential medicines.
- **Drug Development Assistance:** The government provided support for drug development through several schemes, including grants for research and development in new drug development, biotechnology, and traditional medicine systems.
- **Strengthening Regulatory Framework:** Efforts to enhance the regulatory environment for drugs and pharmaceuticals have been ongoing, with the aim of ensuring faster approvals for new drugs and improving the quality of drugs available in the market.

Support for Startups: Various initiatives were announced to support biotech and pharmaceutical startups, including funding, mentorship, and infrastructure support.

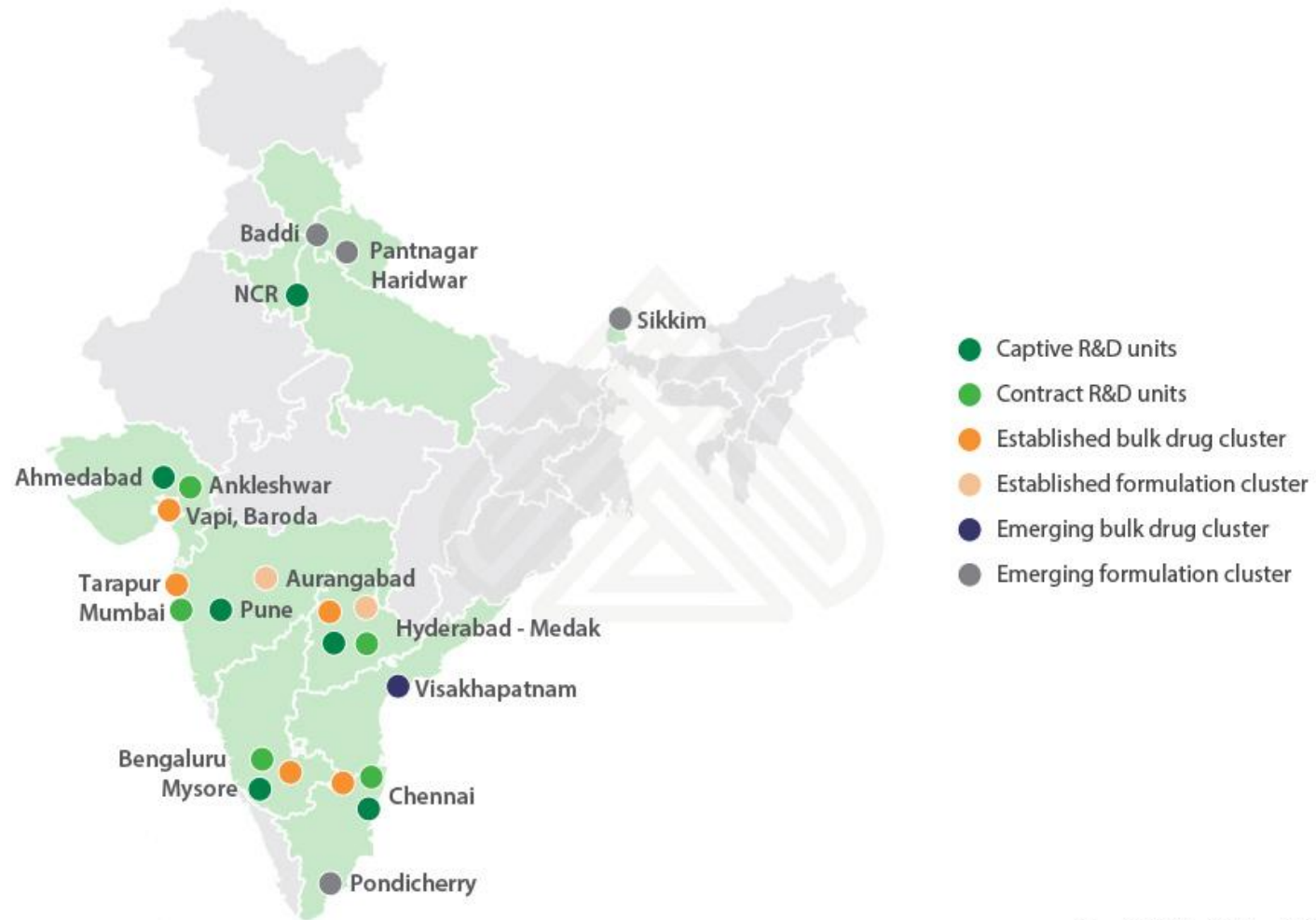
International Collaboration: The Indian government has been engaging in partnerships with other countries and international organizations to enhance technology transfer, research collaboration, and access to global markets.

Focus on Generics and Biosimilars: India has positioned itself as a leading producer of generic medicines and biosimilars, with government support aimed at increasing production and global outreach.

MAJOR PHARMAS IN INDIA

- Maharashtra
- Maharashtra has the highest number of pharmaceutical clusters in India, with 40.
- Gujarat
- Gujarat has 13 pharmaceutical clusters.
- Andhra Pradesh
- Andhra Pradesh has 8 pharmaceutical clusters.
- Uttar Pradesh
- Uttar Pradesh has 6 pharmaceutical clusters.
- Karnataka
- Karnataka has 5 pharmaceutical clusters.
- Tamil Nadu
- Tamil Nadu has 5 pharmaceutical clusters.
- Goa
- Goa has 5 pharmaceutical clusters.

Pharmaceutical Clusters in India



Graphic © Asia Briefing Ltd.

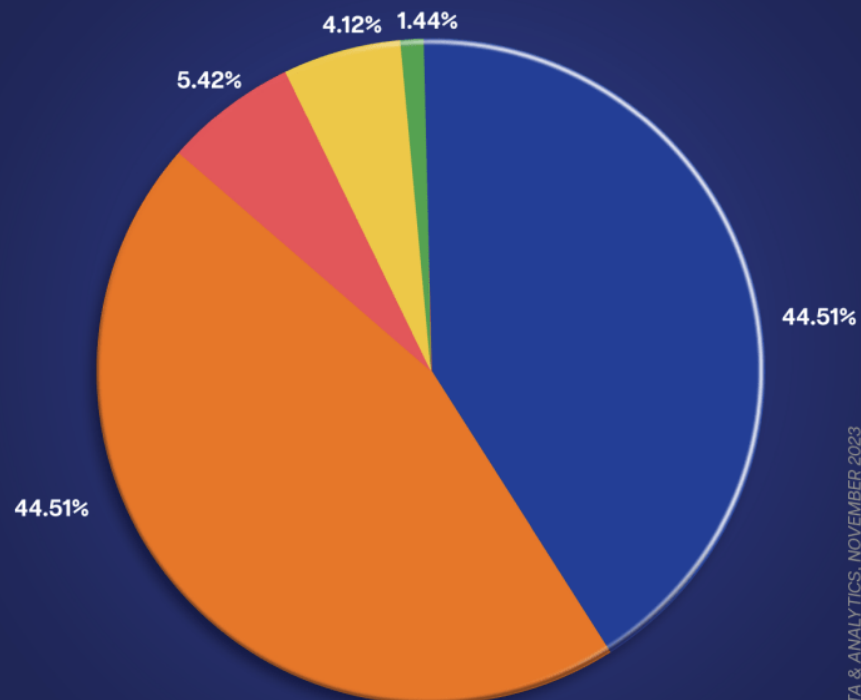
States with large number of Pharma companies



Total : 22000+ Pharmaceutical Companies

PROBE DATA & ANALYTICS, NOVEMBER 2023

Pharmaceutical Industry Segment Distribution

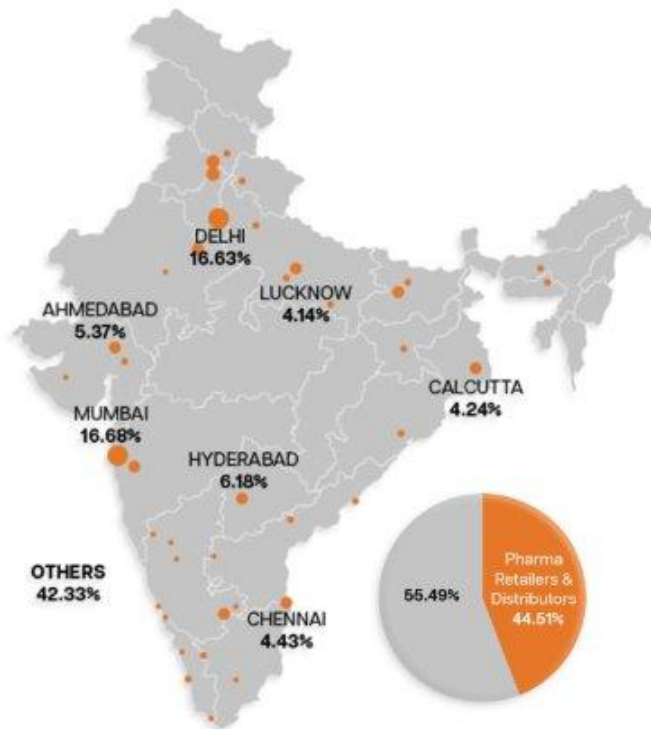


5 Main Pharma Segments

- Animal Pharma
- API Manufacturers
- Contract Research
- Pharma Retailers and Distributors
- Formulations

PROBE DATA & ANALYTICS, NOVEMBER 2023

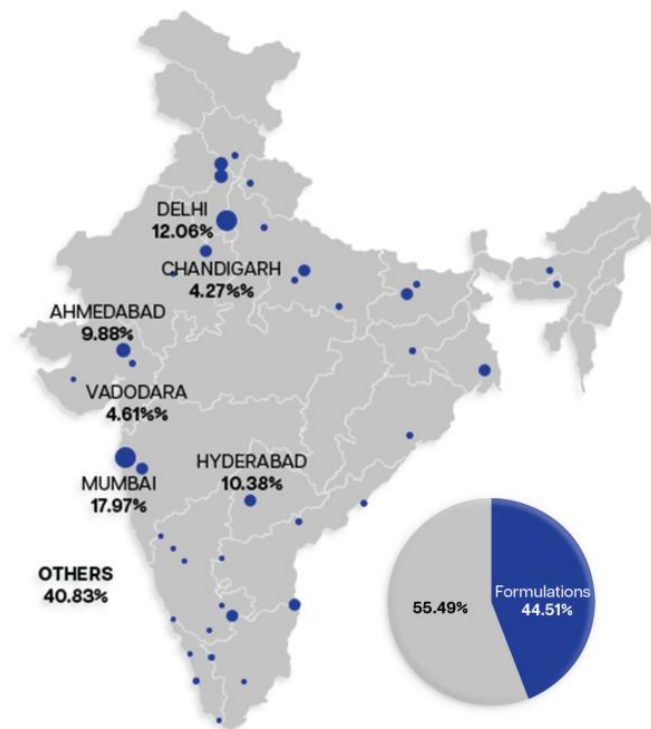
Concentration of Pharma Retailers and Distributors



PROBE DATA & ANALYTICS, NOVEMBER 2023

Pharmaceutical retailers are businesses that sell directly to consumers, while **Pharmaceutical distributors** connect manufacturers and retailers.

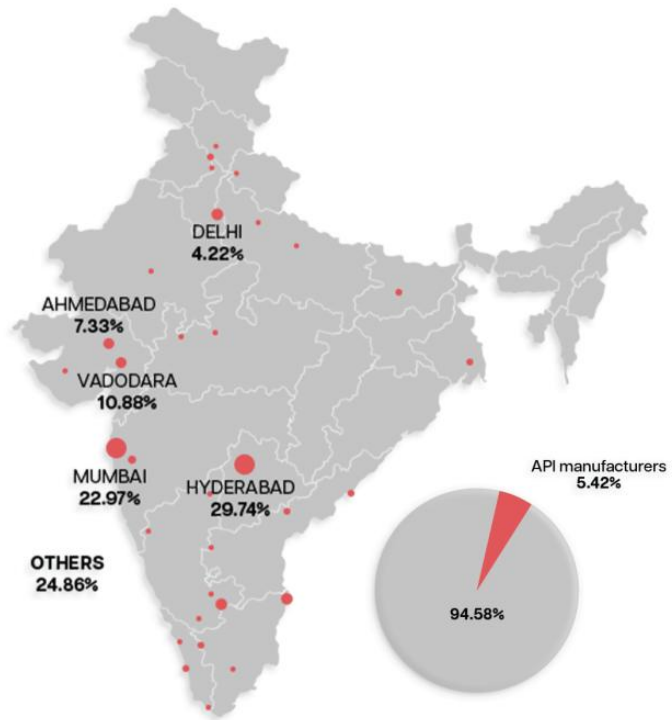
Concentration of Formulation Companies



PROBE DATA & ANALYTICS, NOVEMBER 2023

Pharmaceutical formulation companies are responsible for the development and manufacture of safe and effective drug formulations, suitable for administration.

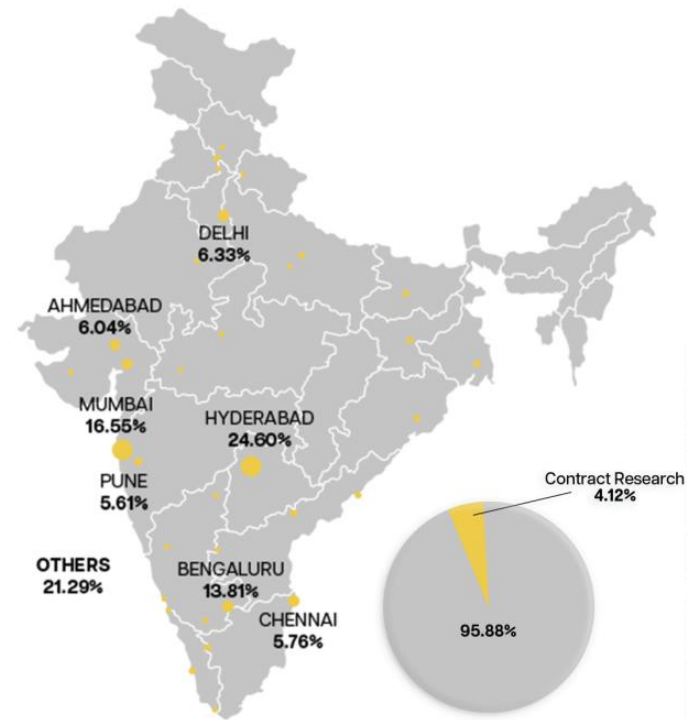
Concentration of API Manufacturers



PROBE DATA & ANALYTICS, NOVEMBER 2023

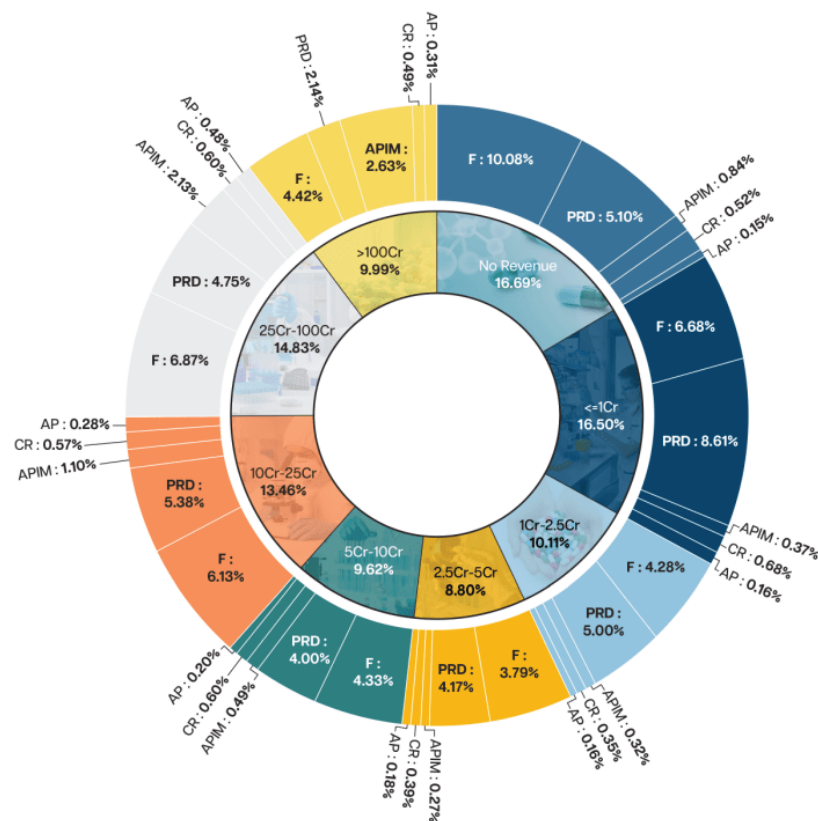
API manufacturers (Active Pharmaceuticals Ingredients) are companies that produce the active chemical components of pharmaceutical drugs, which are the key components of a drug's therapeutic effects.

Concentration of CROs



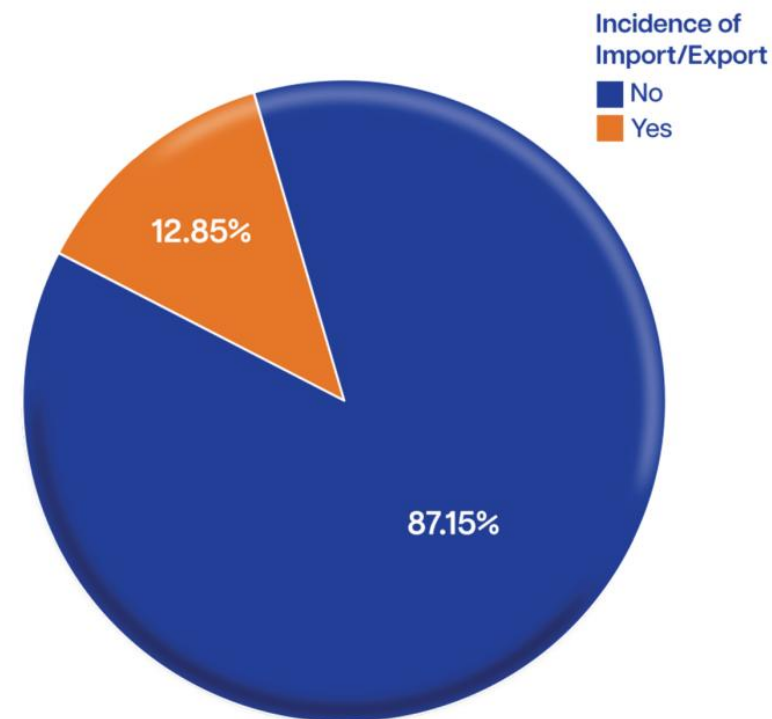
PROBE DATA & ANALYTICS, NOVEMBER 2023

Contract research organizations (CROs) offer crucial research and development services to pharmaceutical, biotech, and healthcare industries on a contractual basis, encompassing drug development, clinical trials, and research.



F : Formulations, PRD : Pharma Retailers and Distributors,
APIM : API Manufacturers, CR : Contract Research, AP : Animal Pharma

Pharma Sector Breakdown across Revenue Ranges



Incidence of Foreign Currency Income/Expenses in India's Pharmaceutical Sector

LARGEST COMPANIES IN INDIA

Rank	Company Name	Market Cap (Rs)	Headquarter
#1	Sun Pharma Limited	4.36 lakh crore	Mumbai, Maharashtra
#2	Divi's Laboratories Limited	1.54 lakh crore	Hyderabad, Telangana
#3	Cipla Limited	1.21 lakh crore	Mumbai, Maharashtra
#4	Torrent Pharmaceuticals Limited	1.17 lakh croe	Ahmedabad, Gujarat
#5	Mankind Pharma Limited	1.15 lakh crore	Delhi
#6	Dr. Reddy's Laboratories Limited	1.10 lakh crore	Hyderabad, Telangana
#7	Zydus Lifesciences Limited	0.99 lakh crore	Ahmedabad, Gujarat
#8	Lupin Limited	0.98 lakh crore	Mumbai, Maharashtra
#9	Aurobindo Pharma Limited	0.73 lakh crore	Hyderabad, Telangana
#10	Alkem Lab Limited	0.65 lakh crore	Mumbai, Maharashtra

LARGEST COMPANIES IN INDIA

By sales and marketing operations within India | 2022

Multinational Pharmaceutical Companies ranked as per active presence of sales, marketing and business in India are as follows:

- | | |
|------------------------|----------------------------------|
| 1. Pfizer | 1. Bristol-Myers Squibb |
| 2. GlaxoSmithKline | 2. Eli Lilly and Company |
| 3. Sanofi Aventis | 3. Abbott |
| 4. Merck | 4. Takeda Pharmaceutical Company |
| 5. Johnson and Johnson | 5. Boehringer Ingelheim |
| 6. Amgen | 6. Astellas |
| 7. Novartis | |
| 8. Roche | |

FOREIGN INVESTMENT

- Per India's Consolidated FDI Policy, 2020 (the “FDI Policy”), foreign direct investment (“FDI”) in the pharmaceutical sector in greenfield (new) projects is permitted up to 100% without the approval of the Department of Pharmaceuticals (the “DoP”). In brownfield (existing) projects, FDI exceeding 74% requires the investor to seek prior approval from the DoP in compliance with the prescribed conditions under the FDI Policy.
- Separately, FDI up to 100% is permitted for the manufacturing of medical devices for both greenfield and brownfield projects without the approval of the DoP.
- foreign direct investment (FDI) in India's pharmaceutical sector has seen significant inflows, with the industry attracting billions of dollars in investments, particularly from companies looking to capitalize on India's strong generic drug manufacturing capabilities; some major players in the Indian pharma market include Sun Pharmaceutical Industries, Dr. Reddy's Laboratories, and Cipla Ltd.

- FDI inflows in 2024
- In April-September 2024, gross inward FDI increased to \$42.1 billion, a significant increase from \$33.5 billion in the same period in 2023
- In the first quarter of 2024-25, FDI inflow reached \$22.5 billion, a 26% increase from the previous year
- Total FDI in 2023-24 was \$71.28 billion
- FDI trends in India
- From 2000 to 2024, India's total FDI inflow was \$991 billion
- The decade 2014-2024 saw an increase of 119% in FDI inflows over the previous decade (2004-2014)
- India's FDI inflows have increased about 20 times from 2000-01 to 2023-24

- FDI and the economy
- FDI contributes to India's GDP growth and forex reserves
- FDI is usually a long-term investment and is largely a non-debt creating capital flow



<https://www.ibef.org/economy/foreign-direct-investment>

FUTURE AHEAD

- The pharmaceutical industry in India is a significant part of the nation's foreign trade and offers lucrative potential for investors.
- Millions of people around the world receive affordable and inexpensive generic medications from India, which also runs a sizable number of plants that adhere to Good Manufacturing Practices (GMP) standards set by the World Health Organization (WHO) and the United States Food and Drug Administration (USFDA).
- Among nations that produce pharmaceuticals, India has long held the top spot.
- Medicine spending in India is projected to grow 9-12% over the next five years, leading India to become one of the top 10 countries in terms of medicine spending.

- Going forward, better growth in domestic sales would also depend on the ability of companies to align their product portfolio towards chronic therapies for diseases such as cardiovascular, anti-diabetes, anti-depressants and anti-cancers, which are on the rise.
- The Indian Government has taken many steps to reduce costs and bring down healthcare expenses.
- The National Health Protection Scheme, which aims to offer universal healthcare, the ageing population, the rise in chronic diseases, and other government programmes, including the opening of pharmacies that offer inexpensive generic medications, should all contribute to boost the Indian pharmaceutical industry.
- Speedy introduction of generic drugs into the market has remained in focus and is expected to benefit the Indian pharmaceutical companies.
- the thrust on rural health programmes, lifesaving drugs and preventive vaccines also augurs well for the pharmaceutical companies.



THANK YOU



POST MARKETING
SURVEILLANCE &
PHARMACOVIGILANCE

Ms. Sonali Correa

Ph.D., M.Sc.



Outline

Introduction

History

Benefits of PMS

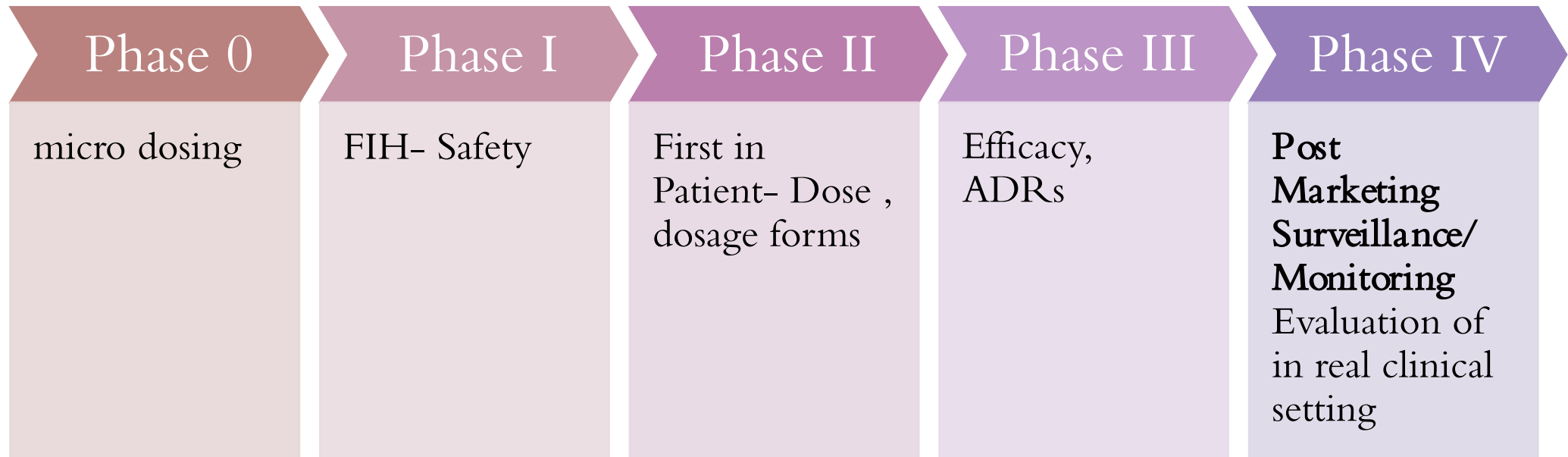
Sources of PMS

Types of Surveillance

Method of Surveillance

Manufacturer PMS system

Phases of Clinical Trials

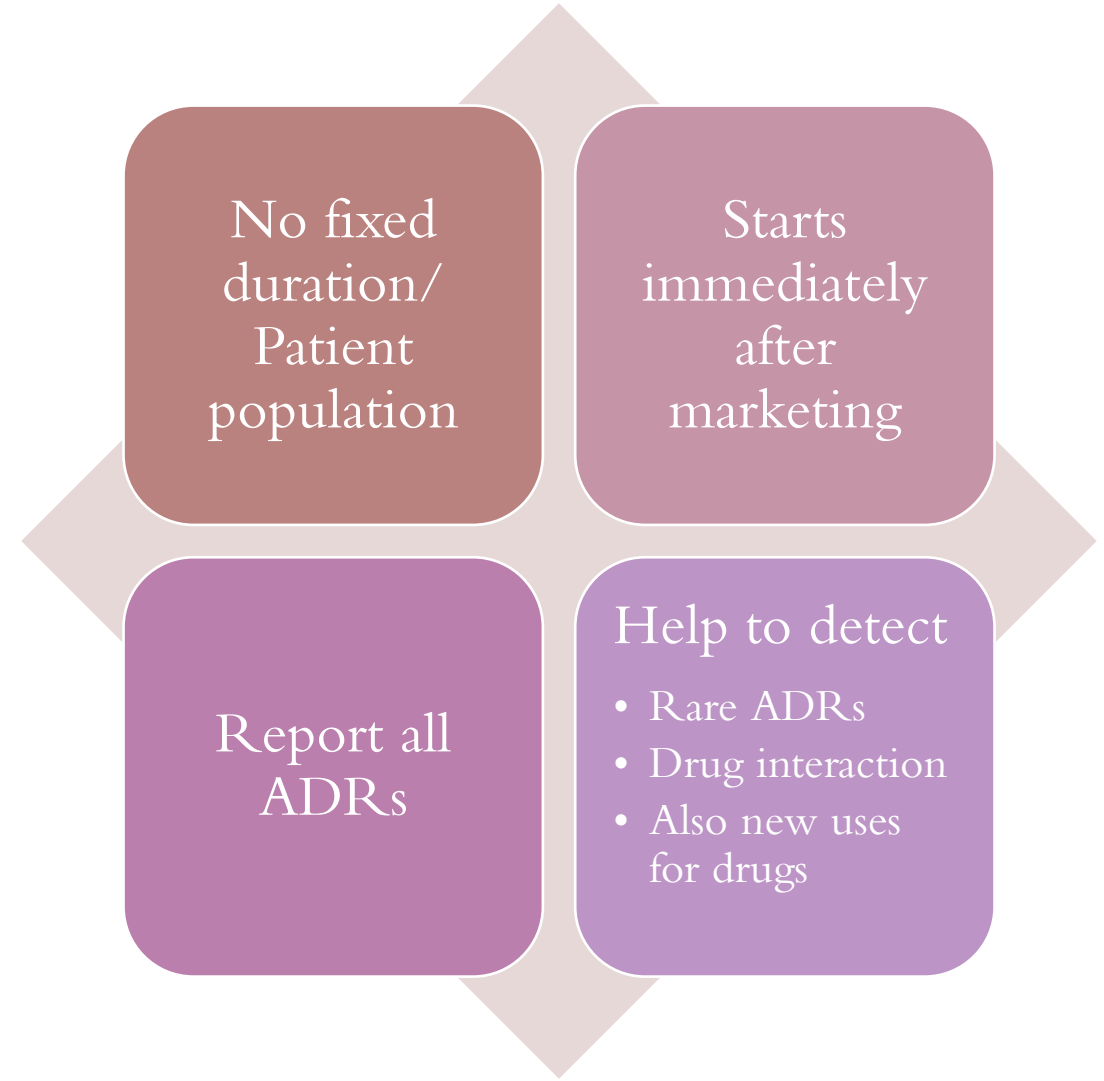


CLINICAL
TRIALS
TESTING IN
HUMANS



PHASE	NUMBER OF PATIENTS	LENGTH	PURPOSE	PERCENTAGE OF DRUGS SUCCESSFULLY TESTED
Phase 1	20 to 100	Several months	Mainly safety	70 percent
Phase 2	100 to 500	Several months to 2 years	Some short-term safety, but mainly effectiveness	33 percent
Phase 3	1,000 to 5,000	1 to 4 years	Safety, effectiveness, dosage	25 to 30 percent

Phase IV PMS



Definition

Post marketing surveillance (PMS), in simple terms, refers to the process of monitoring the safety of drugs once they reach the market after the successful completion of clinical trials.

The primary purpose for the conduct of PMS is to identify previously unrecognized adverse effects as well as positive effects.

Since drugs and medical devices are approved on the basis of clinical trials, which involve relatively **small numbers of people** who have been selected for this purpose – meaning that they normally do not have other medical conditions which may exist in the general population – postmarketing surveillance can **further refine, or confirm or deny, the safety of a drug or device after it is used in the general population by large numbers** of people who have a wide variety of medical conditions.

PMS uses a number of approaches to monitor drug and device safety,

- reporting databases
- prescription event monitoring
- electronic health records
- patient registries
- record linkage between health databases.

These data are reviewed to highlight potential safety concerns in a process known as **data mining**.

Section 505 (0)(3) authorizes the FDA to require certain PMS studies and clinical trials for prescription drugs approved under Section 505 (b) and biological products approved under Section 351

PMS plays a pivotal role in discovery of any undesirable effects as well as provide additional information on the benefit and risk of the drugs.

History

Thalidomide tragedy of the 1960s

Early 1960s, Thalidomide was a widely used for the treatment of nausea in pregnant women. It became apparent in the 1960s that thalidomide treatment resulted in severe birth defects in thousands of children.

The Pre-market approval, Senator Edward Kennedy suggested that a better system was need for monitoring the use and effects of prescription drug after they are marketed.

As a result, the joint commission prescription drugs use was established in 1976, funded largely by the drug industry with the Mandale to design a post marketing surveillance (PMS) system to detect, quantify and describe the unanticipated effects of marketed drugs.

The delayed discovery of the practolol adverse effects spurred effects to improve PMS.

Benefits of PMS system



Detection of Manufacturing problems

- Improvement of medical device quality
- Verification of risk analysis
- intelligence of long term performance
- Chronic complications
- Performance trends
- Performance trends in diverse populations
- Mechanisms of the devices may be misused
- Feedback on indications of use, instructions for use
- Training required for users, use with other products
- Customer satisfaction
- Identification of incident reports (and field safety corrective action reports)

The Need?



PMS is an essential tool that helps to correlate the strength of the exposed drug with that of the adverse events, thereby painting a clearer picture with regard to the types of positive or negative effects that a drug may threaten to pose, over a prolonged period of administration.

Drugs and devices offer opportunities for improved diagnosis and management of disease, they also can carry substantial risks.

Gvt & Regulatory bodies need to balance the goals of expanding therapeutic options with safeguarding public health

In recognition of the importance of medical devices, the WHO established a Medical Device Unit

- focus research and policy on prioritizing access to medical devices in low-resource settings
- Dissemination of innovations
- Training of biomedical personnel to support the use of devices worldwide

Types of PMS



Proactive

- PMS is both
- Customer surveys
- Post CE mark clinical trials, including PMCF
- Manufacturer sponsored device tracking/implant registries
- Expert user groups (focus groups)

Reactive

- Vigilance is reactive
- Customer complaints
- Unsolicited user feedback (other than complaints)
- Maintenance/service reports
- In-house testing (routine)
- Failure analysis
- Social media
- Literature reviews
- Regional or national device registries (non-manufacturer sponsored trials)

PMS Plan



The aim of the PMS System is to actively and systematically gather, record and analyze relevant data on the quality, performance and safety of a device throughout its entire medical device lifetime.

KEEP THE GOAL TOP OF MIND

- As you create a PSUR/PMS report, keep in mind that the primary role of the document is to provide a comprehensive, concise, and critical analysis of the PMS data relating to a device or a group of devices, with the goal of identifying any potential changes to the benefit/risk profile of the device(s).
- The goal is not to create paperwork but rather to improve device(s) and provide a full and accurate picture of a device's post-market performance.

ALIGN YOUR REPORT WITH YOUR PMS PLAN

- As you create a PSUR/PMS report, keep in mind that it must be aligned with the corresponding device's PMS plan, being mindful of any changes required to that PMS plan based on experience and learnings gained throughout the PSUR/PMS report creation process.
- Consider the PMS plan a living document and update it as needed based on the PSUR/PMS report and any other relevant reports.

IDENTIFY AND GATHER SOURCE DATA

- When embarking on the creation of a PSUR/PMS report, you need to identify and gather source data.
- Before doing that, it is essential that you invest the time to fully understand the data and conduct careful up-front planning to ensure that the data can be consistently retrieved, compiled, and presented.
- The creation and use of templates can be very beneficial, but it's important to know that each template must be tailored to the manufacturer's unique process and data sources.

MEET THE MINIMUM DATA REQUIREMENTS OUTLINED IN THE REGULATION.

Ultimately, a PSUR/PMS report should include any data that is relevant, but you must ensure that data referenced in the document includes at a minimum:

- Data regarding sales and usage
- Product complaints
- Reporting of serious incidents
- Field safety corrective actions
- Trend reports
- Relevant data from literature research
- Corrective and preventive actions

PERFORM DATA ANALYSIS

- One of the most common mistakes manufacturers make is simply compiling and presenting the data in the PSUR/PMS report.
- This is a sure way to generate questions from a notified body and prolong the process.
- In addition to presenting the data, you should also analyze it in order to identify any trends or signals that may have an adverse impact on the benefit/risk ratio.

REFERENCE PREVIOUS PSURS/PMS REPORTS

- The work is not done after the initial PSUR/PMS report submission, and notified bodies are paying attention to the actions manufacturers are taking based on the information in these documents.
- You should always review the previous PSUR/PMS report because it is essential that all previous actions required have been carried out or are being carried out.
- Additionally, monitoring of potential adverse trends from the previous PSUR/PMS report is required.

REVIEW LITERATURE ON A REGULAR BASIS

Literature should be reviewed regularly—at least quarterly, but more frequently is better—to ensure that new important safety information is identified and addressed. Bear in mind that there may be information relevant to your device's risks that need to be reviewed

PMS Report

PMS report (Class I)

Class I (low risk) – These are devices that pose almost no risk to patient safety. Class I examples include bandages, wheelchairs, and surgical instruments.

Class IIa (medium risk) – Class IIa medical device examples include surgical clamps, and this classification requires regular assessment by a notified body.

It should contain a summary of the information on the device that is collected and evaluated.

It should also point out the actions that will be taken.

In the MDR, nothing is said about the frequency with which this report needs to be updated, but it needs to be available for Notified Body and Competent Authority.

When it comes to a newly developed product, we advise to create a PMS report at least yearly. If after a few years the data do not show concerns, the frequency could be decreased.

Periodic Safety Update Report (PSUR) (Class IIa/IIb/III)



For higher risk classes, a PSUR is required.

This report is more extensive compared to a PMS report. It should include a rationale and a description of all the corrective and/or preventive actions taken based on the analysed product information.

During the lifetime of a medical device, the PSUR must also set out:

- the conclusions of the benefit-risk determination of the device
- the main findings of the PMCF
- the volume of sales
- an evaluation of the population that uses the device

When the risk class of the device is IIa, the PSUR should be updated at least every two years, while you should update it at least annually if the risk class is IIb or III.

In case the device is an implantable or belongs to risk class III, this report should also be uploaded to the EUDAMED (European Databank on Medical Devices) database, where your Notified Body can consult it.

The Notified Body will then write an evaluation and upload it to the same electronic system, where the report and evaluation are also made available for the Competent Authority.

PSUR of Class IIa devices do not have to be uploaded to EUDAMED for an evaluation, but should be made available for the Notified Body and Competent Authority.

<http://www.jfda.jo/EchoBusV3.0/SystemAssets/PDF/AR/PharmVigilance/checklistforPSUR.pdf>

<https://www.ema.europa.eu/en/human-regulatory/post-authorisation/pharmacovigilance/periodic-safety-update-reports-psurs>

PMCF



Besides product-related technical data, it's also important to keep collecting clinical data of the device, the so called Post Market Clinical Follow-up (PMCF). PMCF is part of the device's PMS plan and is especially needed when the device:

- contains novel innovative medical technology
- presents a high risk during use
- is used at high-risk anatomical locations
- is used for high-risk populations (e.g. children, pregnant women, elderly)

When a PMCF plan is not created, a (good) rationale should be included in the PMS plan.

A PMCF plan needs to be written to make sure the clinical information is gathered, evaluated and updated throughout the whole life cycle of the device.

The plan includes the methods and procedures that will be used, a rationale for the appropriateness of those methods, evaluation of clinical data from equivalent devices and a time schedule.

Findings from the PMCF shall be documented in a PMCF evaluation report, which will be part of the technical documentation of the device. The conclusions of this report could lead to an update of the overall clinical evaluation of the device.

The most important new findings of the PMCF studies need to be introduced in the PSUR as well.

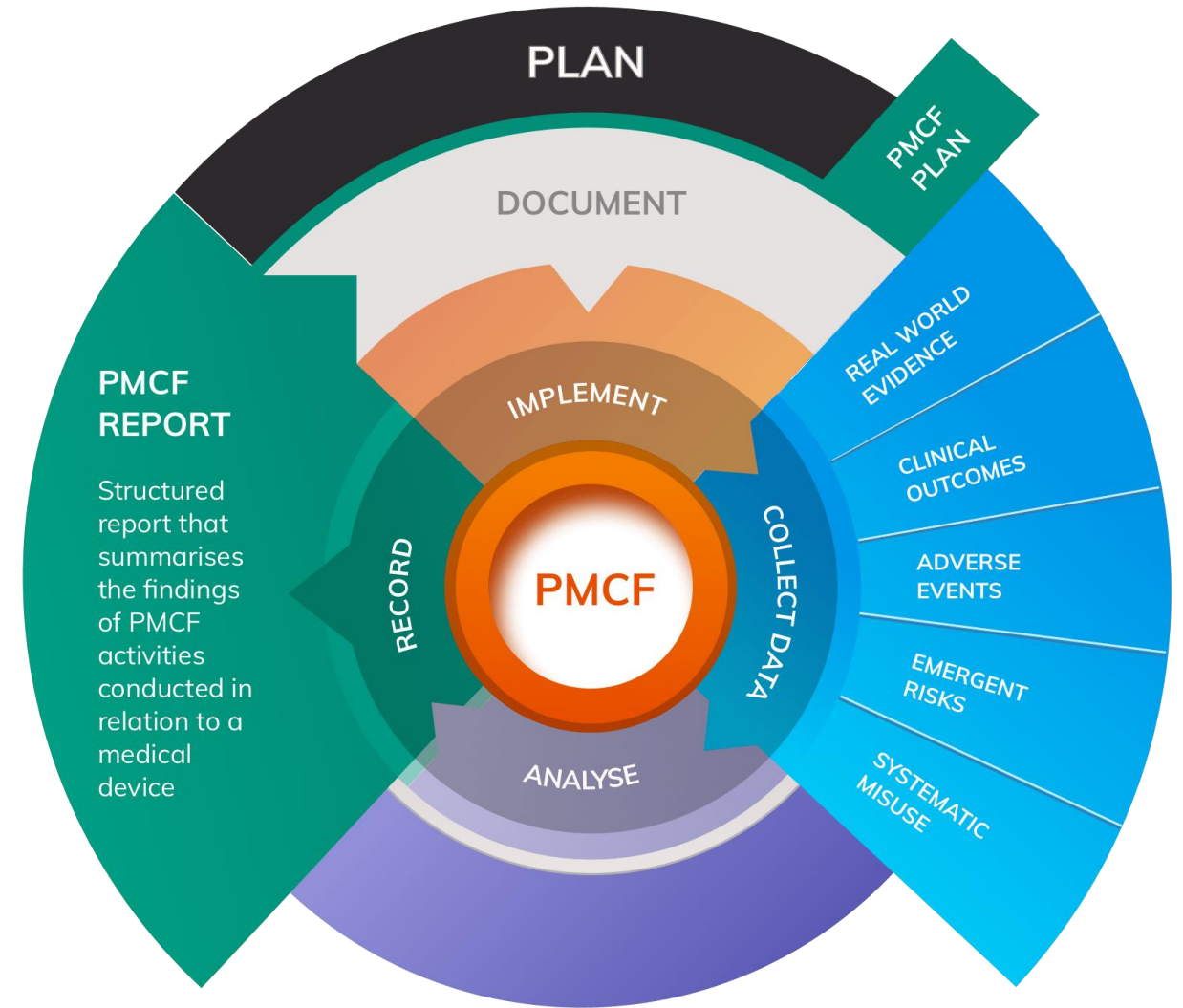
Just like with the PMS and PSUR, any corrective and/or preventive actions must be identified and implemented based on the evaluation results.

PMCF system design should be tailored to the subject device and its associated classification and complexity.

Effective PMCF systems are built to address risk management outcomes alongside the overall objectives for PMCF as specified in the MDR.

A PMCF system must be capable of capturing data on use of a medical device throughout its entire lifetime, meaning that many standard-type clinical investigations will not be suitable for this application.

PMCF systems can range from simple patient or end-user surveys to formal clinical investigations and complex medical device registries that incorporate both clinical and patient-feedback data



A well-written PMCF Plan should include at least:

- Details of general methods to be applied, e.g. gathering clinical data, user feedback and/or literature searching
- Specific methods to be used, e.g. PMCF study/survey/registry
- Rationale for selected general and specific methods
- A reference to relevant parts of the Clinical Evaluation Report (CER) that contain a summary of PMCF data, an appraisal of the PMCF system, and results of data analysis.
- Specific objectives addressed by PMCF
- Evaluation of clinical data from equivalent/similar devices, if relevant
- Reference to any harmonised standards used in developing the PMCF Plan
- Timescale of PMCF activities

The Medical Device Regulation (MDR) 2017/745 greatly enhances the importance of PMCF compared to the MDD. A detailed set of requirements for the design and conduct of PMCF activities is outlined in MDR Annex XIV Part B.

A specific requirement is that all manufacturers document a PMCF Plan to detail the process that will be followed in collecting PMCF data on each of their medical devices

Different submitting regions require different documents like, if to submit in USA, FDA look for **Periodic Adverse Drug Experience Report (PADER/PAER)** from manufacturer

EU, PMS report should be submitted to low risk class I devices and periodic safety update report (PSUR) for Class IIa, Class IIb, and Class III devices.

As a part of technical documentation, these PSUR, PMSR should contain data analysis and a description of the corrective and preventative actions taken and updated by manufacturer regularly based on the relevant regulatory bodies' timelines

FDA 21 CFR part 822 in US summarizes key requirements for PMS of a medical device if the device fall under class II & III. The criteria of Class II & III devices are

- 1) Failure of the device would likely have a serious adverse health consequence
- 2) The device is intended to be implanted in human body for more than one year & a life-sustaining or life-supporting device used outside of a user facility

Guidance documents informing PMCF



The **MEDDEV 2.12-2 Rev. 2** is supported and supplemented by additional guidance documents and ISO standards that provide direction for obtaining relevant, long-term data from any clinical investigation (whether a part of pre- or post-market plans).

EN ISO 14155:2011 defines good clinical practice for the design, conduct, recording, and reporting of clinical investigations carried out in human subjects to assess the safety or performance of medical devices for regulatory purposes

EN ISO 14155:2011 specifies general requirements intended to protect the rights, safety, and well-being of human subjects; ensures the scientific conduct of the clinical investigation and credibility of the results; define the responsibilities of the sponsor and principal investigator; and assist sponsors, investigators, ethics committees, regulatory authorities, and other bodies involved in the conformity assessment of medical devices

Guidance documents



MEDDEV 2.7.1 Rev. 3 outlines the stages of clinical evaluations; the general principles of thorough, objective, and ethical clinical evaluation; the sources of data, appraisal, and analysis of clinical data; and the necessary reporting

MEDDEV 2.7.3 gives guidance on what is reportable in a clinical investigation as 'incidents' are not 'adverse events'.

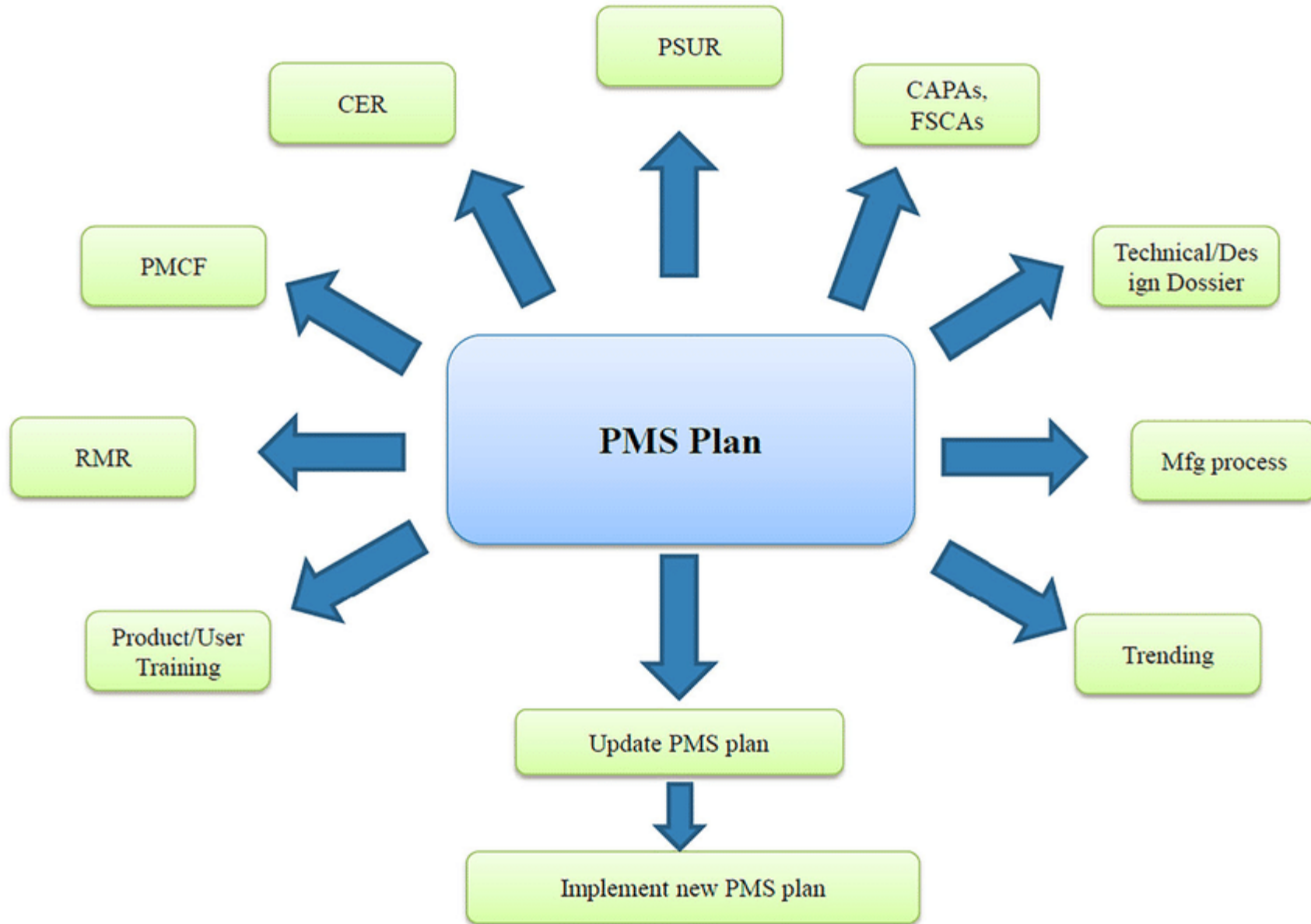
- MEDDEV 2.7.4 focuses mainly on pre-market clinical investigations but provides relevant best practices for any clinical investigation.

Objectives of the post-market surveillance plan



The post-market surveillance plan (PMS plan) is used – as the name suggests – to plan post-market surveillance activities. It must therefore define, on a device-specific level:

- which people and roles
- are responsible at which times and on which occasions
- for which activities,
- which information these people collect and
- how they evaluate this information,
- who they have to communicate this evaluation to and
- which actions or notifications they have to trigger.



Post-market Surveillance under MDR & USA

Few of the regulations and standards for PMS are **MDR 2017/745**, **ISO/TR 20416:2020**, **ISO 13485:2016**, **ISO 14971:2019**. In **MDR 2017/745** – article 15, 83, 84, 85, 86, and annex 3 explains about a person responsible for regulatory requirements, post-market surveillance system for the manufacturer, post-market surveillance plan, post-market surveillance report, PSUR and technical documentation for post-market surveillance system respectively.

ISO/TR 20416:2020, **ISO 13485:2016** & **ISO 14971:2019** tells about the requirements for Post-market surveillance for manufacturers, Quality management systems — Requirements for regulatory purposes, Clause 8 & Application of risk management to medical devices, clause 10.

Postmarketing Surveillance Programs

FDA Adverse Event Reporting System (FAERS)

It is a computerized information database designed to support the FDA's post-marketing safety surveillance program for all approved drug and therapeutic biologic products.

The ultimate goal of FAERS is to improve the public health by providing the best available tools for storing and analyzing safety reports.

Healthcare professionals, consumers, and manufacturers submit reports to FAERS. FDA receives voluntary reports directly from healthcare professionals (such as physicians, pharmacists, nurses and others) and consumers (such as patients, family members, lawyers and others).

The reports in FAERS are evaluated by a multidisciplinary staff safety evaluators, epidemiologists and other scientists in the Center for Drug Evaluation and Research's (CDER) Office of Surveillance and Epidemiology to detect safety signals and to monitor drug safety.

As a result, the FDA may take regulatory actions to improve product safety and protect the public health, such as updating a product's labeling information, sending out a "Dear Health Care Professional" letter, or re-evaluating an approval decision.

<https://www.fda.gov/drugs/surveillance/questions-and-answers-fdas-adverse-event-reporting-system-faers>

MedWatch program

It is for health professionals and the public to voluntarily report serious reactions and problems with medical products, such as drugs and medical devices.

It also ensures that new safety information is rapidly communicated to the medical community thereby improving patient care.

All data contained on the MedWatch form will be entered into the AERS database.

The MedWatch page includes sections on how to report an adverse event, safety information, and publications.

The Division of Drug Marketing, Advertising and Communications page also contains other useful drug advertising and surveillance information.

After a drug is approved and marketed, the FDA uses different mechanisms to assure that firms

- 1) adhere to the terms and conditions of approval described in the application
- 2) that the drug is manufactured in a consistent and controlled manner.

This is done by periodic, unannounced inspections of drug production and control facilities by FDA's field investigators and analysts.

Manufacturers of prescription medical products are required by regulation to submit adverse event reports to the FDA.

The MedWatch site provides information on mandatory reporting by manufacturers. In addition, drug manufacturers must submit either error and accident reports or drug quality reports when deviation from current good manufacturing practice regulations occur.

FDA receives medication error reports on marketed human drugs (including prescription drugs, generic drugs, and over-the-counter drugs) and nonvaccine biological products and devices.

The National Coordinating Council for Medication Error Reporting and Prevention defines a medication error as "any preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of the health care professional, patient, or consumer.

Such events may be related to professional practice, health care products, procedures, and systems, including prescribing; order communication; product labeling, packaging, and nomenclature; compounding; dispensing; distribution; administration; education; monitoring; and use."

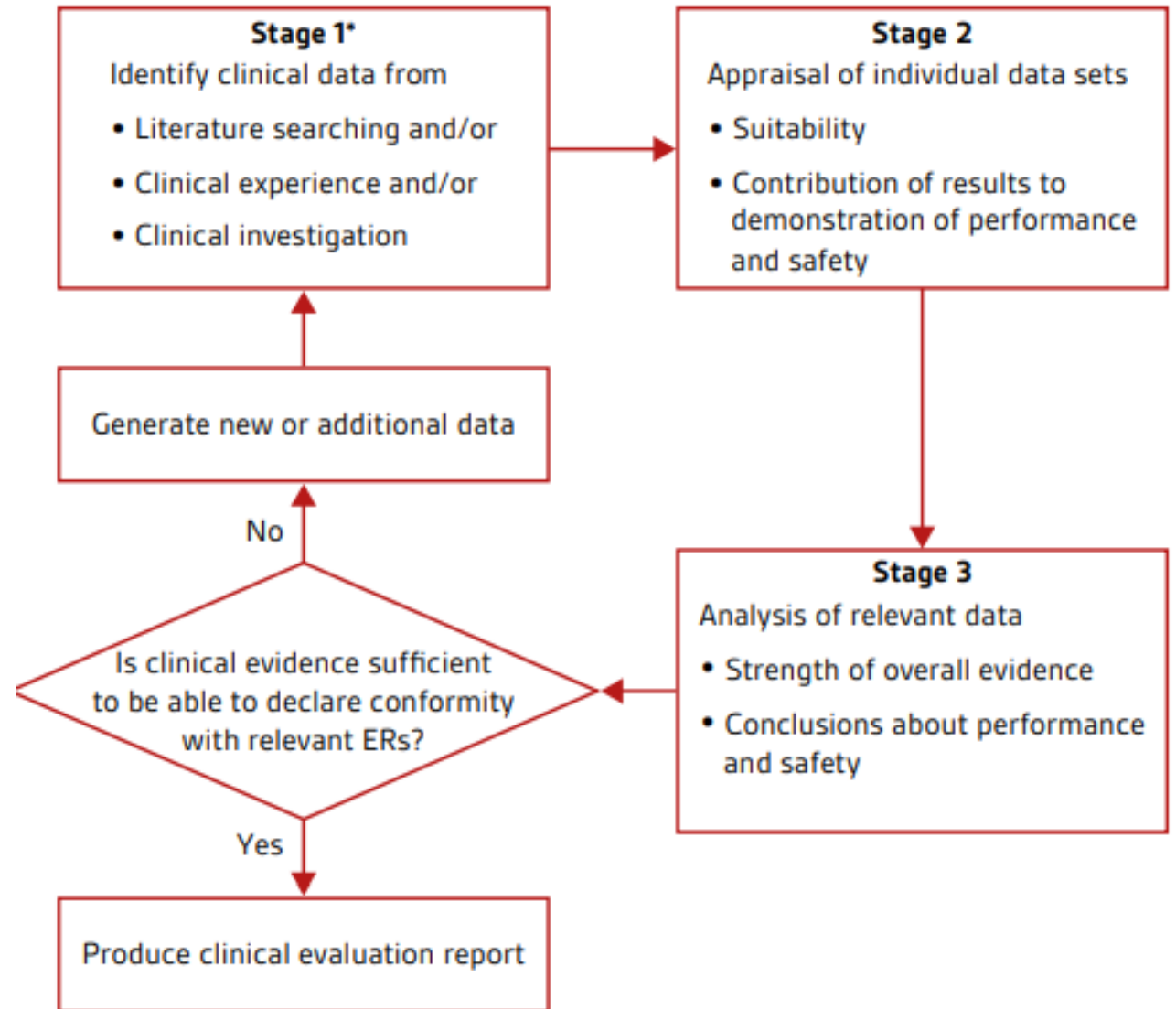
CDER medication errors program staff review medication error reports sent to the USP-ISMP Medication Errors Reporting Program and MedWatch, evaluate causality, and analyze the data to provide feedback to others at FDA.

Therapeutic Inequivalence Reporting

In the past 10 years, FDA's Center for Drug Evaluation and Research has received an increase of reports of drug products that fail to work in patients because the product simply has no effect or is toxic.

These problems are usually attributed to switching brands of drugs. As a result, on Sept. 14, 1988, FDA created in CDER the Therapeutic Inequivalence Action Coordinating Committee (TIACC) to identify and evaluate reports of therapeutic failures and toxicity that could indicate that one product is not equivalent to another similar product

PMS EVALUATION



Methods of PMS



Identification of drugs effects

- Controlled CT
- Spontaneous voluntary recording
 - Cohort studies
 - Case control studies

Controlled CT

- ✓ PMS studies conducted after the launch of a product is part of phase IV development of drug.
- ✓ Some of these studies may be retrospective case-control evaluation
- ✓ These are done to evaluate rare suspected side effects.
- ✓ For eg: when there was a suspicion that use of oral contraceptives may be associated with an increased incidence of thrombophlebitis (clotting of blood in the deep veins) and thromboembolism (blockage of smaller arteries due to detached blood clots) case-control studies were carried out.
- ✓ A group of cases of thromboembolism were compared with age matched controls that were similar to the cases as possible, but without the disease.
 - To minimize bias through such method as randomization and “double-blinding
 - Directly monitor patients for the duration of studies.
 - For evaluating a drug’s efficacy and safety.
 - They are often costly.

Spontaneous reporting

- ✓ A communication from an individual (e.g: health care professional, consumer) to a company or regulatory authority
- ✓ This describes a suspected adverse events
- ✓ But the actual incidence of adverse drug reaction cannot be determined through spontaneous reporting
- ✓ The spontaneous reporting system process
 1. Data acquisition
 - 2.data assessment
 - 3.data interpretation

1. Data acquisition :

which depends largely on the input of information derived from reports submitted by the health professionals who have encountered what they suspect is an ADR

2. Data assessment :

which involves assessment of the individual case reports and assessment of pooled data obtained from various sources such as the international database of the WHO The spontaneous reporting system.

3. Data interpretation

based on the available data and the assessments made, a signal related to the adverse reaction may be generated

REPORTING SYSTEMS



India SUSPECTED ADVERSE
DRUG REACTION REPORTING
FORM

UK YELLOW CARD 1964

Australia BLUE CARD

US MEDWATCH

Postmarketing surveillance: MHRA



PMS in the UK is practiced in the form of the Yellow Card scheme that is jointly operated by the MHRA and the Committee of Human Medicines (CHM).

The Yellow Card scheme is credited as being one of the first PMS schemes aimed at mitigating ADRs

Encompasses the following objectives:

- Monitoring the use of medicines in everyday practice with the aim to identify erstwhile unrecognized ADRs and also changes in the patterns of adverse effects.
- Carrying out risk–benefit analysis for medicines and suggesting suitable actions, if and when necessitated.
- Providing regular updates to healthcare professionals and patients with regard to the safe and efficacious use of medicines.

A yellow card form for reporting suspected adverse drug reactions. The form includes sections for patient details, suspected drug(s), suspected reaction(s), other drugs, and reporter details. It is a standard form used by the MHRA and CHM in the UK for post-marketing surveillance.

Cohort Studies

- ✓ Studies follow a defined group of patient for a period of time.
- ✓ Patient are not randomly assigned , & there is no blinding.
- ✓ Other names of cohort study are Longitudinal study, Incidence study and forward looking study
- ✓ Features of cohort studies
 - Cohorts are identified prior to appearance of disease under investigation
 - The study groups are observed over a period of time to determine the frequency of disease among them
 - The study proceeds from cause to effects

✓ Indications for cohort study

- There is good evidence of an association between exposure and disease, from other studies.
 - Exposure is rare.
 - Attrition of study population can be minimized.
 - Sufficient fund is available.

Consideration during selection of Cohort

- The cohort must be free from disease under study.
- Insofar as the knowledge permits, both the groups should be equally susceptible to disease under study.
- Both the groups must be comparable in respect of all variable which influence the occurrence of disease
- Diagnostic and eligibility criteria of the disease must be defined beforehand

Types of Cohort Studies

- Prospective study
- Retrospective cohort study
- Ambi-directional cohort study

Advantage of Cohort Studies

- Temporality can be established
- Incidence can be calculated.
- Several possible outcome related to exposure can be studied simultaneously.
- Provide direct estimate of risk.
- Since comparison groups are formed before disease develops certain forms of bias can be minimized like misclassification bias.
- Allows the conclusion of cause effect relationship

Disadvantage of Cohort Studies

- Large population is needed
- Not suitable for rare diseases.
- It is time consuming and expensive
- Certain administrative problems like loss of staff, loss of funding and extensive record keeping are common.
- Problem of attrition of initial cohort is common
- Study itself may alter people's behavior

Case Control Studies

- ✓ Case control studies identify patient with the adverse effects to be studied, and compare them with the sample drawn from the same cohort that gave rise to cases.

- ✓ Case-Control Studies

A case control study involves two populations – cases and controls and has three distinct features :

- Both exposure and outcome have occurred before the start of the study.
- The study proceeds backwards from effect to cause.
- It uses a control or comparison group to support or refute an inference.

Advantages

- ❖ Efficient - saves time and money
- ❖ Used for rare diseases, smaller sample sizes
- ❖ Multiple associations with a disease
- ❖ Can generate hypotheses for future study
- ❖ Can be used to examine multiple exposures, such as smoking and asbestos exposure in mesothelioma risk

Disadvantages

- ❖ Susceptible to bias – recall, reporting
- ❖ Prone to methodological errors
- ❖ Observational
- ❖ Works for certain types of conditions

National Implementation

Canada

- Health Canada is the regulatory body which approves drugs, and it has a division called "Marketed Health Products Directorate" (MHPD) which coordinates Canadian postmarketing surveillance

• United Kingdom

- is practiced in the form of the Yellow Card scheme that is jointly operated by the The Medicines and Healthcare products Regulatory Agency (MHRA) and the Committee of Human Medicines (CHM).

• USA

- Postmarketing surveillance is overseen by the Food and Drug Administration (FDA), which operates a system of passive surveillance called MedWatch, to which doctors or the general public can voluntarily report adverse reactions to drugs and medical devices.

• India

- The regulatory framework for conducting clinical trials of drugs is provided under the Drugs and Cosmetics Act, 1940 ("Act") and the Drugs and Cosmetics Rules, 1945 ("Rules").



The principle differences between vigilance and post-marketing surveillance:

PV: the reporting of adverse incidents by manufacturers to the regulatory authorities and their subsequent sharing of key incident data between each other;

PMS: the process by which information on overall performance is captured, analysed and acted upon

Pharmacovigilance



According to the world health organization, "pharmacovigilance also known as drug safety, is the pharmacological science related to the collection, detection, assessment, monitoring, and prevention of adverse effects of the pharmaceutical drugs"

Pharmacovigilance is the process of monitoring the effects of drugs, both new and existing ones.

This includes collecting data, analyzing it, and taking steps to prevent any negative effects.

Pharmacovigilance must happen throughout the entire life cycle of a drug, from when it is first being developed to long after it has been released on the market.

Introduction

PV is critical for the pharmaceutical and health-care industries for drug approval, continuance, and to promote safe use of their products.

It helps the medical practitioners to enhance and communicate their knowledge in the betterment of therapeutic practices.

For patients and the general public, it is the source of all drug safety information and education.

PV is a very important instrument for **collection, assessment, identification, and communication of all the vital drug safety information that can help patients lead a healthy life.**

Data management is a vital part of PV as there is continuous generation of patient safety data.

Adverse event information can be generated from various modes, for example, clinical trials, post marketing programs, spontaneous reports, and literature or legal reports.

This information needs to be collected and reported to the regulatory authorities for analysis.

The role of a case processor is to monitor and track all serious adverse events, serious and medically significant adverse drug reactions (ADRs), and other medical-related product information followed by timely processing and reporting of such information in accordance with the company and regulatory reporting timelines.

The case processor usually has an educational background in one of the life sciences (e.g., nursing, pharmacy, or other allied health professionals); it is also advantageous to have a working knowledge of medical terminologies.

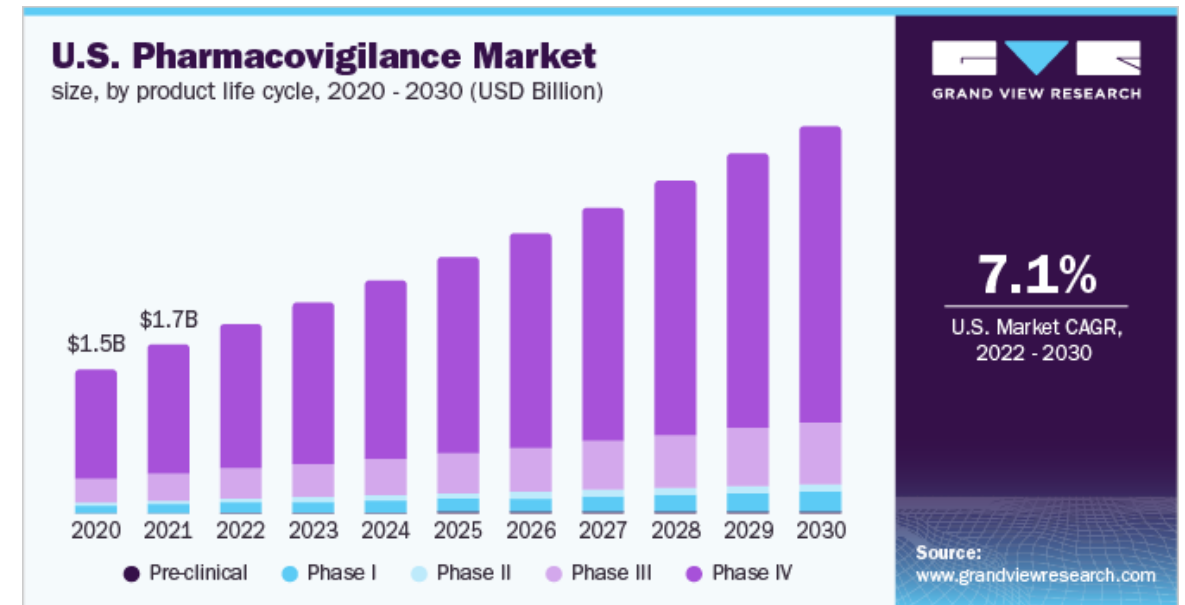
PV Global Scenario



The global pharmacovigilance market size was valued at **USD 7.80 billion in 2022** and is expected to expand at a compound annual growth rate (CAGR) of 10.3% from 2023 to 2030

The major driving factors of this market include **rising drug consumption and drug development rates, rising rates of adverse drug reactions and drug toxicity, and an increasing tendency to outsource** pharmacovigilance services.

The rising prevalence of lifestyle-related diseases, such as diabetes, hypertension, and cardiovascular diseases (CVDs), as a result of sedentary lifestyles, a lack of physical activity, changing lifestyle patterns, and poor diets, leads to increased drug consumption, indicating a high demand for drug monitoring and fueling the market growth.



On the other hand, factors such as high data security risk, a lack of global regulatory harmonization, and a lack of data standards for adverse event collection are expected to pose challenges to the global pharmacovigilance market.

The global pharmacovigilance market is expected to be dominated by the North American region. The number of drug abuse and associated adverse drug reactions in the United States has increased significantly. According to the U.S. Department of Health and Human Services, ADRs account for an estimated one-third of all hospital adverse events and affect about 2 million hospital stays each year in inpatient settings.

According to the Centers for Disease Control and Prevention, about 6.7 percent of hospitalised patients experience a severe adverse drug reaction (ADR), with a mortality rate of 0.32 percent. If these figures are assumed to be true, then there are about 2,216,000 severe ADRs in hospitalised patients per year, which results in over 106,000 deaths.

Aim Of Pharmacovigilance



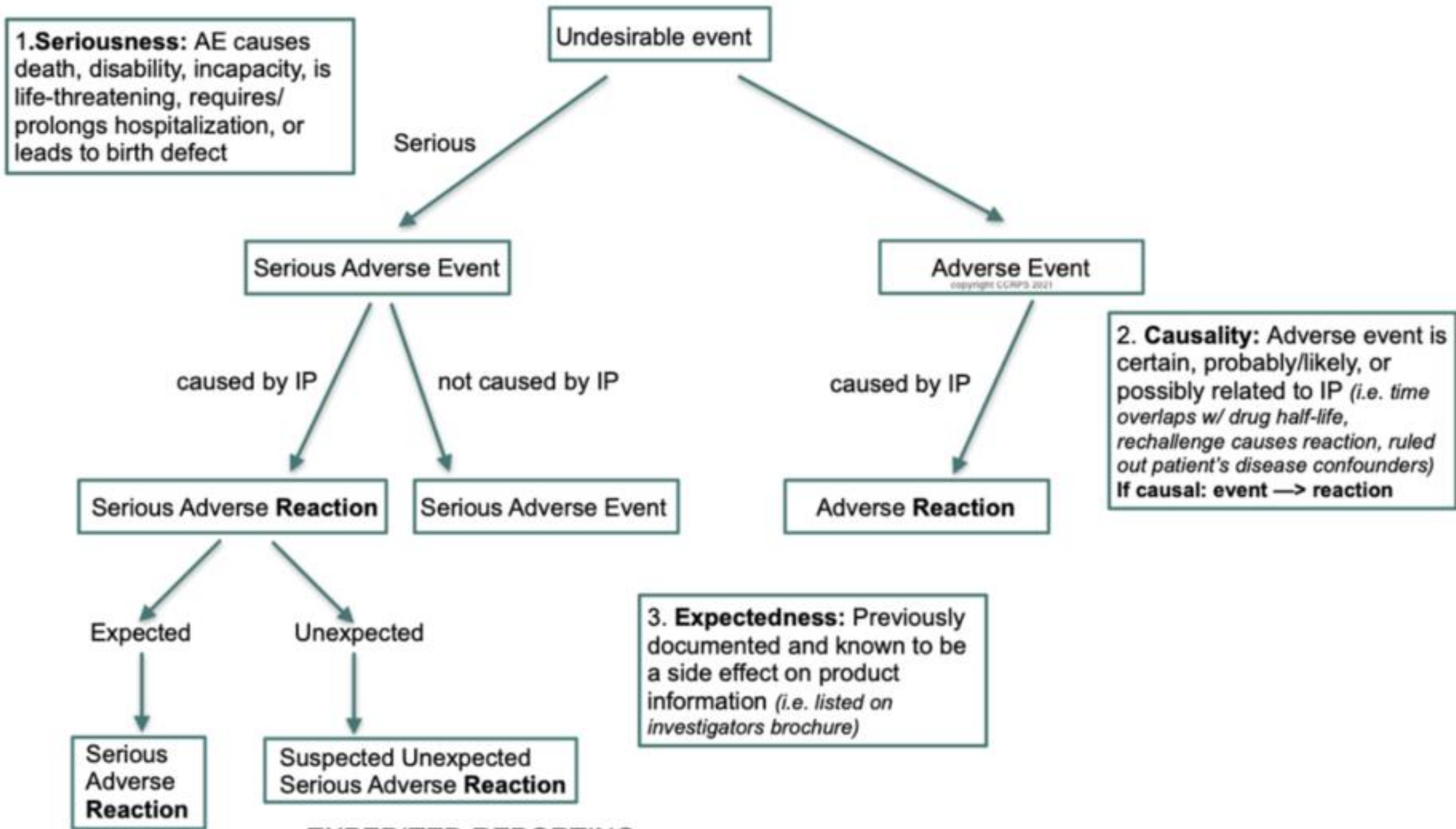
Pharmacovigilance is a vital field, with three key objectives:

surveillance, operations and focus.

Surveillance aims to ensure safety by producing Development Safety Update Reports (DSURs) and Periodic Benefit-Risk Evaluation Reports (PBRER).

Operations consists of monitoring drug progress during preclinical trials as well researching real-world evidence regarding adverse effects reported by patients or healthcare professionals.

Systems focuses on developing strong data management systems for pharmaceutical research protocols while staying compliant with all regulatory rules – an absolute necessity in this ever-changing industry



EXPEDITED REPORTING

**see country-specific regulatory body reporting timeframe*

Objectives of pharmacovigilance

To properly detect the unknown adverse effect and drug interaction

Distribution of information about the adverse effect of drugs and drug interactions so that they occur less frequently

Timely correction of therapeutic errors

Reevaluation of the risk-benefit balance of medicine, so we can effectively use those drugs for a patient having more benefits and fewer side effects

Identifications of mechanisms that cause adverse effects of drugs

Goals of PV



The rational and safe
use of medical drugs

The assessment and
communication of
the risks and benefits
of drugs, on the
market

Educating and
informing of patients

Stakeholders in pharmacovigilance



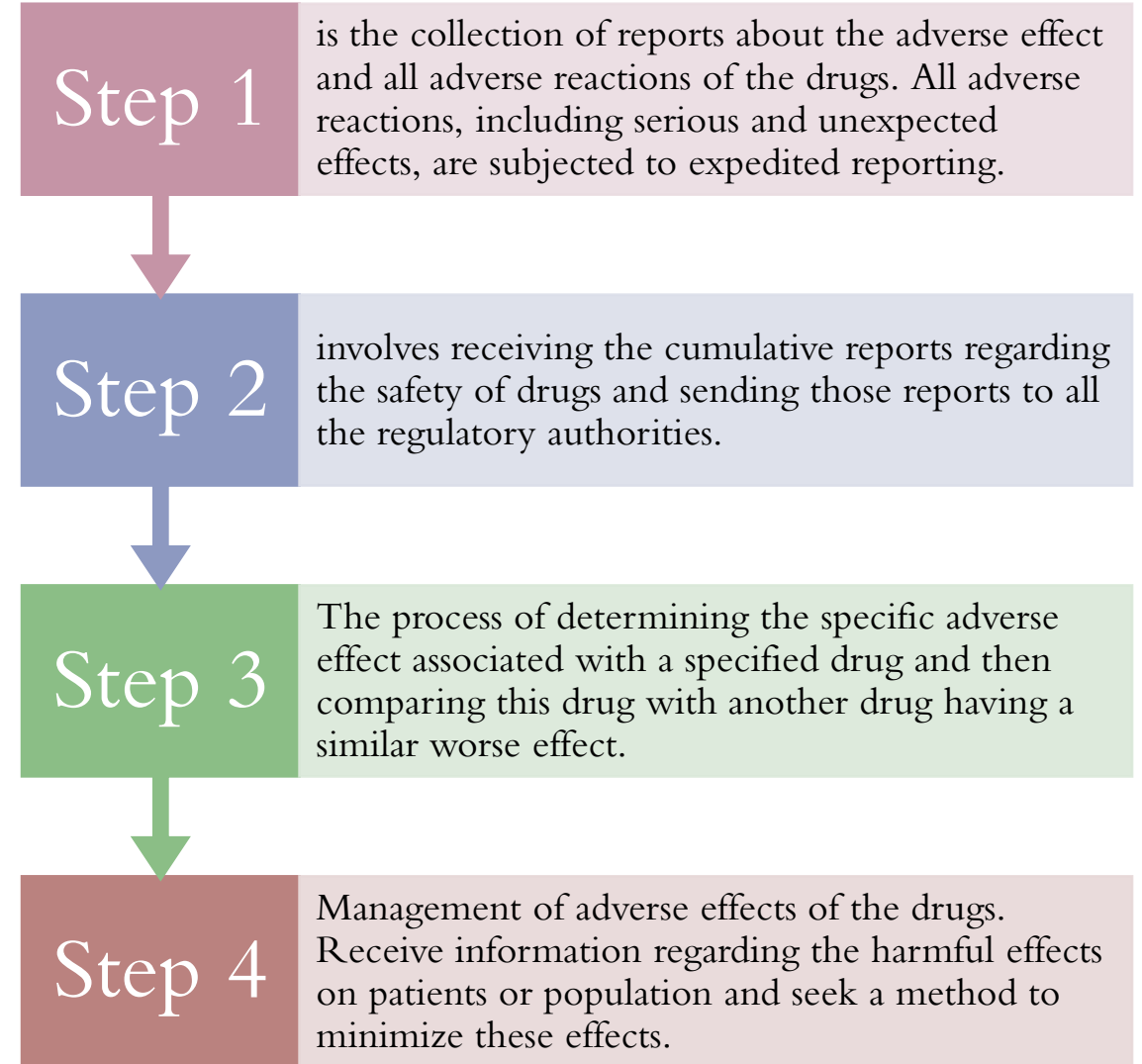
1- Patient

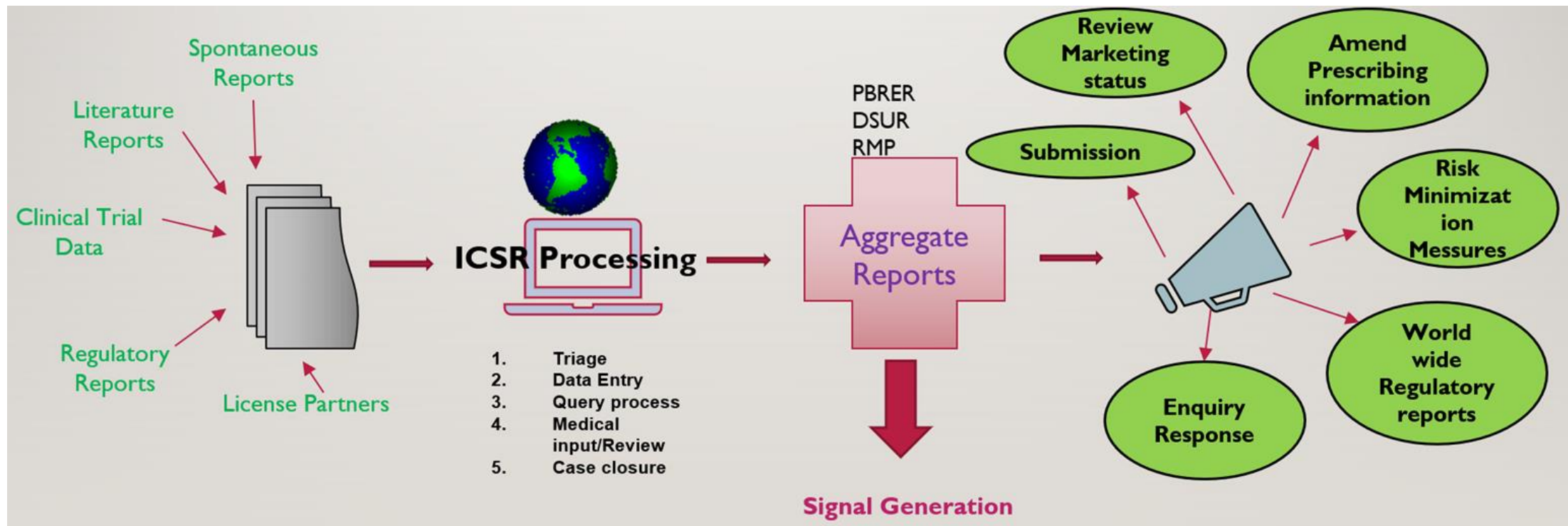
2- Doctor or
pharmacist

3- Marketing
authorization
holder

4- Regulators
such as
authorities,
committee

PV Workflow





Pharmacovigilance Process (Stage 1): Detection

Collection of Individual Case Safety Reports (ICSRs)

Pharmacovigilance starts with safety information that comes from different sources such as solicited and unsolicited reporting.

Solicited sources – Received as a result of targeted data collection (i.e., establish a channel to collect the Adverse effects)

- Clinical trials: Phases I-III and post-marketing (non-interventional and interventional studies) and trials
- Registries – organised systems with a predetermined scientific, clinical, or public health purpose
- Personalized programs for non-registered drug administration

Unsolicited sources – Received without request or Spontaneous in nature

- Consumers and Patients (regardless of medical confirmation)
- Healthcare professionals
- Regulatory authorities
- Literature reports
- License Partners
- Lawsuits against medicinal product: Legal cases
- Internet and other media sources

Pharmacovigilance Process (Stage 2): Assessment

After the collection of adverse events, an assessment of ICSRs is conducted

Steps involved in the assessment ICSR

1. Triage

The triage phase of a potential adverse event report involves establishing the validity of an ICSR. A valid ICSR should mandatorily possess the below mentioned four elements

- A. an identifiable patient
- B. an identifiable reporter
- C. a suspect drug
- D. an adverse event

2. Data Entry

- Every medical product company maintains its own safety database (e.g., Argus, ARISg, etc.)
- After case validation, the Safety associate enters the safety information into the safety database
- The further process in data entry includes
 - Seriousness determination
 - Coding of adverse events using MedDRA
 - Causality assessment
 - Labelling assessment
 - Clear and concise narrative writing

3. Query process

It involves the raising of queries to the reporter if any additional information is required and also to clarify the discrepancies (if any) in the safety data.

4. Medical input/Review

Safety Physician reviews safety information which is the emphasis on causality, seriousness, labelling, and provides the company pharmacovigilance comments.

Medical review is the first level step in signal detection activities.

5. Case closure

The completed ADR report is Submitted to the respective regulatory authorities.

Pharmacovigilance Process (Stage 3): Understanding the drug safety profile

Aggregate data review is performed to understand the drug safety profile using the following documents

- Periodic Benefit-Risk Evaluation Report (PBRER): All spontaneous and clinical trial cases reported during the review period are discussed in detail and benefit-risk analysis is performed

- Specific adverse reaction follow-up questionnaires are used to get the structured information on reported suspected adverse reactions for the important risks.
- Signal Analysis – Every MAH has a proper signal management system and performs the signal analysis.
- Risk Management Plan (RMP)
- Development safety update report (DSUR)

All the above safety documents are periodically submitted to the regulatory authorities.

Pharmacovigilance Process (Stage 4): Prevention of adverse effects

MAH takes action to prevent the Adverse drug reactions by:

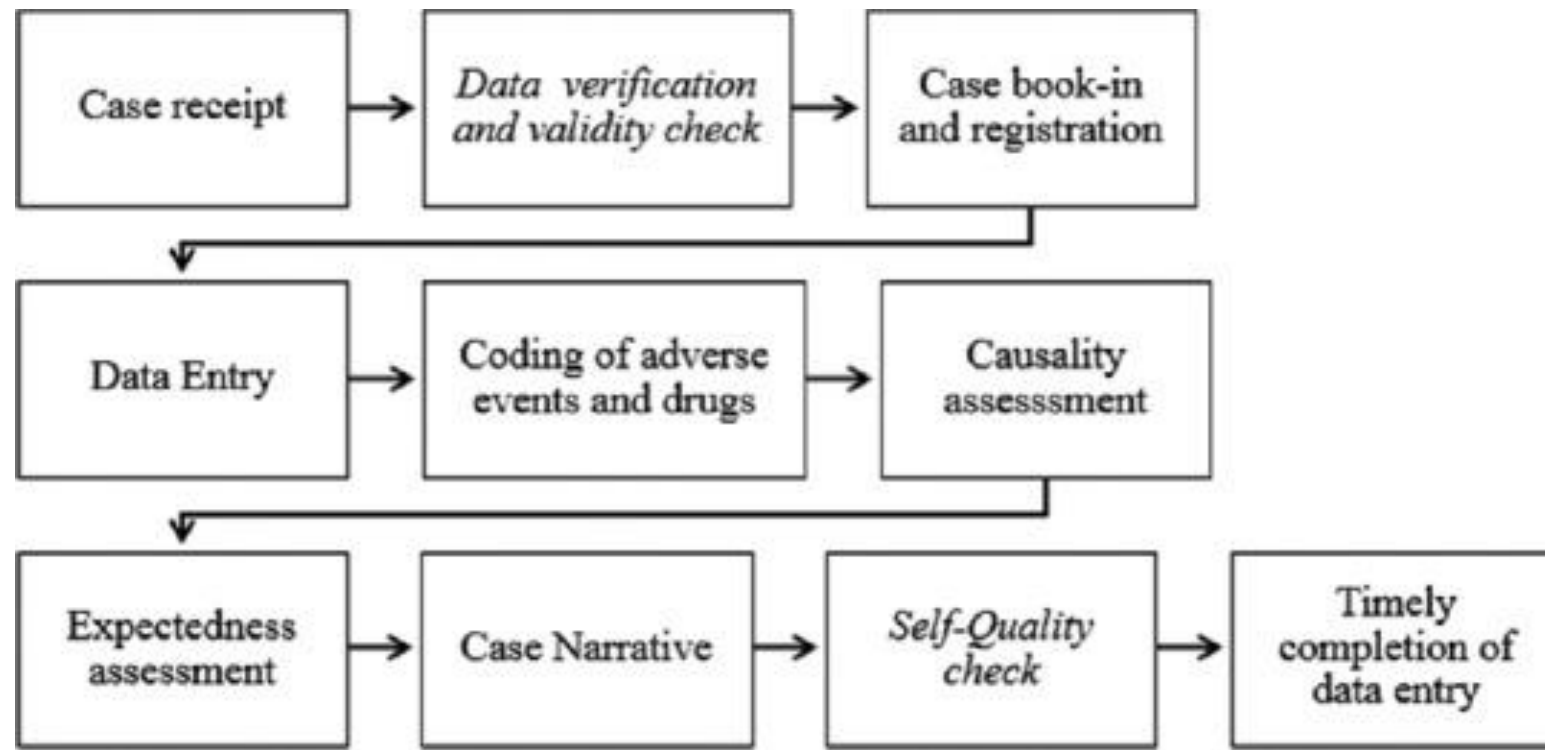
1. Performing Risk Minimisation Activities i.e., to update the Summary of Product Characteristics, Patient Information Leaflet, Labelling, and packaging of the medicine and Legal status of the medicine.
2. Monitoring Risk Minimisation Activities

Pharmacovigilance Process (Stage 4): Prevention of adverse effects

MAH takes action to prevent the Adverse drug reactions by:

1. Performing Risk Minimisation Activities i.e., to update the Summary of Product Characteristics, Patient Information Leaflet, Labelling, and packaging of the medicine and Legal status of the medicine.
2. Monitoring Risk Minimisation Activities

ROLE OF CASE PROCESSOR



Task

Case Receipt

Mailbox management

The first thing a case processor will do is to check the email after logging in to the system.

This allows the case processor to check the number of cases that have been assigned and also allows the associate to go through any feedback or other communication from the stakeholders

Acknowledgment

The case processor will acknowledge the report (every report needs to be acknowledged, more so the valid reports). T

he reporter may continue to send the same report repeatedly till it is acknowledged; hence, this simple action avoids duplication.

Acknowledgment establishes a contact with the reporter for more information, if and when required.

Data verification and validity check

Validity assessment

After receiving the source document (via email, fax, or phone), the case processor will look for the minimum information which is required for a valid safety report, i.e., an identifiable patient, an identifiable reporter, an adverse event/reaction, and a suspect or interacting drug

Triage

Under this step, the case processor prioritizes all the incoming reports as per the receipt date, seriousness, causality, and expectedness assessment in the triage step.

Once the cases are triaged, they can be processed as per the priority assigned to each case.

This is done to ensure that cases which need expedited reporting can be processed and submitted to the regulatory authorities within timelines.

Standard regulatory timelines are as follows:

As per the International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use E2A guideline, fatal or life-threatening unexpected ADRs should be notified to regulatory agencies within 7 days and all other serious, unexpected ADRs **within 15 days**.

According to the General Reporting Requirements of the Individual Case Safety Reports (European Medicines Agency), all serious ADRs should be notified **within 15 days** and all non-serious ADRs should be notified **within 90 days**.

Duplicate search

Due to greater awareness, stringent regulations, and multiple reporting sources, the phenomenon of duplicate reports is common.

Every safety management software has a facility to identify and delete duplicates.

The case processor performs duplicate search for every document received with certain characteristics of a case (gender, age or date of birth, reaction, clinical trial protocol, country, reference number, etc.) to identify whether the same report has been databased previously or not.

This action is of significance for further processing of the case.

The duplicate could actually be follow-up information that could alter the seriousness and hence reporting timeline of the case. Missed out duplicates could send misleading information to signal detection systems.

Book-in and registration

Once the case processor has completed duplicate search for a case in database, there are two ways of processing a case. They are as follows:

Upon positive duplicate search, i.e., when the processor finds a case in the database, the processor can add a follow-up to the existing case and process the information

Upon negative duplicate search, i.e., when the processor does not find any case in the database, the case processor will create a new case to process the information.

Complete data entry

Once the case has been triaged, the case processor will perform the complete data entry.

The case processor will enter all the details from the source documents into the database, which include reporter information, patient information, patient medical history, laboratory details, family and surgical history, clinical course, and treatment details of the patient.

Coding the adverse events and drugs

This step involves coding of the adverse events in the Medical Dictionary for Regulatory Activities (MedDRA).

The case processor can auto code the event term (auto code: when the term has an exact match in MedDRA) or code it manually if the exact match is not available in MedDRA.

The principle is to be talking the same language across countries, companies, and regulatory bodies.

The case processor will characterize the drug as provided by the primary reporter suspect/concomitant/interacting and will code them in the drug dictionaries which have been configured in the company database. MedDRA is a single standardized international medical dictionary which can be used for regulatory communication and evaluation of data pertaining to medicinal products for human use.

A suspect drug is the drug associated with the ADR as determined by the initial reporter.

The associate needs to code the company drugs in company product dictionary and the noncompany drugs in the WHO Drug Dictionary (WHO DD).

The WHO DD is an international classification of medicines created by the WHO Program for International Drug Monitoring and managed by the Uppsala Monitoring Centre. The associate will also add other information for the suspect drug which includes indication, therapy details, and action taken with the suspect drug in response to the event.

Causality assessment

A causality assessment is the relationship between the drug treatment and the occurrence of an adverse event.

The case processor enters the reporter's causality assessment (whether event is related to the drug: yes/no/not reported) in this section.

If the reporter has not provided the causality assessment, the case processor can also assess the case causality. Various approaches have been developed for the structured determination of the likelihood of a causal relationship between drug exposure and adverse events. The case processor can apply the systems which are largely based on the following considerations:

The chronology or association in time (or place) between drug administration and the event

Current knowledge of nature and frequency of adverse reactions due to the suspect molecule, or the pharmacology of the drug

Medical or confounders based on signs and symptoms, laboratory tests, pathological findings and mechanism of action

Likelihood or exclusion of other causes for the same adverse events, often the disease condition or concomitant medication

Expectedness assessment

The case processor goes through the reference safety documents to check whether the ADR which is reported is listed/expected in the reference safety information (RSI) document or not.

The RSI is a list of medical terms detailing the Serious Adverse Reactions (SARs) that are expected for the investigational medicinal products (IMP) and is used by investigators as a reference point when assessing a SAR to determine whether it is a SUSAR.

Various sections of the RSI need to be carefully checked, especially the undesirable effect section. An “unexpected” adverse reaction is one, the nature/severity/specificity/outcome of which is not consistent with the information in the relevant source documents

Case narrative

The case processor describes the story of the entire case in this section.

The case processor can use predefined templates for writing narratives as per the customer requirements.

The associate provides a summary of the events to the readers, ensure completeness, chronology, and sufficient detail in a narrative so that the reader can come to a conclusion.

The inputs given by the medical reviewers can also be incorporated in the safety narrative of the case.

Self-quality check

The associate does a self-quality check for the case before the case is pushed to the next workflow.

The associate ensures the completeness and accuracy of the information entered into the database for all the cases.

This is a very crucial step, especially when the associate is working on Suspected Unexpected Serious Adverse Reaction or fatal cases.

CASE PROCESSING SIGNIFICANCE

For the case processor, every case is different, even within the same study.

In fatal cases, the case processor will have to check many things such as the cause of death and autopsy details,

whereas in other scenarios, the processor will have to check adverse events of special interests and important medical events and report the same to the stakeholders.

The case processor should ensure that there are no grammatical/spelling errors in the case, while routing it to the next workflow.

The case processor shares the best practices, which help other associates to do their job efficiently.

It may seem that the case processor's job could become mundane after a few months; however, for a case processor, every day and every case is a new experience. There is continuous learning from each and every case which will have new information, thereby increasing the learning curve

Good Pharmacovigilance Practice – ICH GCP Guidelines For Pharmacovigilance



E2A: Clinical safety data management: Definitions and standards for expedited reporting

What is e2b in pharmacovigilance? E2B: Electronic reporting of adverse events: Implements electronic reporting R3 E-ICSR Implementation Guide I Link CIOMS I to E2B Guide

E2C: Periodic benefit– risk evaluation report (PBRER): Format and content of safety updates for reporting at intervals to regulatory authorities

E2D: Post-approval safety data management: Definitions and standards for expedited reporting: Guidance for gathering and reporting information

E2E: Pharmacovigilance planning: Helps to make PMS plan, especially in early Phase IV

E2F: Development Safety Update Report (DSUR): Standard for periodic reporting on drugs

M1: Medical dictionary for Regulatory activities (MedDRA)

M4: The Common Technical Document (CTD): Formatting of reports to regulatory bodies (M4 Organization, M4Quality, M4Safety, M4Efficacy)

CIOMS I: The international reporting form (links with ICH E2B)

CIOMS II: Periodic safety update reports manual

CIOMS III: Core data sheets

CIOMS IV: Benefit-risk assessments

CIOMS V: Practical issues in Pharmacovigilance

CIOMS VI: Clinical trial safety data

CIOMS VII: Development safety update reports

CIOMS VIII: Practical Aspects of Signal Detection in Pharmacovigilance

Global
Pharmacovigilance
Laws And Regulations
– IAOCR Directory

Local Pharmacovigilance Regulatory
Bodies

Canada – Health Canada ~2%

USA★ – FDA: Food and Drug
Administration ~33%

European Union★ – EMA: European
Medicines Agency ~17%

JAPAN★ – PMDA: Ministry of Health,
Labour and Welfare ~12%

Korea (South) – Korean Food and Drug
Administration (KFDA) ~1%

Pharmacovigilance Jobs

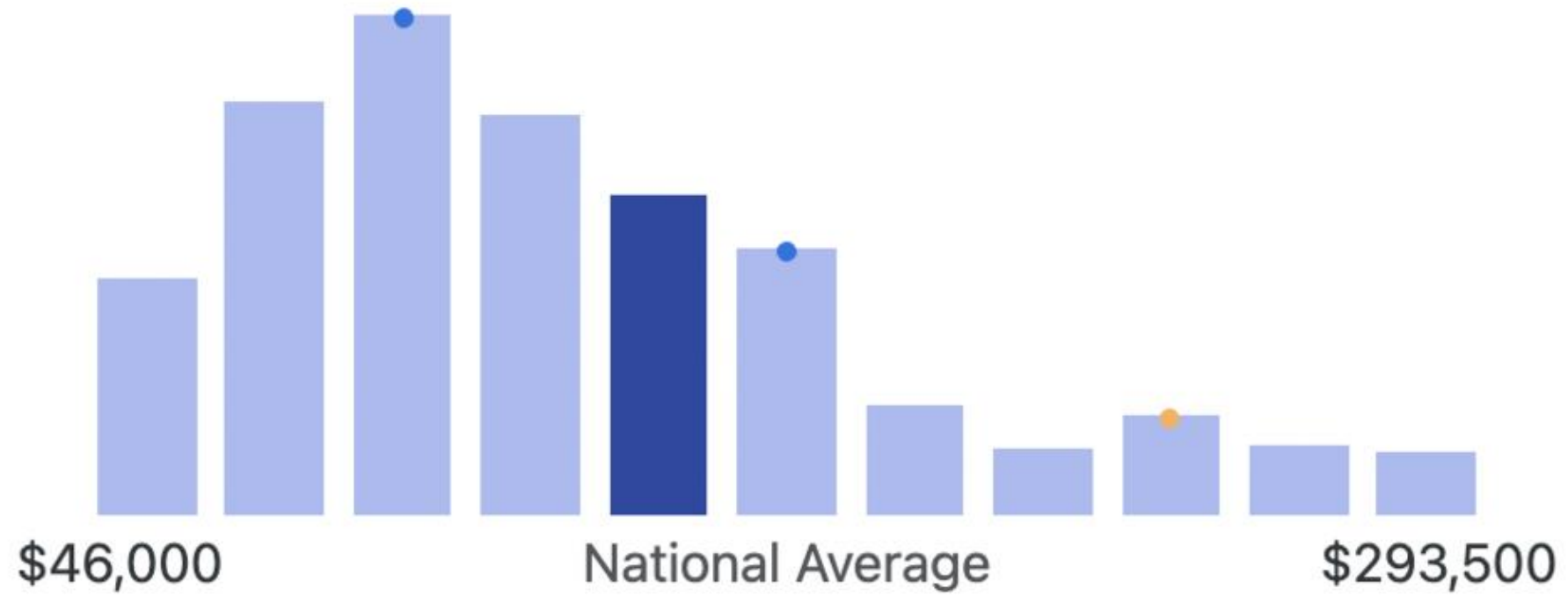
Common titles for pharmacovigilance officer jobs include:

Drug Safety Officer, Pharmacovigilance Officer, PV Officer, Drug Safety Quality Assurance Officer, Clinical Safety Manager, Global Regulatory Affairs & Safety Strategic Lead, Medical Safety Physician/MD/MBBS or IMG, Risk Management and Mitigation Specialist, Clinical Scientist Advisor in Pharmacovigilance and Drug Surveillance, Drug Regulatory Affairs Professional with PV Knowledge and Experience, Senior Regulatory Affairs Associate with PV Expertise and Knowledge, Senior Clinical Trial Safety Associate or Specialist, MedDRA Coder (Medical Dictionary for Regulatory Activities), PV Compliance Reviewer or Auditor, GCP (Good Clinical Practices) Specialist with PV Knowledge and experience.

Pharmacovigilance Jobs Entry Level

Remote Pharmacovigilance Jobs

Salary



Case Study PMS

Post-Marketing Surveillance Study of Aripiprazole in Patients With Autism

<https://beta.clinicaltrials.gov/study/NCT03179787?tab=results>

Case Study PV

CAR-T Cells

The case study presented at World Drug Safety Congress US (Boston) involved an exploratory analysis of CAR-T Cell therapies. The study involved searching the literature from January 2022 to August 2022 using biogit MLM-AI

<https://www.biogit.com/post/pharmacovigilance-case-studies-from-the-scientific-literature>

Conclusion

PMS is an essential tool that helps to correlate the strength of the exposed drug with that of the adverse events, thereby painting a clearer picture with regard to the types of positive or negative effects that a drug may threaten to pose, over a prolonged period of administration.

US or EU market standards are quickly harmonizing and showing more focus on PMS and risk management. Any manufacturer to sell their medical device in US & EU must act quickly to understand PMS requirements, to develop a plan and to build a strong compliant PMS system.

PMS is a part of good business strategy, it alerts manufacturer to react quickly on potential flaws in the design or implementation of the device. This effective PMS system with continuous improvements and outcomes build up a trust to the end user and shows manufacturer's commitment towards user safety.

The aim of pharmacovigilance is to protect public health by identifying, evaluating and minimizing safety issues to ensure that the overall benefits of medicines outweigh the risks.

The PV data processing cycle starts with data collection in computerized systems followed by complete data entry which includes adverse event coding, drug coding, causality and expectedness assessment, narrative writing, quality control, and report submissions followed by data storage and maintenance.

The global pharmacovigilance market size was valued at USD 7.80 billion in 2022 and is expected to expand at a compound annual growth rate (CAGR) of 10.3% from 2023 to 2030.

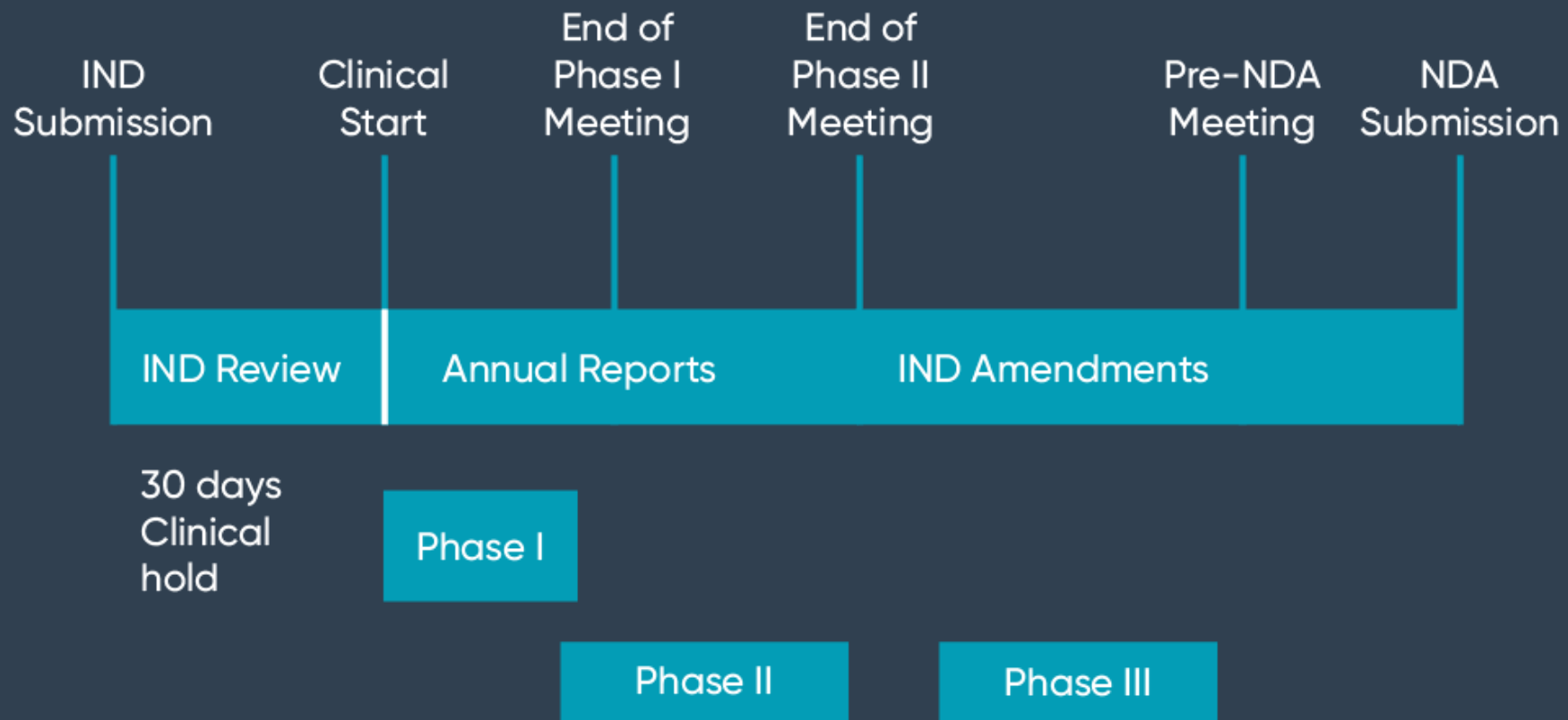


NDA Application & Approval

Ms. Sonali Correa

Ph.D., M.Sc.

Drug Development Timeline



Pre-NDA Review Meeting

- Prior to the submission of a New Drug Application (NDA), the FDA and sponsor meet to discuss and agree upon the content and formatting of the NDA.
- The sponsor should leave the meeting with a good understanding of their readiness for filing and their likelihood of approval

Definition

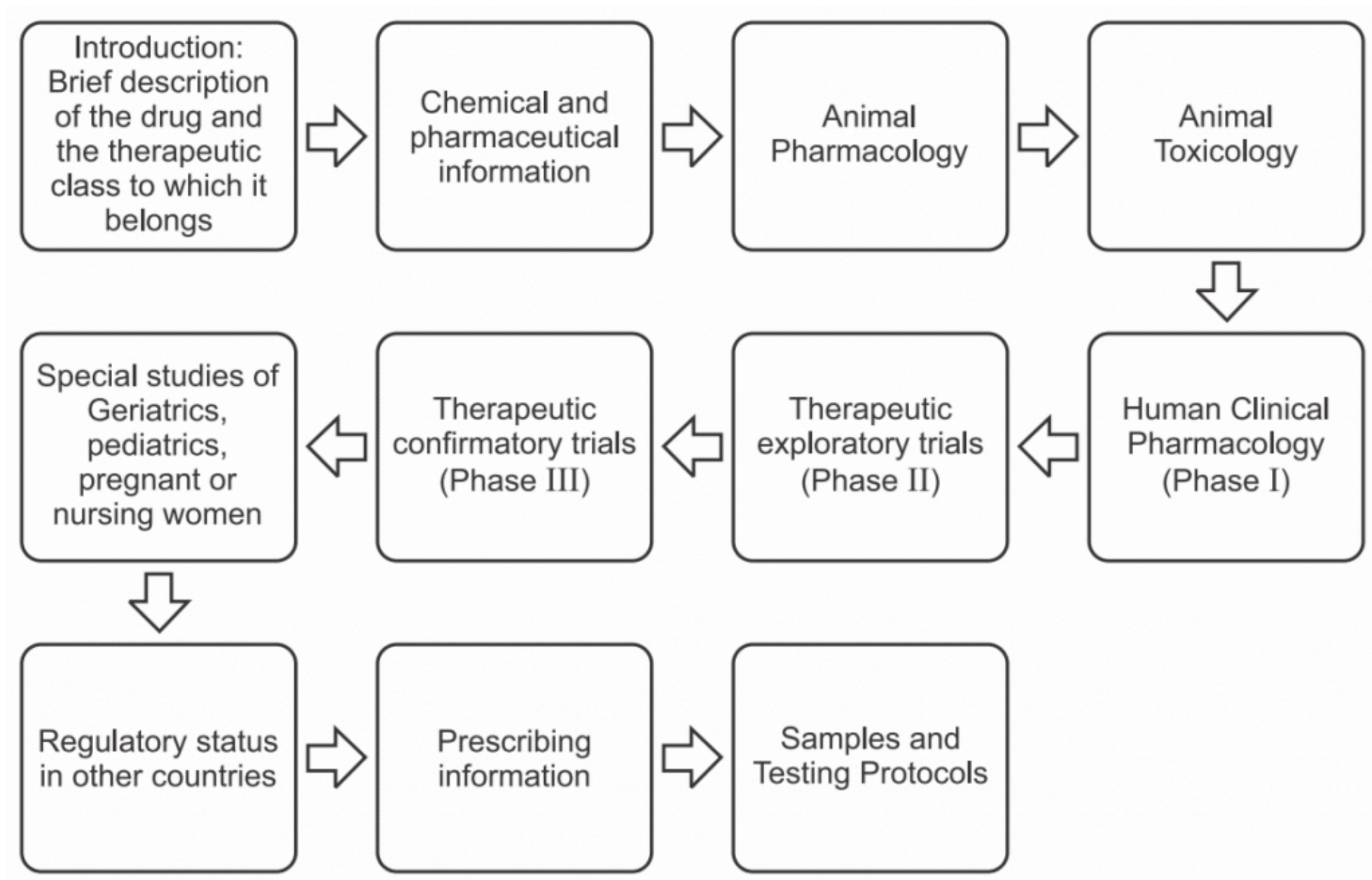
- NDA application is the vehicle through which drug sponsors formally propose that the FDA approve a new pharmaceutical for sale and marketing in the U.S
- For decades, the regulations and control of new drugs in the US has been based on the New Drug Application (NDA)
- Since 1938, every new drug has been the subject of an approved NDA before U.S. commercialization.
- The data gathered during the animal studies and human clinical trials of an Investigational New Drug (IND) become part of the NDA.
- The documentation required in an NDA is supposed to tell the **drug's whole story**, including what happened during the clinical tests, what the ingredients of the drug are, the results of the animal studies, how the drug behaves in the body, and how it is manufactured, processed and packaged.

Objective

Whether the drug is safe and effective in its proposed use(s), and whether the benefits of the drug outweigh the risks.

Whether the drug's proposed labelling (package insert) is appropriate, and what it should contain.

Whether the methods used in manufacturing the drug and the controls used to maintain the drug's quality are adequate to preserve the drug's identity, strength, quality, and purity.



Drug information required in a New Drug Application (NDA)



Kalkine Image (Source: FDA)

Guidance Documents for NDAs

Guidance documents represent the Agency's current thinking on a particular subject.

These documents are prepared for FDA review staff and applicants/sponsors to provide guidelines to the processing, content, and evaluation/approval of applications and also to the design, production, manufacturing, and testing of regulated products.

They also establish policies intended to achieve consistency in the Agency's regulatory approach and establish inspection and enforcement procedures.

Bioavailability and Bioequivalence Studies Submitted in NDAs or INDs and General Considerations

Changes to an Approved NDA or ANDA

Changes to an Approved NDA or ANDA: Questions and Answers

Container Closure Systems for Packaging Human Drugs and Biologics

Format and Content of the Microbiology Section of an Application,

Format and Content of the Clinical and Statistical Sections of an Application

Summary for New Drug and Antibiotic Applications--Format and Content of the Summary for New Drug and Antibiotic Applications

Formatting, Assembling and Submitting New Drug and Antibiotic Applications,

GUIDELINE FOR SUBMITTING SUPPORTING DOCUMENTATION IN DRUG APPLICATIONS FOR THE MANUFACTURE OF DRUG PRODUCTS

NDAs: Impurities in Drug Substances

Format and Content of the Human Pharmacokinetics and Bioavailability Section of an Application

Format and Content of the Nonclinical Pharmacology/Toxicology Section of an Application

Providing Clinical Evidence of Effectiveness for Human Drug and Biological Products

Drug Master Files: Guidelines

FDA IND, NDA, ANDA, or Drug Master File Binders

PET Drug Applications - Content and Format for NDAs and ANDAs – 2011

Bioavailability and Bioequivalence Studies Submitted in NDAs or INDs and General Considerations

- The Food and Drug Administration (FDA) is announcing the availability of a draft guidance for industry entitled “Bioavailability and Bioequivalence Studies Submitted in NDAs or INDs—General Considerations” (draft BA and BE guidance for NDAs).
- The draft guidance provides recommendations to sponsors and/or applicants planning to include bioavailability (BA) and bioequivalence (BE) information for drug products in investigational new drug applications (INDs), new drug applications (NDAs), and NDA supplements.
- This draft guidance revises those parts of the March 2003 guidance entitled “Bioavailability and Bioequivalence Studies for Orally Administered Drug Products—General Considerations” relating to BA and BE studies for INDs, NDAs, and NDA supplements.

Changes to an Approved NDA or ANDA

- This guidance provides recommendations to holders of new drug applications (NDAs) and abbreviated new drug applications (ANDAs) who intend to make post approval changes in accordance with section 506A of the Federal Food, Drug, and Cosmetic Act (the Act) and § 314.70 (21 CFR 314.70).
- The guidance covers recommended reporting categories for post approval changes for drugs other than specified biotechnology and specified synthetic biological products. It supersedes the guidance of the same title published November 1999.

Recommendations are provided for post approval changes in:

- | | |
|---------------------------------|-------------------------------|
| (1) components and composition, | (5) container closure system, |
| (2) manufacturing sites, | (6) labeling, |
| (3) manufacturing process, | (7) miscellaneous changes |
| (4) specifications, | (8) multiple related changes |

Changes to an Approved NDA or ANDA: Questions and Answers

- This document provides questions and answers relating to the guidance on Changes to an Approved NDA or ANDA (the guidance).
- The questions are based on those posed to CDER by applicants. The questions and answers are presented using subject headings that correspond to the table of contents in the guidance.

Container Closure Systems for Packaging Human Drugs and Biologics

- This document is intended to provide guidance on general principles for submitting information on packaging materials used for human drugs and biologics.
- This guidance supersedes the FDA Guideline for Submitting Documentation for Packaging for Human Drugs and Biologics, issued in February 1987 and the packaging policy statement issued in a letter to industry dated June 30, 1995 from the Office of Generic Drugs.
- This guidance is not intended to describe the information that should be provided about packaging operations associated with drug product manufacture.
- Approaches which differ from those described in this guidance may be followed, but the applicant is encouraged to discuss significant variations in advance with the appropriate CDER chemistry review staff or CBER review staff.
- This is to prevent applicants or sponsors from spending unnecessary time and effort in preparing a submission that the FDA may later determine to be unacceptable.

Format and Content of the Microbiology Section of an Application

- Antimicrobial drugs differ from other classes of drugs in that they are intended to affect microbial, rather than patient, physiology.
- For this reason, the technical reports of in vivo and in vitro effects on microorganisms are critical for establishing effectiveness.

Format and Content of the Clinical and Statistical Sections of an Application

- This guideline is intended to assist an applicant in presenting the clinical and statistical data required as part of an application under 21 CFR 314.50.
- The guideline describes an acceptable format for organizing the clinical and statistical sections and presenting the clinical and statistical information and accompanying statistical documentation of a clinical trial.
- With respect to documenting the results of individual studies, the guideline describes a fully integrated clinical and statistical report rather than two separate reports.
- Paragraphs (d)(5), (d)(6), and (f) of 21 CFR 314.50 provide a general outline for this submission

Summary for New Drug and Antibiotic Applications-- Format and Content of the Summary for New Drug and Antibiotic Applications

- This guideline is intended to assist drug firms in preparing the summary to a new drug application required under 21 CFR 314.50(c).
- The summary section is to be included in the archival copy and in each technical section of the review copy of the application.
- The guideline is issued under 21 CFR 10.90.
- A person may, but is not required to, rely upon the guideline in preparing the summary section of an application.
- When a different approach is chosen, a person is encouraged to discuss the matter in advance with FDA to prevent the expenditure of money and effort on preparing a submission that may later be determined to be unacceptable.
- Each full application is required under 21 CFR 314.50(c) to contain summary, ordinarily 50 to 200 pages in length, that integrates all of the information in the application and provides reviewers in each review area, and other agency officials, with a good general understanding of the drug product and of the application.
- The summary should discuss all aspects of the application and should be written in approximately the same level of detail required for publication in, and meet the editorial standards generally applied by referred scientific and medical journals.

Formatting, Assembling and Submitting New Drug and Antibiotic Applications

- The Federal Food, Drug, and Cosmetic Act (the act) provides that a new drug may not be introduced or delivered for introduction into interstate commerce unless the Food and Drug Administration (FDA) has approved a new drug application (NDA) for it (21 U.S.C. 355).
- FDA approves an application for a new drug only if the applicant demonstrates by adequate scientific evidence that the drug is safe and by substantial evidence that the drug is effective for the conditions prescribed, recommended, or suggested in proposed labeling for the product.
- The act defines substantial evidence of effectiveness as "evidence consisting of adequate and well-controlled investigations, including clinical investigations "that are conducted by qualified experts. Additionally, to obtain approval, an applicant is required to show that the methods used in, and the facilities and controls used for, the manufacturing, processing, and packaging of the drug are adequate to preserve its identity, strength, quality, and purity.

NDAs: Impurities in Drug Substances

- This guidance recommends that applicants refer to Q3A Impurities in New Drug Substances (January 4, 1996, 61 FR 371) when seeking **guidance on identification, qualification, and reporting of impurities in drug substances that are not considered new drug substances**.
- Although Q3A was developed by the ICH to provide guidance on the information that should be provided in a NDA in support of impurities in new drug substances that are produced by chemical syntheses, the Agency believes that the guidance provided there on identification, qualification, and reporting of impurities should also be considered when evaluating impurities in drug substances produced by chemical syntheses that are not considered new drug substances
- This recommendation applies to applicants planning to submit NDAs and supplements for changes in drug substance synthesis or process.
- It also applies to holders of Type II drug master files (DMFs) that support such applications. Applicants should note that this recommendation would not apply to DMFs cited in an NDA or supplement if the DMF information has been deemed acceptable for that dosage form, route of administration, and daily intake prior to the publication of the final version of this guidance

Format and Content of the Human Pharmacokinetics and Bioavailability Section of an Application

- Biopharmaceutic studies are required by Part 320, Title 21, and by the Waxman-Hatch Amendments to the Federal Food, Drug and Cosmetic Act (the act).
- These amendments define bioequivalence and require generic drugs to show bioequivalence to the previously approved dosage form.
- The regulations define important biopharmaceutic terms and establish acceptable procedures for determining the bioavailability of drug products.
- The new drug application (NDA) rewrite regulations require a separate biopharmaceutic review section in the NDA (21CFR 314). This guideline is intended to assist applicants to prepare the biopharmaceutics section of the NDA.

Format and Content of the Nonclinical Pharmacology/Toxicology Section of an Application

- This guideline describes an acceptable format for organizing and presenting the pharmacology and toxicology data required under 21 CFR 314.50(d)(2), and any related data, in the nonclinical section of the application.
- These recommendations pertain only to organization of existing data.
- This guideline is not intended to affect documentation required by the Good Laboratory Practices (GLP) reporting regulations under 21 CFR 58.185, nor does it describe the specific study requirements for particular therapeutic uses or regimens: these are addressed in other guidelines.

Providing Clinical Evidence of Effectiveness for Human Drug and Biological Products.

- This document is intended to provide guidance to applicants planning to file new drug applications (NDAs), biologics license applications (BLAs), or applications for supplemental indications on the evidence to be provided to demonstrate effectiveness.
- This document is also intended to meet the requirements of subsections 403(b)(1) and (2) of the Food and Drug Administration Modernization Act (the Modernization Act) of 1997 for human drug and biological products (P.L. 105-115).
- Subsection 403(b)(1) directs FDA to provide 3 guidance on the circumstances in which published matter may be the basis for approval of a supplemental application for a new indication.
- Section III of this guidance satisfies this requirement by describing circumstances in which published matter may partially or entirely support approval of a supplemental application. Subsection 403(b)(2) directs FDA to provide guidance on data requirements that will avoid duplication of previously submitted data by recognizing the availability of data previously submitted in support of an original application to support approval of a supplemental application.
- Section II of this guidance satisfies this requirement by describing a range of circumstances in which related existing data, whether from an original application or other sources, may be used to support approval of a supplemental application.

Guideline for Drug Master Files (DMF)

- A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs.
- The submission of a DMF is not required by law or FDA regulation. Drug Master Files are provided for in 21 CFR 314.420
- A DMF is submitted solely at the discretion of the holder.
- The information contained in the DMF may be used to support an Investigational New Drug Application (IND), a New Drug Application (NDA), an Abbreviated New Drug Application (ANDA), another DMF, an Export Application, or amendments and supplements to any of these.
- A DMF is NOT a substitute for an IND, NDA, ANDA, or Export Application.
- It is not approved or disapproved. Technical contents of a DMF are reviewed only in connection with the review of an IND, NDA, ANDA, or an Export Application.

PET Drug Applications - Content and Format for NDAs and ANDAs_2011

- On December 9, 2009, FDA issued a final current good manufacturing practice (CGMP) regulation for the production of positron emission tomography (PET) drugs.
- Under the requirements of section 121 of the Food and Drug Administration Modernization Act (FDAMA), within two years following this publication date, an NDA or ANDA must be submitted for any PET drug marketed for clinical use in the United States.
- This guidance is intended to assist applicants in preparing new drug applications (NDAs) or abbreviated new drug applications (ANDAs) for fludeoxyglucose F 18 injection, ammonia N 13 injection, and sodium fluoride F 18 injection used in PET imaging for the indications cited in Section III (below).
- FDA approval of an NDA or ANDA will make it possible to market these PET drugs for clinical use according to the requirements of the Federal Food, Drug, and Cosmetic Act (the Act).

NDA forms

- Form FDA-356h. Application to Market a New Drug, Biologic, or An Antibiotic Drug For Human Use
- Form FDA-356h instructions
- Form FDA-3397. User Fee Cover Sheet
- Form FDA-3331. New Drug Application Field Report
- Impurity of Drug substances
- Required specifications for NDA, BLA and ANDA drug master files binders
- Refusal to file

Form 356h: Application to Market a New Drug, Biologic, or An Antibiotic Drug For Human Use

- To apply to market a new drug, biologic, or an antibiotic drug for human use, you need to complete Form 356h.
- Form 356h includes information such as applicant information, proposed indication, establishment information, and more
- Filling form 356h
- For each site, please include the establishment name, address, registration (FEI) number, Master File (MF; Drug Master File (DMF) or Biologic Master File (BMF)) number (for facilities used under a MF), and establishment DUNS number. Indicate whether or not the establishment is new to the application, if applicable.

Form FDA 3397

- Form FDA 3397, the PDUFA user fee cover sheet, is designed to provide the minimum necessary information to determine whether a fee is required for review of an application, to determine the amount of the fee required, and to help FDA track payments
- <https://www.fda.gov/industry/fda-user-fee-programs/prescription-drug-user-fee-amendments#:~:text=Form%20FDA%203397%2C%20the%20PDUFA,to%20help%20FDA%20track%20payments>.

Form FDA-3331

- Use Form FDA 3331a to submit **Field Alert Reports (FARs) for CDER- or CBER-regulated drug products that are approved under a new drug application (NDA) or abbreviated new drug application (ANDA).**

NDA and ANDA

- An Abbreviated New Drug Application (ANDA) contains data which is submitted to the FDA for the review and potential approval of a **generic drug product**
- New Drug Application (NDA) is the application through which sponsors formally propose the approval of a **new pharmaceutical drug**.
- In 2020, the FDA approved or tentatively approved 948 ANDAs, including 72 first generics and only approved 53 novel drugs, either as new molecular entities (NMEs) under NDAs, or as new therapeutic biologics under Biologics License Applications (BLAs).
- Historically each year, there are around 20 times more generic drugs approved than novel drugs approved.

Approval Pathways: 505(b)(1) NDA, 505(j) ANDA, & 505(b)(2) NDA

- There are three approval pathways for NDAs and ANDAs which are: 505(b)(1) NDA, 505(j) ANDA, and 505(b)(2) NDA.
- These refer to the particular parts of Section 505 of the Federal Food, Drug, and Cosmetic (FD&C) Act which respectively covers the approval of innovator drugs, generic drugs, and drugs that share key similarities with approved drugs but differ in other potentially crucial ways.
- The approval pathways for non-innovator drugs are abbreviated and are meant to encourage a quicker time to market.
- They allow applicants to use existing knowledge instead of performing a full collection of safety and efficacy studies themselves, as would be required for an innovator drug.

Abbreviated New Drug Applications (ANDAs)

- When a Sponsor submits a generic drug for marketing approval, they submit an ANDA instead of a full 505(b)(1) NDA.
- Generic drug applications are called “abbreviated” because they do not require new pre-clinical (animal) and clinical (human) data to establish safety and effectiveness.
- In an ANDA, the applicant is claiming that their drug is a duplicate of an already-approved drug.
- Within the ANDA, the innovator drug, commonly referred to as the “Reference Listed Drug” (RLD), is specified. Because the FDA has already approved the RLD to be safe and effective, the goal of an ANDA is to demonstrate “sameness” with the RLD. Sameness is demonstrated via a bioequivalence assessment.
- According to Section 505(j)(8)(B)(i) of the FD&C Act, a generic drug is determined to be bioequivalent to the listed drug if the rate and extent of absorption of the generic drug do not show a significant difference from those of the RLD.

- Drugs approved under an ANDA must be therapeutically equivalent to the RLD.
- This means that if an ANDA-approved drug is swapped with the RLD, patients should experience the same clinical effect and safety profile.
- The active ingredient, dosage form, route of administration, and strength must all be the same, and the product labeling is usually the same.
- In addition, the sameness requirement also extends to the inactive ingredients in a generic drug product. Inactive ingredients that are not considered exception excipients (such as preservatives, buffers, and antioxidants) must be qualitatively and quantitatively (Q1/Q2) the same as those in the RLD.
- Because a generic drug is intended to act as a duplicate of the RLD, no new safety or efficacy studies are performed and only small confirmatory studies are allowed to support the ANDA.
- **If additional pre-clinical or clinical data are needed to support safety or efficacy, then the ANDA route is not appropriate and a 505(b)(2) NDA should be used.**

505(b)(2) New Drug Applications (NDAs)

- Section 505(b)(2) of the FD&C Act allows the FDA to approve marketing applications for non-innovator drugs that do not otherwise qualify for approval under a 505(j) ANDA.
- Under this statute, the approval may rely in whole or in part on published literature and/or on the FDA's findings of safety and/or effectiveness for a previously approved drug.
- Certain findings and data submitted in support of a 505(b)(2) NDA will stem from investigations that were not conducted by or on behalf of the applicant and for which the applicant has not obtained a right of reference.
- Changes to approved drugs that would require a 505(b)(2) NDA include differences in dosage form, strength, route of administration, formulation (i.e., excipients), dosing regimen, active ingredient (e.g., different salt or enantiomer), or indication (i.e., repurposed drugs).
- Applications for a new combination product would also be included, even if the individual active ingredients have been previously approved. Changing a drug from a prescription indication to an over-the-counter indication would also require a 505(b)(2) NDA.

- Additional information may be found in the FDA draft guidance, "Applications Covered by Section 505(b)(2).“
- Because the **goal of a 505(b)(2) application** is to present a complete body of evidence to demonstrate that the drug is safe and effective, if there are aspects of the drug that cannot be adequately supported by published literature, then the applicant is responsible for conducting appropriate investigations to bridge the formulations.
- These investigations may include nonclinical or clinical studies or a combination of both.
- Data from these bridging investigations are then submitted to the FDA for review as part of the NDA. It is important to develop a strong strategy for 505(b)(2) approval. The NDA review includes:
 - Medical Review(s)
 - Chemistry Review(s)
 - Environmental Assessment
 - Pharmacology Review(s)
 - Statistical Review(s)
 - Microbiology Review(s)
 - Clinical Pharmacology/Biopharmaceutics Review(s)
 - Risk Assessment and Risk Mitigation Review(s)

Code Of Federal Regulations (CFR)

- The final regulations published in the Federal Register (daily published record of proposed rules, final rules, meeting notices, etc.) are collected in the CFR.
- Section 21 of the CFR contains all regulations pertaining to food and drugs. The regulations document all actions of all drug sponsors that are required under Federal law.
- **21CFR Part 314 - Applications for FDA Approval to Market a New Drug or an Antibiotic Drug.**

CDER's Manual of Policies and Procedures (MaPPs)

- These documents are approved instructions for internal practices and procedures followed by CDER staff to help standardize the new drug review process and other activities.
- MaPPs define external activities as well.
- All MaPPs are available for the public to review to get a better understanding of office policies, definitions, staff responsibilities and procedures.
- MaPPS of particular interest to NDA applicants:
 - Review of the Same Supplemental Change to More than One NDA or ANDA in More Than One Review Division
 - NDAs and BLAs: Filing Review Issues
 - Action Packages for NDAs and Efficacy Supplements
 - Refusal to Accept Application for Filing From Applicants in Arrears
 - Requesting and Accepting Non-Archivable Electronic Material for CDER Applications

Prescription Drug User Fee Act (PDUFA)

- On November 21, 1997, The President signed the Food and Drug Administration Modernization Act of 1997.
- This legislation includes authorization for FDA to continue to collect three types of user fees from applicants who submit certain new drug and biological product applications.
- FDA was first authorized to collect user fees under the Prescription Drug User Fee Act (PDUFA) of 1992.
 - Prescription Drug User Fee Act Related Documents

Advisory Committees

- Advisory committees provide independent advice and recommendations to the FDA on scientific and technical matters related to the development and evaluation of products regulated by the Agency.
- CDER requests advice from advisory committees on a variety of matters, including various aspects of clinical investigations and applications for marketing approval of drug products.
- Committee members are scientific experts such as physician-researchers and statisticians, as well as representatives of the public, including patients.
- Although the committees provide recommendations to the Agency, final decisions are made by FDA.
- **FDA Advisory Committees**
- **CDER Advisory Committees**
- **CFR 21 Part 14 Public Hearing Before a Public Advisory Committee.**
 - Detailed description of advisory committees from the Code of Federal Regulations.

- **Guidance for Industry: Advisory Committees.**

- Includes information on membership, conflict of interest, scheduling, and action on recommendations.

- **Advisory Committee Meeting Calendar.**

- Several dates have been set aside by CDER advisory committees for possible future meetings. The subject matter and location of the meetings (if they are held) will be published in the Federal Register in the month prior to the meeting date

Originator (NDA) versus Generic (ANDA) Review Process Requirements

NDA Requirements

1. Labelling
2. Pharmacology/Toxicology
3. Chemistry
4. Manufacturing
5. Controls
6. Microbiology
7. Inspection
8. Testing
9. Animal Studies
10. Clinical Studies
11. Bioavailability

ANDA Requirements

1. Labelling
2. Pharmacology/Toxicology
3. Chemistry
4. Manufacturing
5. Controls
6. Microbiology
7. Inspection
8. Testing
9. Bioequivalence



Requirements for NDA

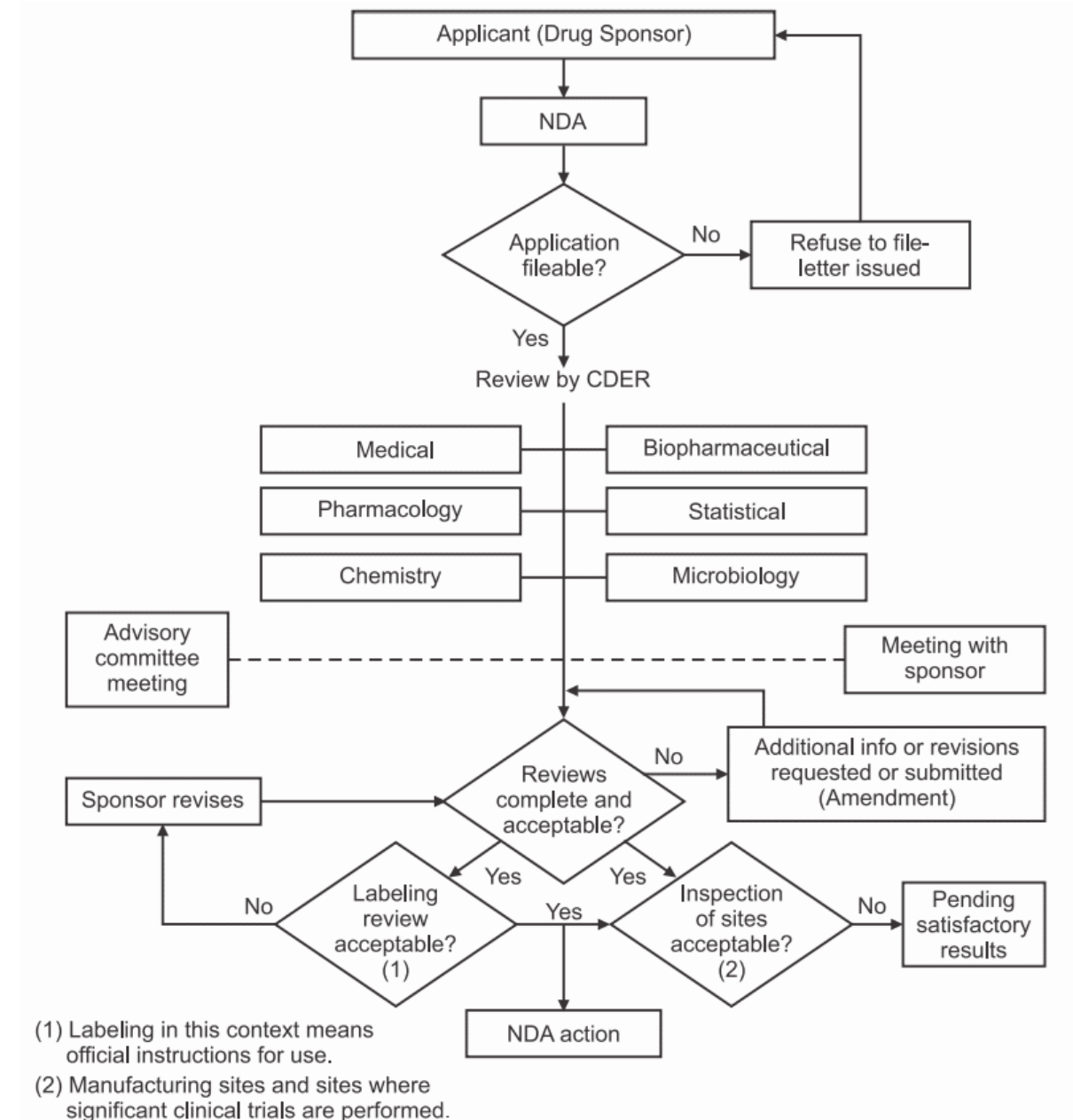
- Format requirement: No of copies :2
- **1) Archival copy**
 - Archival copy is a complete copy of an application submission.
 - -Reference copy for FDA (i.e. retained by FDA)
 - -Locate information not contained in review copy.
- **2) Review copy:**
 - - Review copy of an application is divided into 5 or 6
 - containing technical and scientific information separately
 - bound.
 - - It contains copy of cover letter, application form, overall
 - summary ,index ,specific review section.

- 1) A copy of the FDA cover letter.
- 2) A copy of the application form.
- 3) A copy of index to the entire application
- 4) A copy of the overall summary ,
- 5) A copy of a reference or authorisation letter to access NDAs , DMFs , etc

DOCUMENTS	FOLDER COLOUR	FORM NUMBER
Archival copy	Light blue	FD 2626
Chemistry,manufacturing, and controls section	Red	FD 2626a
Non-Clinical pharmacology and toxicology section	Yellow	Fd 2626b
Human pharmacokinetics and bioavailability section	Orange	FD 2626c
Microbiology section	White	Fd 2626d
Clinical data section	Light brown	Fd 2626e
Statistical section	Green	Fd 2626f
Field copy	Maroon	Fd 2626h

NDA Review process

- On submission of application, the FDA has **60 days to conduct a preliminary review** which will assess whether the NDA is “**sufficiently complete to permit a substantive review**”.
- If everything is found to **be acceptable**, the FDA will decide if the NDA will get a **standard or accelerated review** and communicate the acceptance of the application and their review choice in another communication known as **the 74-day letter**



- There really is no predicting how eventful your review will be.
- Throughout the course of the Agency's review, there are likely to be issues that come up, but the number and significance of these requests will vary from application to application.
- Ensure that all of the components of your NDA were accurate and complete, and you reviewed any previous correspondence with the Agency to make sure that all issues were addressed.
- FDA may request clarification on where information resides in your NDA, clarification on information you already provided, or they may request additional analyses.
- Should also be prepared for an inspection of the facility or facilities where the drug will be manufactured, which may result in additional requests.
- Regardless of how many information requests you receive during the NDA review process, it is important that you make sure you understand what FDA is asking for and that you respond with information that is clear, comprehensive and transparent.
- This will help the reviewers understand your response and minimize the need for additional follow-up questions or requests. It is the sponsor's responsibility to respond to these requests as soon as possible, as timely and complete responses help ensure that you don't inadvertently push out your action date.

- **Drug Labeling**

- The FDA reviews the proposed post-market labeling for the drug and ensures that it communicates appropriate information to both healthcare professionals and consumers/patients.

- **Facility Inspection**

- The FDA inspects the facilities where the drug will be manufactured to determine a facilities' compliance with regulations and the Food, Drug and Cosmetic Act.

- **Drug Approval**

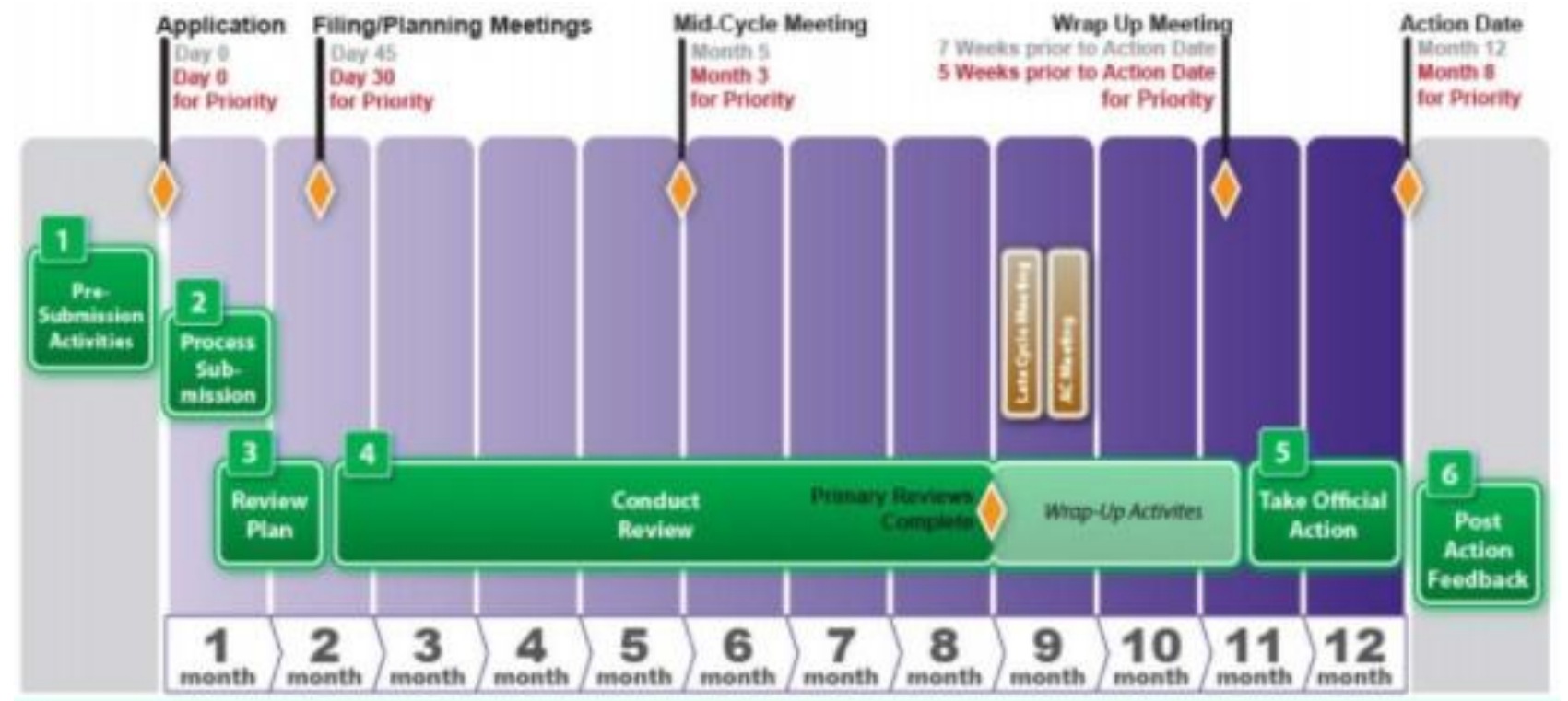
- The FDA approves the NDA, denies the NDA (very rare) or issues a complete response letter (CRL) that provides detailed information about why the application cannot be approved in its current form, as well as instructions regarding the additional steps the sponsor needs to take to achieve approval. Sponsors that receive CRLs have up to 12 months to address the deficiencies cited in the CRL.

Label negotiation during the NDA review process

- This very critical step takes place at the very end of the review, in many cases in the last two weeks.
- Tensions often run high during this period, since many parties have an interest in the final wording of the label.
- You should have developed strong, data-driven statements about your product prior to submission, but you need to be prepared to defend those statements now.
- Because timing and responsiveness is critical at this stage, all important members of your team should be available to review versions of labels and participate in calls on very short notice

NDA submission and review timeline

- The NDA review process is long and complex, so make sure you're aware of what to expect during each step so you have plenty of time to prepare in advance.
- The FDA has provided the following NDA timeline to help visualize the entire process.



Fast Track, Breakthrough Therapy, Accelerated Approval, Priority Review

- Speeding the availability of drugs that treat serious diseases are in everyone's interest, especially when the drugs are the first available treatment or if the drug has advantages over existing treatments. The Food and Drug Administration has developed four distinct and successful approaches to making such drugs available as rapidly as possible:
 - Priority Review
 - Breakthrough Therapy
 - Accelerated Approval
 - Fast Track

Fast Track

- Fast track is a process designed to facilitate the development, and expedite the review of drugs to treat serious conditions and fill an unmet medical need.
- The purpose is to get important new drugs to the patient earlier.
- Fast Track addresses a broad range of serious conditions.
- E.g. AIDS, Alzheimer's, heart failure and cancer are obvious examples of serious conditions. However, diseases such as epilepsy, depression and diabetes are also considered to be serious conditions.
- **Filling an unmet medical need is defined as providing a therapy where none exists or providing a therapy which may be potentially better than available therapy.**

- If there are available therapies, a fast track drug must show some advantage over available therapy, such as:
- Showing superior effectiveness, effect on serious outcomes or improved effect on serious outcomes
- Avoiding serious side effects of an available therapy
- Improving the diagnosis of a serious condition where early diagnosis results in an improved outcome
- Decreasing a clinical significant toxicity of an available therapy that is common and causes discontinuation of treatment
- Ability to address emerging or anticipated public health need

- A drug that receives Fast Track designation is eligible for some or all of the following:
- More frequent meetings with FDA to discuss the drug's development plan and ensure collection of appropriate data needed to support drug approval
- More frequent written communication from FDA about such things as the design of the proposed clinical trials and use of biomarkers
- Eligibility for Accelerated Approval and Priority Review, if relevant criteria are met
- Rolling Review, which means that a drug company can submit completed sections of its Biologic License Application (BLA) or New Drug Application (NDA) for review by FDA, rather than waiting until every section of the NDA is completed before the entire application can be reviewed. BLA or NDA review usually does not begin until the drug company has submitted the entire application to the FDA
- Fast Track designation must be requested by the drug company.
- Once a drug receives Fast Track designation, early and frequent communication between the FDA and a drug company is encouraged throughout the entire drug development and review process.

Breakthrough Therapy

- Breakthrough Therapy designation is a process designed to expedite the development and review of drugs that are intended to treat a serious condition and preliminary clinical evidence indicates that the drug may demonstrate substantial improvement over available therapy on a clinically significant endpoint(s).
- For purposes of Breakthrough Therapy designation, clinically significant endpoint generally refers to an endpoint that measures an effect on irreversible morbidity or mortality (IMM) or on symptoms that represent serious consequences of the disease:
- An effect on an established surrogate endpoint
- An effect on a surrogate endpoint or intermediate clinical endpoint considered reasonably likely to predict a clinical benefit (i.e., the accelerated approval standard)
- An effect on a pharmacodynamic biomarker(s) that does not meet criteria for an acceptable surrogate endpoint, but strongly suggests the potential for a clinically meaningful effect on the underlying disease
- A significantly improved safety profile compared to available therapy (e.g., less dose-limiting toxicity for an oncology agent), with evidence of similar efficacy

- A drug that receives Breakthrough Therapy designation is eligible for the following:
- All Fast Track designation features
- Intensive guidance on an efficient drug development program, beginning as early as Phase 1
- Organizational commitment involving senior managers
- FDA may suggest that the sponsor consider submitting a request if: (1) after reviewing submitted data and information (including preliminary clinical evidence), the Agency thinks the drug development program may meet the criteria for Breakthrough Therapy designation and (2) the remaining drug development program can benefit from the designation.
- A Breakthrough Therapy designation request should be received by FDA no later than the end-of-phase-2 meetings if any of the features of the designation are to be obtained.
- FDA will respond to Breakthrough Therapy designation requests within sixty days of receipt of the request.

Accelerated Approval

- When studying a new drug, it can sometimes take many years to learn whether a drug actually provides a real effect on how a patient survives, feels, or functions.
- A positive therapeutic effect that is clinically meaningful in the context of a given disease is known as “clinical benefit”. Mindful of the fact that it may take an extended period of time to measure a drug’s intended clinical benefit, in 1992 FDA instituted the Accelerated Approval regulations.
- These regulations allowed drugs for serious conditions that filled an unmet medical need to be approved based on a surrogate endpoint. Using a surrogate endpoint enabled the FDA to approve these drugs faster.
- A surrogate endpoint used for accelerated approval is a marker - a laboratory measurement, radiographic image, physical sign or other measure that is thought to predict clinical benefit, but is not itself a measure of clinical benefit.
- an intermediate clinical endpoint is a measure of a therapeutic effect that is considered reasonably likely to predict the clinical benefit of a drug, such as an effect on irreversible morbidity and mortality (IMM).

- The FDA bases its decision on whether to accept the proposed surrogate or intermediate clinical endpoint on the scientific support for that endpoint. Studies that demonstrate a drug's effect on a surrogate or intermediate clinical endpoint must be “adequate and well controlled” as required by the FD&C Act
- Using surrogate or intermediate clinical endpoints can save valuable time in the drug approval process.
- **For example**, instead of having to wait to learn if a drug actually extends survival for cancer patients, the FDA may approve a drug based on evidence that the drug shrinks tumors, because tumor shrinkage is considered reasonably likely to predict a real clinical benefit. In this example, an approval based upon tumor shrinkage can occur far sooner than waiting to learn whether patients actually lived longer. The drug company will still need to conduct studies to confirm that tumor shrinkage actually predicts that patients will live longer. These studies are known as phase 4 confirmatory trials.

- Where confirmatory trials verify clinical benefit, FDA will generally terminate the requirement.
- Approval of a drug may be withdrawn or the labeled indication of the drug changed if trials fail to verify clinical benefit or do not demonstrate sufficient clinical benefit to justify the risks associated with the drug (e.g., show a significantly smaller magnitude or duration of benefit than was anticipated based on the observed effect on the surrogate).

Priority Review

- Prior to approval, each drug marketed in the United States must go through a detailed FDA review process.
- In 1992, under the Prescription Drug User Act (PDUFA), FDA agreed to specific goals for improving the drug review time and created a two-tiered system of review times – Standard Review and Priority Review. A Priority Review designation means FDA's goal is to take action on an application within 6 months (compared to 10 months under standard review).
- A Priority Review designation will direct overall attention and resources to the evaluation of applications for drugs that, if approved, would be significant improvements in the safety or effectiveness of the treatment, diagnosis, or prevention of serious conditions when compared to standard applications.

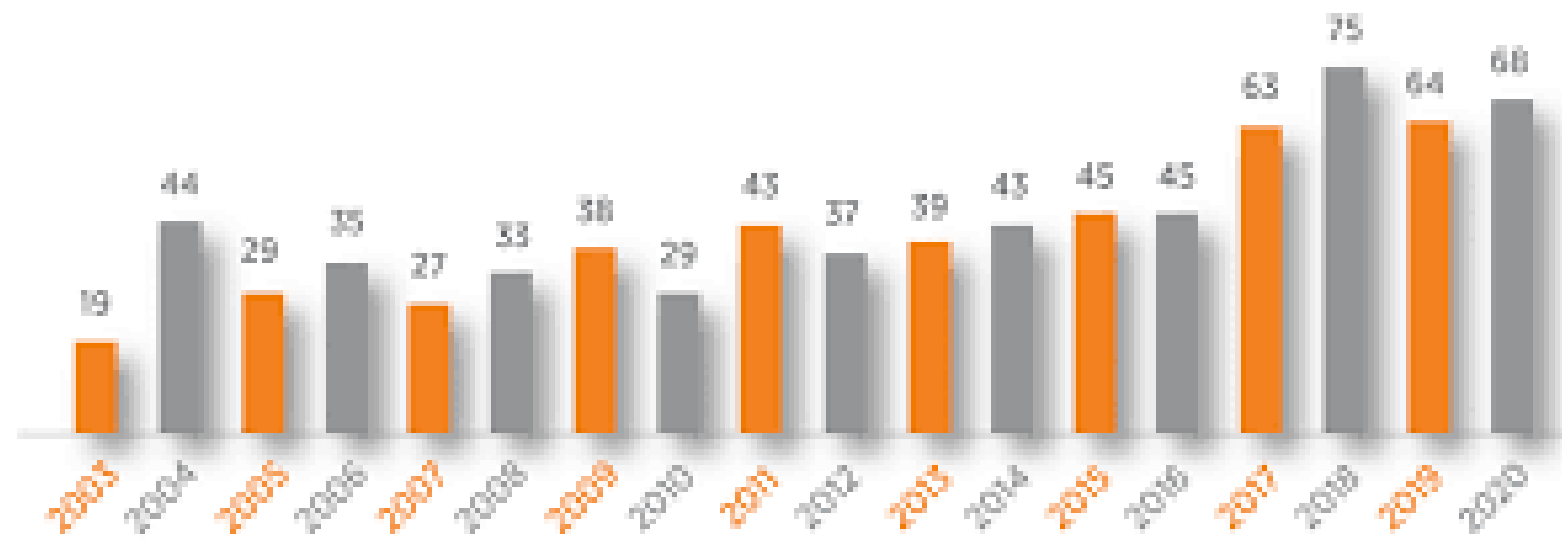
- Significant improvement may be demonstrated by the following examples:
- evidence of increased effectiveness in treatment, prevention, or diagnosis of condition;
- elimination or substantial reduction of a treatment-limiting drug reaction;
- documented enhancement of patient compliance that is expected to lead to an improvement in serious outcomes; or
- evidence of safety and effectiveness in a new subpopulation.
- FDA decides on the review designation for every application.
- However, an applicant may expressly request priority review as described in the Guidance for Industry Expedited Programs for Serious Conditions – Drugs and Biologics.
- It does not affect the length of the clinical trial period. FDA informs the applicant of a Priority Review designation within 60 days of the receipt of the original BLA, NDA, or efficacy supplement.
- Designation of a drug as “Priority” does not alter the scientific/medical standard for approval or the quality of evidence necessary.

Final approval

- After the final approval, the drug becomes accessible for prescription by doctors.
- However, drugs might not come to the market with immediate effect because of patents disagreements, manufacturing issues, or controlled substance designation from the DEA.
- Usually, the pricing is revealed after approval. However, the FDA generally does not consider the pricing of drug or any economics as part of the approval process.
- In contrast, many other nations consider the economic effects of new drugs in their society.



505(b)(2) NDA Approvals by Year Through 2020



Conclusion

- According to the FDA, 30% or less of initial drug candidates proceed through the complete multi-year drug development process, concluding with an NDA approval, after successful completion of the clinical studies and then the review of the clinical data.
- New Drug Applications are usually regulated by the Center for Drug Evaluation and Research (CDER) of the FDA.
- The filing of an NDA, after successful completion of clinical trials, represents a crucial breakthrough for a new drug. An NDA filing is a critical milestone for any organization, and investors keep a close watch on the process. The fate of any pharmaceutical company can change when the FDA passes an NDA approval of a new drug. This step is more crucial for small-cap healthcare companies, as their market value can change once their drug candidate receives approval.

- The 505(b)(2) NDA and 505(j) ANDA pathways are both intended to provide an efficient means for approving non-innovator drug products by harnessing existing information to support the approval package.
- Each of these abbreviated pathways includes its own nuances and regulatory requirements. Understanding which pathway applies to your drug is critical for making good drug development decisions and ensuring that your product gets approved and on the market in a timely manner.
- After the submission of the NDA, the probability of a drug getting approved is very high. Accordingly, healthcare companies that file NDA witness a rise in their share prices even before the response from the FDA.
- If it is not complete, the review team can refuse to file the NDA. If it is complete, the review team has 6 to 10 months to make a decision on whether to approve the drug.

Thank you



Clinical Research and Clinical Practice: A Critical Link

Dr. Sonali Correa

Defining Clinical Research and Clinical Practice

- Clinical Research: "systematic investigation of the etiology, diagnosis, treatment, and prevention of disease"
- Systematic investigation designed to assess the safety and efficacy of drugs, treatments, interventions, or diagnostic tools.
- Focuses on generating new knowledge or validating existing knowledge.
- May include different types: observational studies, randomized controlled trials, cohort studies, and clinical trials.

Defining Clinical Research and Clinical Practice

- Clinical Practice: "the application of clinical knowledge and skills to the care of individual patients"
- The application of medical knowledge to treat patients based on established protocols and guidelines.
- Emphasis on patient care and individual treatment plans.
- Involves diagnosing, treating, and managing patients' health conditions.

Goals

- **Clinical Research:**

- To discover new information that can be generalized to a wider population.
- Aims to advance medical science and improve healthcare practices through evidence-based findings.

- **Clinical Practice:**

- To provide high-quality care tailored to individual patient needs and preferences.
- Focused on improving patient outcomes, quality of life, and overall well-being.

Methodology

- Clinical Research:
 - Employs rigorous scientific methods, including hypothesis testing, statistical analysis, and peer review.
 - Often involves control groups and randomization to reduce biases and validate results.
- Clinical Practice:
 - Relies on the integration of evidence-based guidelines, clinical judgment, and patient values.
 - Adaptability is key; practitioners modify approaches based on individual patient responses.

Participants

- Clinical Research:
 - Involves volunteers who are recruited based on specific inclusion and exclusion criteria.
 - Participants may not represent the broader population; focus is often on specific subgroups (e.g., age, disease state).
- Clinical Practice:
 - Encompasses a diverse patient population with varying demographics, conditions, and comorbidities.
 - Involves ongoing relationships with patients based on trust and communication.

Regulatory Affairs

- Clinical Research:
 - Subject to strict regulatory oversight (e.g., Institutional Review Boards, Institutional Ethics Committees).
 - Must comply with ethical standards, informed consent regulations, and Good Clinical Practice (GCP) guidelines.
- Clinical Practice:
 - Governed by medical boards, licensing authorities, and professional ethics.
 - Practitioners are responsible for adhering to accepted standards of care and best practices in their specific fields.

Outcomes

- Clinical Research:
 - Outcomes are measured in terms of statistical significance and generalizability of findings.
 - Aims to produce publishable results that can influence policy, guidelines, and future research.
- Clinical Practice:
 - Outcomes are measured in terms of patient health, satisfaction, and quality of life.
 - Focus on immediate impacts and individualized care rather than broader statistical outcomes.

CR vs CP Map

